LingAfriq: Senior Audit and Elevation Plan

Prepared for: Tope (Fabian Umole) · **Date:** 17 October 2026 · **Scope:** Flutter app (`lingafriq-mobile-app-clone`, 1,039 Dart files, ~306k lines), Node/TypeScript backend (`node-backend`, 1,120 TS files, 76 route files, 107 models), React admin (`lingafriq-admin`), Python voice service (`lingafriq-voice-service`), curriculum bundles and the pasted Rive avatar plan.

Method. Static review of a read-only copy of the source (keys and media were stripped from most of the copy before review; a few secrets were still in it, listed in §1). I mapped routes, screens, providers and backend routes, then checked each feature against its code. Every finding below cites `file:line` or a reproducible data measurement. Where I could not verify something (a missing media file, production config, the live server) it is marked **[Unverified]**. Nothing was run against production. No code was modified.

How to read this. §0 is the verdict. §1 is the P0 list (fix before anything else). §2 is the feature inventory with a status for each feature. §3 to §15 are the deep sections Tope asked for. §16 is the roadmap. **§17 is the new information architecture (every feature gets a home) and §18 is the complete Feature Catalogue with a deep-dive for every user-facing feature, grouped by hub.** No feature is cut anywhere in this plan; duplicate implementations are consolidated only when the feature itself survives. The companion document **`LINGAFRIQ_AI_MODEL_STACK.md`** gives the commercially licensed model choices, coverage per language, integration plan and costs.

---

0. The verdict in one page

LingAfriq has **enormous surface area and very little load-bearing substance.** The repo contains **319 status Markdown files** in the app root alone (`ALL_FIXES_COMPLETE_FINAL.md`, `PRODUCTION_READY_100_PERCENT_COMPLETE.md`, `WORLD_CLASS_IMPLEMENTATIONS_COMPLETE.md`…) and roughly 150 more in the backend. The code tells a different story:

1. **The curriculum, the product's core, is mostly placeholder or template output.** Three of the four curriculum bundles contain literal placeholders (`"word": "word1a"`, `"meaning": "example meaning"`, `"question": "Sample MCQ"`, `"answer": "a"`). My script found **24,234 of 48,847 vocabulary entries (49.6%) across the bundles are placeholders.** The "authentic" A1–C2 file, the best content in the repo, has **only 85 to 111 unique vocabulary items per language**, recycled across 126 lessons per language. **126 of 126 dialogues per language repeat lines.** The **same two grammar sentences appear in 1,546 lessons**, and one of them is false for most of the languages it is attached to (details in §6). No curriculum file references a single audio file.
2. **Pronunciation and tone feedback, the main reason to choose an African-language app over a phrasebook, is mostly fake.** Pronunciation Duel never opens the microphone and returns a random 75–95% (`screens/games/pronunciation_duel_game.dart:105-129`). Karaoke's pitch score is `60 + rng.nextInt(35)` (`pronunciation_karaoke_screen.dart:155`). The tutor's "Rhythm" score is `score * 0.9` (`tutor_pronunciation_mode_screen.dart:114`). The voice service's tone endpoint is labelled `# Placeholder` and reuses text similarity (`lingafriq-voice-service/main.py:262-274`).
3. **Private conversations are not private.** Any signed-in user can join any socket room (`node-backend/src/services/socket.service.ts:263-268`), and private rooms are named `private_<smallerId>_<largerId>` (`:106-109`), so they can be guessed. A client can also claim to be any user id (`:249-260`) and then receive that user's snaps and streak events (`:84-90`). Read receipts, presence and reactions go to **every connected user** through `io.emit` (`:817, 830, 840, 852`).

4. **Children are in scope, but child safety is not.** Onboarding collects `'child'` as an age category (`unified_onboarding_screen.dart:745`). The app ships private DMs, voice/video calls, Snapchat-style disappearing media and a public X-style feed. There is **no age gate, no parental consent and no guardian controls anywhere** (grep for `parental|coppa|guardian|age.?gate` finds nothing relevant). Chat "moderation" is a **4-word English list plus 7 African words matched by substring** (`services/toxicityFilter.ts:9-30`). It flags the Yoruba word *ode* inside "code" and "mode" and flags the English word "were", and it only records a score; nothing is blocked.
5. **Secrets are committed.** A live-format **Firebase Admin SDK service-account private key** for project `lingafriq-15d2a` sits at `node-backend/sample.json`, and `.gitignore` does not cover it. An **Apple `.p8` private key** (`AuthKey_QWGCD23R6B.p8`) is in the mobile repo root and is also not ignored. The parent reported Admin SDK JSONs inside `node-backend/dist/services/` on the Mac. `build.yml` compiles **five paid AI provider keys into the app binary** (`.github/workflows/build.yml:99-113`), where anyone can extract them.
6. **The economy and leaderboards can be cheated with one HTTP call.** `POST /api/gamification/currency/award` ("admin/system use") lets any signed-in user mint unlimited currency for themselves (`routes/gamification/currency.route.ts:100-125`). `POST /accounts/update_points/` adds any positive number of points (`controllers/accounts.controller.ts:360-378`). `POST /api/gamification/xp/award` allows 10 × 1,000 XP per minute (`routes/gamification/xp.route.ts:22,37`).
7. **Updates log every user out and repeat onboarding.** Any semver change, even 1.6.1 → 1.6.2, clears tokens, credentials and onboarding (`providers/auth_provider.dart:54-75`). Retention resets with every release.
8. **A huge, ambitious feature set with no spine connecting it to the course.** There are WhatsApp-style (`wa-*`), Snapchat-style (`snap-*`) and X-style (`x-*`) surfaces, a Clubhouse-style audio room, villages, tribes, classrooms, micro-mentors and a culture magazine (`my_app.dart:277-480`). Each has real potential, but none is yet tied into the learning loop. On top of that, **88 finished screens are unreachable** from `main.dart` (330 files by reachability; 183 files that no file imports at all), including the moderation tools, native-speaker verification, offline downloads, lesson recap, the guardian dashboard and the Tone Forge game: the very tools this product most needs. §17.5 revives 25 of them.

The opportunity is real and unclaimed. No app in 2026 teaches African languages with tone-accurate audio, native-validated content, real pronunciation and tone feedback, and culturally grounded AI. LingAfriq already has the right instincts: the Intron proxy, tone service scaffolding, a learner model, persona registries with primary-source fields, CEFR framing and a village metaphor. The plan below is to **harden, connect and deepen every feature**: one coherent learning loop that every feature plugs into, one trustworthy social layer, one design system, and content that can survive a native speaker reading it. Nothing is cut; every feature gets a home in the new IA (§17) and its own elevation (§18).

---

1. P0: fix now (security, safety, integrity, data loss)

#	Finding	Evidence	Fix	Effort
P0-1	Firebase Admin SDK private key committed	`node-backend/sample.json:4-5` (service_account, `firebase-adminsdk-us33e@lingafriq-15d2a`); `.gitignore` covers `*firebase-adminsdk*.json` but not `sample.json`; `messagingService.ts:29` loads a key JSON from `src/services/`; parent reports `dist/services/*firebase-adminsdk*.json` on the Mac	**Revoke the key in GCP IAM today**, scrub git history (`git filter-repo`), move to Workload Identity or a secret manager, add gitleaks/trufflehog pre-commit and CI gates	S

#	Finding	Evidence	Fix	Effort
P0-2	Apple `.p8` key in repo root	`lingafriq-mobile-app-clone/AuthKey_QWGCD23R6B.p8` (PEM private key), not in `.gitignore`	Revoke in the Apple Developer portal, purge from history, store as a CI secret only	S
P0-3	Paid AI keys compiled into the app	`build.yml:99-113` passes `GROQ_API_KEY`, `HUGGINGFACE_TOKEN`, `STABILITY_AI_KEY`, `REPLICATE_API_KEY`, `LEONARDO_API_KEY` as `--dart-define`; read in `lib/services/env_config.dart:43-60`; direct calls in `lib/providers/persona_providers.dart:16-45`, `ai_chat_provider_groq.dart:1404,1762,1967`, `ai_chat_provider.dart:44-58`	Rotate all five. Delete every direct provider call from the client. All AI goes through an authenticated, metered backend gateway (§7)	M
P0-4	Any user can listen to any private chat	`socket.service.ts:263-268` (`join_room` joins any room); `:106-109` predictable room ids	Server-side membership check on every join (room ACL from DB). Derive room ids server-side and make them opaque (UUIDv7)	S
P0-5	Socket identity spoofing	`socket.service.ts:249-260` overwrites `userId` from the client payload; `:84-90` routes per-user events by that map	Delete `user_connected`. Identity comes only from the verified JWT	S
P0-6	Global metadata leak	`io.emit` for delivered/read/presence/reactions/status/story (`:817,830,840,852,861,881`); full online-user list to everyone on every connect (`:915-921`), which is also O(N^2)	Emit to room or participant sockets only. Presence is limited to contacts and opt-in	S
P0-7	Self-mint currency	`routes/gamification/currency.route.ts:100-125`: `requireSignin` only, amount unbounded (`:44`)	Admin/service-only. Grants come from server-side event rules (lesson completed → reward), never from a client amount	S
P0-8	Self-award points/XP	`controllers/accounts.controller.ts:360-378`; `routes/gamification/xp.route.ts:22,37`	Server computes XP from verified events with idempotency keys (`sourceId` unique per user) and daily caps	S
P0-9	Open LLM proxy	`controllers/aiCompletion.controller.ts:65-79`: client supplies `systemPrompt`, any `role` (incl. `system`), `max_tokens` unbounded; route has only `requireSignin` (`routes/ai.route.ts:12`)	Server-owned prompt templates selected by `mode` enum; clamp tokens; per-user quotas and cost budgets; strip client `system` roles	M
P0-10	No child safety or age assurance	`unified_onboarding_screen.dart:745` collects `child`; DMs, calls, snaps, public feed exist; no consent flow	Age gate before account creation. Under-13: no DMs/calls/snaps/public feed, guardian consent (NDPA 2023, GDPR Art. 8, COPPA if US users). 13–17: restricted defaults. See §13	M
P0-11	Moderation is cosmetic	`services/toxicityFilter.ts:9-73` (11 terms, substring match, score only)	Layered moderation (§9): classifier + per-language lexicons with word boundaries + image/video hash matching + human queue + block/report UX	M

#	Finding	Evidence	Fix	Effort
P0-12	"Disappearing" snaps never disappear	soft delete only (`controllers/snap.controller.ts:104-111`); `media_url` is any client-supplied URL (`:52-68`)	Hard-delete blobs on expiry. Media only via signed upload URLs to your own bucket. Reject external URLs. Under-18: disable snaps	S
P0-13	Persona missions silently fail in release	`persona_providers.dart:22-45` uses the client Groq key; `build-and-release.yml:216,234` injects no AI key; `historical_roleplay_controller.dart:322-329` returns "I hear you. Let us continue." on any error	Route through backend. Show an honest error state with Retry	S
P0-14	Persona safety filter blocks harmless learners	`roleplay_safety_filter.dart:24-30,158-160`: substring matching blocks "**skill**" (contains *kill*), "**whatever**" (contains *hate*), and **"discrimination" when talking to Mandela**	Replace with a model-based policy classifier plus persona-specific rules. Historical violence is discussable at an age-appropriate level	S
P0-15	Persona told to deny being AI	`node-backend/src/services/historicalPersonality.service.ts:314` "Do not mention… being an AI model"	Always disclose ("AI recreation based on sources"). EU AI Act Art. 50 transparency duties apply from 2 Aug 2026; minors are in the audience	S
P0-16	Curriculum is placeholder or template	§6 measurements	Freeze marketing claims. Pull placeholder bundles from the build. Start the content pipeline (§6)	L
P0-17	False tone and grammar teaching	`assets/data/game_content.json` word 1: "High tone on àárọ̀ marks respect" (àárọ̀ carries low tones; respect comes from the honorific *Ẹ*). The phonetic guide equals the tonal note for **1,409/1,409** words. "Affixes carry tense and person" is attached to Yoruba, Pidgin, Twi and others; "SVO default" is attached to Somali and Amharic (both SOV)	Pull until native-reviewed (§6 pipeline)	M
P0-18	Fake pronunciation scores	`pronunciation_duel_game.dart:105-129` (no recording, random score), `pronunciation_karaoke_screen.dart:155`, `tutor_pronunciation_mode_screen.dart:114`	Remove the random scores today. Show "practice mode, no scoring" until real scoring ships (§7)	S
P0-19	Pronunciation backend lies with HTTP 200	`services/advancedPronunciation.service.ts:26-28` defaults to `localhost:8001-8003`; on failure returns `200 {overall_score:0…}` (`:211-224`), so the client's Whisper fallback never runs (`tutor_pronunciation_mode_screen.dart:90-95`)	Return 503 with `retryable:true`. Client falls back or shows "unavailable"	S
P0-20	Every release logs everyone out	`providers/auth_provider.dart:54-75`	Delete the semver reset. Use schema-versioned migrations for local data	S
P0-21	Onboarding answers corrupted	`unified_onboarding_screen.dart:354` stores `Map.toString()`; `providers/onboarding_provider.dart:496-503` `fromJson` fails silently and resets	`jsonEncode`, typed model, migration for existing corrupted values	S
P0-22	Offline queue drops data	`services/offline/offline_handler.dart:120-152` marks ops `completed` and removes them while `_syncOperation` is a stub ("For now, simulate sync"); used by `services/lesson/lesson_item_service.dart:22`	Single durable outbox (§13), delete the stub	M

#	Finding	Evidence	Fix	Effort
P0-23	Push spam, duplicated	`workers/notificationScheduler.ts:17-80`: every 2 h, no "already sent today" check, server timezone, N+1 queries over all users; scheduler starts both in-process (`app.ts:216-221`) and in the separate PM2 worker (`ecosystem.config.js:27-28`) → up to **24 "streak at risk" pushes a day** if both PM2 apps run **[Unverified on server]**	One scheduler (leader-elected job queue), per-user local-time window, max one reminder per day, quiet hours	S
P0-24	Non-commercial model licences in a paid app	`node-backend/mms-tts-service/main.py:46` uses `facebook/mms-tts-*` (CC-BY-NC 4.0); `coqui-xtts-service/app.py:64` uses XTTS-v2 (Coqui Public Model License, non-commercial); `NLLB_API_URL` in `secrets_manager.dart:85` (NLLB-200 is CC-BY-NC)	Legal review. Swap to the commercial replacements in `LINGAFRIQ_AI_MODEL_STACK.md` §1.1 (Intron/Spitch/Azure TTS, custom isiXhosa voice, Google/Azure MT, MADLAD-400)	M
P0-25	Login enumeration, no lockout	`controllers/auth.controller.ts:120,135` ("User not Found" vs "Password is not a match"); limiter is 10/min per IP only	Generic error; per-account progressive delay; breached-password check; passkeys (§13)	S
P0-26	Unofficial Google Translate scraping in prod	`services/freeGoogleTranslate.service.ts:7` (`@vitalets/google-translate-api`)	Remove. ToS and IP-ban risk, and quality is poor for tonal orthography	S
P0-27	Purchases not verified server-side	No RevenueCat webhook or receipt route in `routes/subscription.route.ts`; RC keys are read from `.env`/secure storage (`revenuecat_service.dart:21-25`) and not injected by the release workflow **[Unverified: may be provided another way]**	RevenueCat webhooks → entitlement table; backend enforces premium gates	M

P0 count: 27. Most of them are one to three days each. P0-1 to P0-9 should be done this week.

---

2. Feature inventory and verified status

Status key: **Works** (wired end to end, quality aside) · **Partial** (wired, with material defects) · **Broken** (fails in normal use) · **Fake** (shows outcomes it does not compute) · **Stub** (scaffold only) · **Dead** (not reachable from the app).

#	Feature	Entry point(s)	Status	Headline defect (evidence)
1	Startup / splash	`main.dart`, `screens/splash/splash_screen_material3.dart`	Partial	About 15 sequential awaited inits before `runApp` (`main.dart` 40-200); 1.2 s artificial delay (`splash…:28`); 30 s timeout falls back to Login, so offline users are kicked out (`:39-48`); Sentry starts after everything else, so startup crashes are not captured
2	Onboarding	`OnboardingScreenMaterial3` → `UnifiedOnboardingScreen` (9 steps)	Partial	9 question screens before any learning; "learning style" step (VARK myth) (`:93`); answers stored as `Map.toString()` (`:354`); 3 other full onboarding flows are dead (`enhanced_*` 2,530 lines, `kijiji_*` 1,629, `modern_*` 750)
3	Placement test	`placement_test_screen.dart` via dialog after onboarding	Partial	Offered in an `AlertDialog` after 9 screens; `placement_quiz_screen.dart` is a duplicate

#	Feature	Entry point(s)	Status	Headline defect (evidence)
4	Auth	`world_class_login_screen.dart`, `world_class_signup_screen.dart`, email verify, reset	Partial	Enumeration (P0-25); 7-day HS256 token, and the "refresh" endpoint just re-signs an unexpired access token (`auth.controller.ts:320-364`); `is_admin` in the JWT payload (DB re-checked server-side, good); no passkeys/OAuth; 4 legacy auth screens remain
5	Home / dashboard	`tabs_view_material3.dart` (Home, Courses, AI, Social, Profile) + 36-item drawer	Partial	Feature dump IA; two dashboards (`modern_dashboard_screen.dart` unreachable, consolidated into Today)
6	Courses / curriculum map	`'curriculum'` → `LessonsMapEntryScreen`; `services/curriculum_service.dart`	Partial	Master index is loaded first from the placeholder `lingafriq_FINAL_curriculum/master_index.json` (`curriculum_service.dart:113-118`); "Coming Soon" for missing levels (`curriculum_screen_material3.dart:367`)
7	Lesson flow	`'lesson-flow'`, `screens/lesson*`	Partial	10 declared stages, but content is template-driven (§6) and there is no audio
8	Grammar hub	`grammar-hub`, `grammar_exercise_screen.dart` (1,300 lines)	Partial	Grammar copy is generic (§6)
9	Vocabulary / flashcards / SRS	`my-vocabulary`, `flashcard-review`, `local_database_service.dart:226`	Partial	**Four separate memory models**: SM-2 locally (`local_vocabulary.dart:13`), HLR in the learner model (`learning/core/hlr_forgetting_curve.dart`), backend `aiChatSRS` and `gameSRS` models. None of them is FSRS
10	Listening / writing practice	`listening-practice`, `writing-practice`	Partial	No audio corpus ships (§6)
11	Tutorials / features guide	`features_guide`, `game_onboarding_overlay.dart`	Partial	Feature tours rather than skill tutorials
12	Quizzes (language, history, random, unit)	`language_quiz/`, `history_quiz/`, `random_quiz/`, unit quizzes	Partial	Unit quizzes in placeholder bundles are `"Sample MCQ"` / `"a"`; no item analysis or calibration
13	Games (≈45 screens)	`screens/games/*`, `games/*`	Partial/Fake	Random pronunciation scores; tone trainer forces a 3-tone H/M/L model on all languages (`tone_trainer_game.dart:63`); finished game code is unreachable (Tone Forge and the mastery results screen are revived; older templates and loaders are consolidated into GameKit, §17.5)
14	Pronunciation / speak lab / tone	`tutor_pronunciation_mode_screen.dart`, `speak-engine-lab`, `tone-rhythm-trainer`; backend `advancedPronunciation.service.ts`; `services/tone-service/app.py`	Fake/Broken	P0-18/19; tone service falls back to "Install librosa" (`app.py:241-248`) and compares tone strings position by position

#	Feature	Entry point(s)	Status	Headline defect (evidence)
15	Polie AI chat	`polie_workspace_screen.dart` (6,485 lines), `ai_chat_provider_groq.dart` (2,516)	Partial	Direct Groq calls with a client key, or a non-streaming backend fallback (`:1302-1330`); 4 dead chat screens
16	AI tutor modes (story, grammar, dialogue, assess, translation, pronunciation)	`screens/tutor/*`	Partial	Prompt-only, with no grounding in curriculum items; JSON parsing with no schema validation on the server
17	Roleplay scenarios	`roleplay_scenario_selection_screen.dart`, `ai/modes/roleplay_engine.dart`	Partial	Same client-key issue
18	Historical Personas + missions	`historical-personas` → `PersonalitySelectionScreen` (backend), `persona_missions/*` (client Groq)	Partial/Broken	Two persona systems (27 client personas in `historical_persona_registry.dart`, backend seeds in `scripts/extendedHistoricalPersonas/*`); missions fail silently (P0-13); no retrieval grounding, `sources: []`, `confidence: 0.9` hard-coded (`historicalPersonality.service.ts:195`); factual slips (Nkrumah listed as `English/Twi`, `personas-e01.ts:20`; he was Nzema)
19	Live translate	`live_translate_screen.dart`, socket `live_translate_event`	Partial	Built on the unofficial Google Translate scraper (P0-26) **[Unverified which path is live]**
20	Point-and-say (OCR / object detection)	`point-and-say`, ML Kit	Partial	ML Kit labels are English; the translation path is the same issue as #19
21	Culture magazine / heritage archive / living dictionary	`cultural-hub`, `flb-heritage-archive`, `living-dictionary`	Partial	`culture_magazine_screen.dart` is an unreachable older version (consolidated, §17.5); backend scraper (`cultureScraper.service.ts`) raises copyright questions
22	Tribes	`tribe-selection`, `my-tribe`, `tribal-duel`, `inter-tribe-leaderboard`	Partial	Duplicate routes (`tribe`, `tribe-hub`, `tribe-discovery` all → selection); the channel-subscription handlers in `socket/handlers.ts:107-145` are never registered (only `attachSocketInstance` is used, `socket.service.ts:192`), so tribe/leaderboard live updates have no subscribers, while `join_room` lets anyone join any `tribe:*` room
23	Villages	`villages-hub`, `language-village`, market/cafe/elder hut	Partial/Fake	`flashcard_focus_screen.dart:26-34` shows **mock participants** as live co-learners
24	Classrooms (live, LiveKit)	`classroom-lobby`, `classroomV2.route.ts`	Partial	`live_classroom_screen.dart` and `classroom_chat_livekit_screen.dart` are unreachable older versions (consolidated into the live screen, §17.5); no teacher tooling for assignments/grading in the app
25	Global chat	`chat.route.ts:69`, socket `send_message`	Partial	No moderation (P0-11); rooms by language name

#	Feature	Entry point(s)	Status	Headline defect (evidence)
26	Private chat (WhatsApp-style)	`private-chat-inbox`, `wa-*`	Broken (privacy)	P0-4/5/6; no E2EE; `private_chat_screen_material3.dart` unreachable (consolidated)
27	Snaps / stories (Snapchat-style)	`snap-*`	Broken (safety)	P0-12; streak counts are client-supplied (`socket.service.ts:888-905`)
28	X-style feed	`x-*`, `feed.route.ts`	Partial	Public by default (`feed.route.ts:23`); no minor protections
29	Social audio rooms	`social_audio/*`, `socialAudio.route.ts`	Partial	No recording consent or moderation tooling **[Unverified]**
30	Friends, connections, friend quests	`connections`, `friend-quests`	Partial	`friendQuest.controller.ts:325` takes `amount` from the body
31	Micro-mentors	`micro-mentor-hub`, staff reports	Partial	Paid/peer tutoring without safeguarding (background checks, session recording policy)
32	UGC / voice contribution	`ugc`, `contribute_voice`	Partial	`native_speaker_contribution_screen.dart` and `lesson_item_verification_screen.dart` are unreachable (revived in §17.5/§18.6); no contributor consent or licensing terms for voice data
33	Gamification (XP, 3 currencies, leagues, hearts, badges, streaks, magic items, ancestral tree)	`routes/gamification/*` (15 route files)	Broken (integrity)	P0-7/8; 3 currencies (`ngwenya`, `cowries`, `ancestral_beads`) with no clear sinks
34	Passport / proficiency credential	`passport-proctored`, `passport-credential`	Partial	A "credential" built on template content and random scoring is a liability
35	Avatars (Rive)	`lib/avatars/*` (2,594 lines of scaffolding), `assets/rive/`	Not built yet	**The Rive avatars have not been made yet** (confirmed by Tope, 17 Oct 2026), so there are no visuals to judge. The code scaffolding should be replaced rather than finished: `avatar_engine.dart:68-90` references 12 `.riv` files that don't exist, wires deprecated state-machine inputs (`:106-219`), and the three avatar engines disagree on parameters. §4 is a forward design and production brief
36	Profile / settings	`settings_screen_material3.dart` (1,286 lines)	Partial	Not audited line by line
37	Notifications	FCM, `workers/notificationScheduler.ts`	Broken	P0-23; streak-loss copy ("Your N-day streak is at risk!")
38	Subscriptions	`revenuecat_service.dart`, `subscription.route.ts`	Partial	P0-27
39	Offline / sync	7 competing systems: `offline_handler`, `background_sync_service`, `sync_v2_service`, `persisted_outbox_service`, `backend_sync_provider`, plus 2 feature queues	Broken	P0-22; AES via the `encrypt` package default mode (SIC/CTR) with no authentication tag (`cache_encryption.dart:25-28`)

#	Feature	Entry point(s)	Status	Headline defect (evidence)
40	Localization (UI)	`l10n/*.arb` (20 locales)	Partial	Only **103 keys** exist; **1,104 hard-coded `Text('…')` literals** vs **35 localization lookups**; Yoruba strings lack tone marks and subdots (`app_yo.arb`: "E kaabo si LingAfriq", "Tesiwaju", "Ko")
41	Accessibility	—	Partial	471 `Semantics(` uses (good start); no audit of text scaling, contrast, captions for audio, or reduced motion
42	Admin panel	`lingafriq-admin` (React 18, Vite 4, Redux)	Partial	No native-review workflow, no content versioning, 0 tests; session token in `sessionStorage` (`store/reducers/userReducer.ts:12-40`)
43	Voice services	`lingafriq-voice-service/main.py`, `node-backend/{whisper-asr,mms-tts,tts-mms,coqui-xtts,voice}-service`, `intron.route.ts`	Partial	No auth on the voice service endpoints; the **Intron proxy is never called by the app** (0 references in `lib/`); placeholder tone endpoint; licensing (P0-24)
44	Platform: CI/CD, observability, tests	`.github/workflows/*`, Sentry, Jest	Partial	`flutter analyze --no-fatal-warnings`; 64 Flutter test files for 306k lines; 52 backend tests; single PM2 fork instance with in-memory presence and rate limits; workers duplicated

Feature areas audited: 44 (this table). The complete per-feature catalogue, **151 user-facing features each with its own deep-dive**, is in §18. Unreachable code: **183 Dart files / 49,074 lines are never imported** by any file; a stricter reachability scan from `main.dart` finds **330 unreachable files, 88 of them screens**. These are not deleted: §17.5 revives 25 screen-level features, consolidates older duplicates into the surviving feature, and moves 3 developer examples out of the build.

---

3. Product thesis, information architecture and design system

3.1 Design read (Playbook law 1)

LingAfriq is a daily, voice-first practice companion that gets diaspora, heritage and curious learners speaking African languages correctly, tones included, and gives them a place to use the language with real people. Every screen should serve one of three jobs: **learn it** (comprehensible input + retrieval), **say it** (pronunciation, tone, conversation), **live it** (community, culture, people). Every existing feature is mapped to at least one of these jobs (§18); a feature that doesn't yet serve one is redesigned until it does, not removed.

3.2 What is wrong today

- **Breadth without a spine.** `my_app.dart:277-480` registers roughly 120 named routes, including WhatsApp-, Snapchat- and X-style surfaces. The 36-item drawer (`screens/tabs_view/app_drawer/app_drawer_material3.dart`) is a symptom: features compete for attention instead of feeding one learning loop, and none has yet been taken to its full potential. The fix is placement and depth (§17, §18), not removal.
- **Seven design systems at once:** `utils/african_theme.dart`, `app_colors.dart`, `design_system.dart`, `modern_griot_design_system.dart`, `pan_african_design_system.dart`, `polie_design_tokens.dart`, `stitch_theme_extensions.dart`, plus `theme/` and `pan_african_ui.dart`. There are **740 hard-coded**

`Color(0x…)` literals and **10+ font families** (Nunito, Plus Jakarta Sans, Playfair, Lato, Philosopher, IM Fell English, Cinzel, JetBrains Mono…). Bundled Dosis is almost unused.
- `google_fonts` **fetches fonts at runtime.** On a first offline launch the fonts fall back, and stacked Yoruba diacritics (ọ, ẹ) can mis-render in fallback fonts **[Unverified on device]**.
- **Every state does not exist** (Playbook law 8). Errors are caught and swallowed in most screens, for example `catch (e) { /* fall through */ }` in the tutor and the persona fallbacks above. Users see wrong data rather than an error state.

3.3 Target IA (five tabs + Me; every feature kept)

Tab	Contains	Brings together
Today	One adaptive session (10–15 min) mixing review, new input, speaking and a cultural moment; challenges, quests, events, streak	Home, dashboard variants, daily goals, daily challenges, quests, seasonal events
Learn	CEFR-aligned Path, lessons, Words & Review, Grammar, Listening, Writing, Tone Gym, Culture shelf, Passport, Downloads	Courses, curriculum, authentic path, learning path, grammar, vocabulary, magazine, heritage
Speak	Polie with all tutor modes, Scenarios, Historical Personas & Missions, Shadowing, Code-Switch and Respect coaches	AI tab, tutor modes, persona missions, speak-engine lab
Play	The Arcade: every game on skill shelves, Traditional Games, Multiplayer	All three game catalogues, cultural and traditional games
Village	Map, Circles, Square feed, Chats, Status, Clips, Radio, School, Elder Hut, Corrections, Create & Contribute	Social hub, tribes, villages, wa-, *snap*-, x-*, social audio, classrooms, mentors, UGC
Me (avatar, top-left)	Avatar, progress, achievements, leagues, family, Plus, settings, **All features** directory	Profile, achievements, badges, leaderboards, settings, features guide

Global **Search**, the **Lens** camera (Point & Say, Live Translate, Import Media) and the long-press **Word Card** (Living Dictionary) are available on every tab. Games are a full destination (Play) **and** are surfaced inside Today and Path as practice formats, so they update the learner model. The full nav map, sitemap and unlocking rules are in §17; every feature's place is in §18.

3.4 Design system: one source of truth

- **Tokens** in one package (`lib/design/`): colour roles in OKLCH ramps (Playbook law 3, one accent with one meaning), a 4-pt spacing scale, radius, elevation, motion curves (ease-out `cubic-bezier(0.23,1,0.32,1)`, ≤300 ms for product UI, Playbook §6.2–6.3). Generate them from a single JSON shared with the admin panel and the Rive files (colours bound through Rive data binding, §4).
- **Typography for African orthographies.** Pick faces by **glyph coverage**, not taste alone. Test every candidate with: `Ẹ kú àárọ̀ · Ọ̀ṣun · Ị́ bịa · ɓakɔ dan yaya · ɛyɛ · ŋ ɲ ɔ̃ · ሰላም · ጌዜ · ᏔᎧᏣ · ⵜⴰⵎⴰⵣⵉⵖⵜ · 𞤤𞤢𞤪𞤢𞤲𞤣𞤫` . Recommended: **SIL Andika** (designed for literacy, unambiguous letterforms and diacritics) or Noto Sans for target-language text; **Noto Sans Ethiopic / N'Ko / Tifinagh / Adlam** for those scripts; one characterful display face verified against the string. Bundle all of them (no runtime fetch). Target-language text never below 18 sp, with leading ≥1.5 so tone marks don't collide.
- **A signature visual language for tone: the "Tone Lens".** Every target-language word can show a thin pitch contour above it (H/M/L/rising/falling/downstep), derived from orthography and validated audio. It looks the same in lessons, chat, games, personas and the dictionary. It is LingAfriq's equivalent of Duolingo's owl: instantly recognisable and actually pedagogical. It must never be colour-only (accessibility); use contour shape plus optional colour.
- **Cultural art direction without cliché.** The Playbook flags the generic "tasteful" palettes, and kente stripes on every card are the same trap. Commission a **pattern library per culture** (adire, uli/nsibidi motifs, kente, bogolan, Ndebele geometry, Ethiopian tilf) used sparingly as unit and region identity, credited to the artists, with a cultural consultant's sign-off.

- **States kit:** skeletons matching layout, empty states that point forward, error anatomy (what happened, why, what to do; Playbook §15.4), offline state, "AI unavailable" state, and "content under review" state.
- **Motion:** celebrations only in the rare tier (lesson complete, level up). Answer feedback ≤160 ms with haptic plus sound. Honour reduced motion (gentler, not zero; Playbook §6.8).
- **Kill the slop tells** (Playbook §9): emoji as icons (the notification copy uses 🔥), fake-precise numbers (the "confidence 0.9" chips), "World-class"/"Elite" copy in UI strings.

---

4. Avatar system: design and production brief (avatars not yet built)

Status: the Rive avatars have **not been made yet**, so this section judges no existing visuals. It is a brief for when Tope, or Cursor working with Tope, builds them. It takes Tope's pasted Rive plan (hierarchy, 0–100 expression numbers, Triggers (Boolean), the Idle…LessonComplete states, the 200×200 artboard, the hair library, Solos/visibility toggles, ViewModel data binding) as the starting point. It keeps what is strong, flags what to change *before* any artwork is drawn, and specifies the contract, pipeline, acceptance tests and build order. The only thing that exists today is Dart scaffolding (`lib/avatars/*`), which should be rebuilt against this contract.

4.1 What to keep from the pasted plan

A modular hierarchy (head, face parts, hair, body, clothing), a state list close to what a tutor needs (Idle, Listening, Thinking, Speaking, Correct, Incorrect, Celebrating, Encouraging, Hint, LevelUp, LessonComplete), idle micro-motion (breathing, blink variance, eye darts), anticipation, overshoot and follow-through, and a nod to data binding. That is a solid start.

4.2 What to change before building (decide now; it is cheap before artwork exists and expensive after)

1. **Don't build on the deprecated contract.** The plan is built around state machine *inputs* (`Triggers (Boolean)`, `Numbers: emotion, energy, confidence`). As of rive 0.14.x, Rive marks state-machine input and text-run access as deprecated in favour of **View Model data binding** (rive 0.14.10 changelog). The existing scaffolding pins `rive: ^0.12.4` (`pubspec.yaml`) and `avatar_engine.dart:106-219` wires `SMINumber` / `SMITrigger`, so replace it rather than extend it. Plan for **rive ≥0.14.11 with `rive_native`, `RiveWidgetController`, `DataBind`** from day one.
2. **"Triggers (Boolean)" is a category error.** Triggers and booleans are different property types. `correctAnswer`, `celebrate` and the others are **Triggers**; `isListening` and `isSpeaking` are **Booleans**, or better, one **Enum**.
3. **0–100 magic numbers for categorical things.** "Eyes: Neutral (0), Happy (33), Excited (66), Curious (100)" and "Mouth: Neutral 0 … Frown 100" encode *categories* as a continuous number. Blending between 33 and 66 produces in-between faces nobody designed, and the codes are undocumented. Per the Rive skill guidance: **closed sets are Enums** (`expression: neutral|happy|proud|curious|concerned|surprised|thinking|encouraging`). **Continuous dimensions are normalised Numbers 0..1** (`intensity`, `energy`, `gazeX/Y` in −1..1). The specs already disagree three ways, so settle one contract before drawing: `assets/rive/README.md` says `emotion` 0–5 and `confidence` 0–1; Tope's plan says 0–100; `avatar_engine.dart` adds `mouthOpenness/lipRounding/mouthShape`.
4. **No lip-sync contract.** "Jaw movement for speech" is not lip-sync. A speaking tutor in a *pronunciation* app needs **viseme-accurate mouths**. The current `lip_sync_engine.dart:184-230` maps *letters* to visemes ("simplified version, production would use phoneme analysis"). Plan the real pipeline from

day one (§4.3), because letter mapping fails on digraphs (Yoruba *gb*, *kp*), nasal vowels, Zulu clicks (c/q/x) and Amharic syllabary.

5. **One monolithic state machine.** Body states, facial expression, gaze, blink and mouth all fight for the same timeline. Use **layers**: L1 body/gesture, L2 expression, L3 gaze/head, L4 blink (autonomous), L5 mouth/viseme (driven every frame during speech). Layers must not key the same properties (Rive skill: "avoid layers unintentionally animating the same property").
6. **No nested artboards / components.** Hair, headwear and garments should be **nested artboards swapped through Artboard properties** in data binding, not visibility toggles of every variant (file size and draw-call bloat). The plan's "Rive Solos or Visibility toggles" approach does not scale past about 10 options.
7. **No cultural design bible.** The plan's hair library is "Afro, Crop, Braids, Ponytail, Locs, Modern". There are no named regional styles (Fulani braids, Shuku, Bantu knots, Irun kiko threading, cornrows, Ethiopian *albaso*), no headwear (gele, fila, kufi, tignon, duku, Maasai beadwork, Hausa *hula*), no textiles, no skin-tone system (needs at least 10 tones with correct undertones and lighting), no scarification or tribal marks policy (cultural sensitivity: opt-in, reviewed), and no disability representation (glasses, hearing aids, wheelchair-seated pose, vitiligo, albinism, which matters in African contexts).
8. **64 px legibility is ignored.** A 200×200 artboard scaled to 64 px loses brows and pupils. Silhouette and expression must read at 32, 48 and 64 px. Make a dedicated **"chip" artboard**: simplified face, exaggerated brows and mouth, 2-px minimum strokes at 64 px, no thin hair strands, and test it as a 1-bit silhouette.
9. **Add a runtime, performance and accessibility contract.** The plan needs: file-size budget per artboard, draw-call budget, renderer choice (Rive Renderer vs Flutter/Impeller), offscreen pause, battery behaviour, reduced motion (idle loops stop, blinking continues), and screen-reader semantics ("Polie looks proud").
10. **Add economy and identity design.** Avatars are a retention engine only if they express identity and progress. Decide up front what is earned, what is bought, what shows mastery, and what is shared (§4.4).

4.3 Production contract (View Models) to build against

```
File: lingafriq_cast.riv
Artboards: Character (full), CharacterChip (64px), Bust (chat/persona), + nested: Hair/*,
Headwear/*, Top/*, Bottom/*, Accessory/*, Prop/*
State Machine: CharacterSM with layers:
  L1 Body      (activity enum)
  L2 Face      (expression enum + intensity)
  L3 Gaze/Head (gazeX, gazeY, headTilt)
  L4 Blink     (autonomous; blinkRate)
  L5 Mouth     (viseme enum + jawOpen, driven per frame)

ViewModel CharacterVM
  activity:   Enum(idle, listening, thinking, speaking, celebrating, encouraging, hinting,
sleeping)   app→rive
  expression: Enum(neutral, happy, proud, curious, concerned, surprised, playful, focused)
app→rive
  intensity:  Number 0..1  (default 0.6)
app→rive
  energy:     Number 0..1  (time-of-day + streak mood)
app→rive
  gazeX/Y:    Number -1..1 (look at the word being taught, at the mic button, at the user)
app→rive
  viseme:     Enum(sil, PP, FF, TH, DD, kk, CH, SS, nn, RR, aa, E, I, O, U, + lab-velar KP/GB,
click C/Q/X, nasal N~)  app→rive
  jawOpen:    Number 0..1  (from audio RMS, smoothed)
```

```
app→rive
  correct, incorrect, celebrate, levelUp, wave, nod, shrug: Trigger
app→rive
  skinTone:   Color;  hairColor: Color; outfitPrimary/Secondary: Color
app→rive
  hair, headwear, top, bottom, accessory, prop: Artboard properties
app→rive
  reducedMotion: Boolean
app→rive
  tapped:     Trigger
rive→app
```

Lip-sync pipeline: the TTS or recorded audio comes with **phoneme timings**, either from forced alignment (Montreal Forced Aligner with per-language dictionaries, or a CTC aligner on an Omnilingual-ASR/wav2vec2 encoder) or from the TTS engine's own alignment. Phonemes map to visemes through a **per-language table** (extend `PhonemeVisemeMap`), so the app schedules `viseme` changes against the audio clock and adds `jawOpen` from audio RMS for liveliness. Fallback for unaligned audio: RMS-only jaw. Never use letter-based mapping. Tooling and licences (Omnilingual CTC aligner, Apache-2.0; Rhubarb Lip Sync, MIT) are in `LINGAFRIQ_AI_MODEL_STACK.md` §3.8.

4.4 Cast, customisation and economy

- **Cast:** Polie (guide), plus the village archetypes already used in onboarding (Elder, Weaver, Griot, Timekeeper, Rhythm Master, Pathfinder), plus per-language hosts (a Lagos host, an Accra host, a Nairobi host, a Durban host…). Each has a voice and a personality bible. The **learner's own avatar** uses the same rig.
- **Customisation that means something:** items unlock by **demonstrated skill**, not just XP. Examples: finish the Yoruba "Greetings & respect" can-do → unlock *fila* and *gele*. Pass a tone checkpoint → "Talking Drum" prop that animates to your best phrase. Region-authentic outfits are reviewed by the cultural consultant, and each item has a 1-line cultural note ("Aso-oke: hand-woven by Yoruba weavers, worn at ceremonies").
- **No pay-to-look-better pressure on kids;** cosmetics bought with real money are adult-only and never loot boxes (§12).
- **Social use:** stickers and reactions generated from the rig (the avatar saying a learned phrase with correct mouth shapes) for Village chat. Shareable "I can now say…" cards render the avatar speaking the phrase, with audio.

4.5 Flutter integration

- Upgrade to `rive ^0.14.11`, call `RiveNative.init()` in `main.dart`, load files once into a cache, and use one `RiveWidgetController` per visible character. Pause offscreen; drop to the static chip in long lists.
- An `AvatarDirector` service (Riverpod) translates **learning events** (answer graded, mic open, AI streaming, persona emotion tag) into View Model writes, so screens never touch Rive directly. Build it fresh; retire the three scaffolding engines in `avatars/*`.
- Budgets: ≤300 KB per character file with nested artboards streamed (assets via Rive's out-of-band asset loading), ≤2 ms frame cost on a 2020 mid-range Android (e.g., Tecno/Infinix class, which matters for the African market), 60 fps target and graceful at 30.
- Tests: golden tests of chip renders at 32/48/64 px; a contract test that fails CI if a View Model property named in Dart is missing from the `.riv` (load the file and enumerate properties).

4.6 Build order, deliverables and acceptance criteria (for Tope / Cursor)

Step	Deliverable	Done when
1. Character bible	1 page per character: personality, voice, region, palette, silhouette sketches at 64 px; cultural consultant sign-off	Silhouettes are distinguishable as 1-bit shapes at 48 px; reviewer approves dress and hair references
2. Rig spike (one character, Polie)	`lingafriq_cast.riv` with Character + CharacterChip artboards, layers L1–L5, the full `CharacterVM` from §4.3 (enums, not 0–100 codes)	Every property in §4.3 exists and is bound; enumerating the View Model in a Dart test matches the contract
3. States	Idle, listening, thinking, speaking, correct, incorrect, celebrating, encouraging, hint, levelUp, lessonComplete as `activity` values and triggers	Each state reads correctly at 64 px in a 5-person hallway test; transitions ≤ 300 ms; no layer keys another layer's properties
4. Lip-sync	Viseme enum incl. KP/GB, click C/Q/X, nasal; sidecar timing JSON from forced alignment	Native viewers rate sync "natural" on ≥ 80% of 20 clips per Stage-1 language
5. Customisation	Hair/headwear/garments as nested artboards (start with 6 hair styles from the expanded library, 4 headwear, 10 skin tones)	File ≤ 300 KB; swapping an item causes no frame drop on a Tecno/Infinix-class device
6. Integration	`rive ^0.14.11` + `rive_native`, `AvatarDirector`, reduced-motion path, semantics labels	≤ 2 ms/frame on the reference device; offscreen pause verified; screen reader announces expression changes
7. Cast expansion	Village archetypes, then per-language hosts	Each passes steps 3–6

Keep a single `.riv` source in version control with a changelog, export from one Rive file version, and add a CI contract test (step 2) so art and code can't drift.

---

5. Onboarding and first-session magic

5.1 Current state

- **Order:** splash (1.2 s forced delay) → 9-step "village" questionnaire → placement *dialog* → login/sign-up → (email verification) → home. A first-time user answers about 9 screens of questions and creates an account **before hearing one word of the language**.
- **Steps** (`unified_onboarding_screen.dart:58-123`): Welcome, Elder (native language), Weaver (target language), Elder's Question (age + why), Path Chooser (goal), **Rhythm Master (learning style)**, Timekeeper (schedule), Griot (tone and gamification), Naming Ceremony (profile).
- **Defects:** answers serialised with `Map.toString()` (P0-21); "learning styles" have no evidence base (matching instruction to a self-reported style does not improve outcomes; Pashler et al. 2008 and later reviews), so the step wastes time and sets false expectations; age is collected with no consequence (P0-10); every release resets onboarding (P0-20); three alternative onboarding flows sit in the repo unused.

5.2 Target flow: "speak in 60 seconds, account later"

1. **Language pick, one screen.** Show the languages with a native speaker greeting audio on each tile (tap to hear "Ẹ kú àbọ̀", "Karibu", "Sannu"). Detect a heritage link with one optional chip: "My family speaks this".
2. **Instant first win (≤60 s).** Polie greets the learner in the language with the Tone Lens visible. The learner taps to hear, then **says it back**. The real scorer gives honest, encouraging feedback on the one syllable that matters ("your *kú* was mid; lift it high"). The learner has now *spoken a tonal greeting correctly*. That moment is the hook, and no competitor does it well for African languages.

3. **Micro-placement, adaptive and disguised as a conversation (2–3 min, skippable).** Three or four items selected by item response theory (IRT): recognise a greeting, pick a reply, understand a short audio. Heritage speakers often understand far more than they can say, so measure **listening and speaking separately**.
4. **Goal and time, one screen.** "Why?" (heritage/family, travel, partner, work, school, curiosity) and "How much time a day?" (5/10/15/20). The why drives content themes (family vocabulary first for heritage learners).
5. **Age assurance before anything social** (neutral date-of-birth picker, not leading). Under-13 goes to the guardian flow (§13.3).
6. **Save progress = create account** (passkey, Google, Apple or email magic link). Prompt only after the first lesson is finished: "Save your progress and your streak."
7. **Notifications ask in context** after lesson 1: "Want a nudge at 7:30 pm? That's when you practised today."
8. **Avatar creation is optional and later** (unlocked after lesson 2), framed as part of the village story.

5.3 Details that make it feel premium

- **Persist every step** locally (typed, `jsonEncode`) so a kill or crash resumes exactly where the learner left off.
- **Honest copy.** Replace "Who are you, and why do you seek knowledge?" (`:774`) with plain questions; keep the village warmth in the art and voice, not in riddles. Grade-7 reading level (Playbook §15.4).
- **Localised onboarding.** French, Portuguese, Arabic and Swahili UI for continental learners (Francophone Africa is a huge market for Wolof, Lingala, Bambara).
- **Measure it:** funnel events per step, time-to-first-spoken-word, D1/D7 retention by entry path, placement accuracy vs later performance.

---

6. Pedagogy and curriculum

6.1 Curriculum audit (measured, not sampled)

Bundle	Shipped in app?	Content	Verdict
`lingafriq_FINAL_curriculum/languages/*.json` (7 langs)	Yes (`pubspec.yaml` assets); master index loaded first (`curriculum_service.dart:113-118`)	1 lesson per level, `"vocab":[{"word":"example","meaning":"sample"}]`, grammar `"Sentence structure","Verb tense usage"` for Yoruba	**Placeholder**
`curriculum_bundle/curriculum/*` (7 langs, A1–B1)	Yes, fallback	`"word1a"`, `"example meaning"`, `"Sample MCQ"`, answer `"a"`; Yoruba A1/A2/B1 files are 52,978/52,979/52,979 bytes (templated)	**Placeholder**
`curriculum_expanded_bundle/*`	Duplicated inside `lingafriq_full_curriculum_bundle/` (shipped)	`"Yor_word1a"`, `"Example sentence with Yor_word1a."`; also ships `.py` import scripts and a `.dart` file *as app assets*	**Placeholder**
`assets/data/lingafriq_authentic_curriculum_a1_c2.json` (15 keys, 14 langs + duplicate `pidgin`)	Yes, preferred	Real phrases with correct-looking diacritics, but see below	**Thin template**

The "authentic" file in numbers (script over all lessons): - 126 lessons per language, but only **85 (Lingala) to 111 (Yoruba) unique vocabulary items** per language. A real A1 alone needs about 500–800 items; B2 needs about 4,000; C2 is beyond 10,000 word families. - **Every dialogue repeats lines** (126/126 for each language). Example, Yoruba A1 U1 L1: A: "Mo ń kọ́ èdè Yorùbá" / B: "Orúkọ mi ni" (an unfinished "My name is...") / A: "Ó dára", then the same three lines shuffled. This is not a conversation. - **Grammar is two generic**

sentences pasted into 1,546 lessons: "Pattern: subject → verb → object" and "Listen for the verb stem; affixes carry tense and person". Somali and Amharic are **SOV**. Yoruba, Nigerian Pidgin, Twi and Ewe mark tense and aspect with **preverbal particles**, not affixes, and Yoruba verbs don't inflect for person. Teaching this is actively harmful. - `cultural_notes` at unit level is a template ("Yoruba cultural immersion for Discovery & Sound."). - **Three identical 11.3 MB copies** (`…a1_a2_b1.json`, `…a1_c1.json`, `…a1_c2.json` share md5 `20c36058…`), about 34 MB of APK bloat, plus the duplicated bundles. - **Zero audio references** in any curriculum file. `audio_manifest.json` lists 1,363 entries pointing at `assets/audio/...` and `cdn.lingafriq.com/audio/v2/...` with **non-ASCII diacritic filenames** (e.g., `yoruba-mo-ń-kọ-èdè-yorùbá.mp3`). That breaks across NFC/NFD normalisation (macOS/iOS file systems, some CDNs). Note also that the file name drops the tone mark on *kọ́* while the text keeps it, so file-name and text keys will not match reliably. `assets/audio` is not in `pubspec.yaml` assets **[Unverified: media was stripped from the copy]**. - Sample label errors to send to native review: Twi C1 "Idioms" subtitled *Akontabuo* (which means accounting/calculation, not proverbs; proverbs are *mmɛ/ɛbɛ*), Twi A1 *Akyiade* for "Greetings" **[needs native check]**.

Game content (`assets/data/game_content.json`, 1,409 words across 14 languages, about 85–118 each): `phonetic_guide` is a copy of `tonal_note` for **all 1,409**, so there is no IPA or syllable breakdown. The first entry teaches a false tone rule (P0-17). 3 topics are truncated mid-parenthesis.

UI translations: `app_yo.arb` has 62 of 103 keys and no Yoruba tone marks or subdots ("E kaabo si LingAfriq" should be along the lines of "Ẹ káàbọ̀ sí LingAfriq"; "Tesiwaju" → "Tẹ̀síwájú"). A Yoruba app that writes Yoruba without diacritics tells Yoruba speakers it isn't serious.

Offline phrase packs (`services/offline_phrase_pack_manager.dart:561`, `services/translation/offline_translation_service.dart:554`): "Omo", "Eniyan", "Okunrin" without diacritics (Ọmọ, Ènìyàn, Ọkùnrin).

6.2 Learning science the app should be built on

Principle	What it means for LingAfriq	Current
Comprehensible input (Krashen; input slightly above level)	Every unit is anchored in authentic-style audio stories and dialogues at i+1, with graded readers and listening first	Template dialogues
Retrieval practice and spacing	Every item has a memory state; sessions are built from due items + new items; **FSRS-6** scheduler (open-source, benchmarked well ahead of SM-2) running on device with server sync, parameters optimised per learner after about 1,000 reviews	4 competing SRS models
Interleaving and variability	Mix item types (listen, say, choose, build, translate), voices (male/female/elder/child), and contexts for the same item	Each game is siloed
Desirable difficulty with high success	Aim for about 80–85% success; adapt in real time	Fixed
Output and interaction (Swain, Long)	Speaking from session 1; negotiated meaning with Polie; recasts rather than red marks	Text-heavy
Explicit focus on form, minimally	Short grammar "aha" cards with contrastive examples, then practice. Grammar is taught as *patterns*	Generic sentences
Tone and orthography as first-class skills	Tone perception (minimal pairs: Yoruba *ọkọ́* hoe / *ọkọ̀* vehicle / *ọkọ* husband), tone production with pitch feedback, diacritic typing practice with a tone-aware keyboard, downstep for Igbo, Hausa vowel length and tone (usually unwritten, so taught via audio), Zulu/Xhosa clicks and depressor consonants, Amharic fidel syllabary, Wolof vowel length	Tone trainer forces a 3-tone model on all languages, mid tone written as ā (non-standard for Yoruba)
Can-do outcomes	Every unit maps to CEFR / ACTFL can-do statements ("I can greet an elder appropriately in the morning and respond to their greeting"); learners self-assess and are tested on them	Objectives are templates

Principle	What it means for LingAfriq	Current
Mastery learning	A unit is "mastered" when its can-dos are demonstrated (speaking + listening + reading) and its items reach a stability threshold, not when screens are clicked	Unlock rule `previous_lesson_completed`
Feedback quality	Immediate, specific, actionable ("lower the second syllable"); elaborated feedback beats right/wrong	Random scores in places
Cultural pragmatics	Greetings by time and status, honorifics (Yoruba *ẹ* vs *o*), kinship terms, prostration/kneeling customs, proverb usage; taught as *communicative competence*	Mostly notes
Motivation (self-determination theory)	Autonomy (choose themes), competence (visible skill growth), relatedness (family/community). The heritage angle is the strongest motivator: "say it to your grandmother"	Streak pressure

6.3 Learner model (one, not four)

A single **Learner Knowledge Graph** per user and language: - **Nodes:** lexemes (with senses), grammar patterns, phonological skills (each tone pattern, each difficult phoneme), pragmatic skills, can-do statements. - **Per node:** FSRS memory state (stability, difficulty, retrievability), production vs recognition strength, last evidence, error taxonomy counts (tone error, segmental error, word order, register). - **Evidence sources:** lesson items, games, Polie conversations (graded turns), persona missions, pronunciation attempts. Each event goes into the knowledge graph through an idempotent event log (`learning_event` with `client_event_id`). - **Uses:** session builder, Polie's prompt context ("learner struggles with low-tone on àárọ̀; recycle it"), persona mission targets, teacher dashboards, credential evidence. - **Privacy:** on-device first; raw audio is not stored unless the learner opts in.

6.4 Languages: Stage 1, then as many as possible, as fast as possible

Goal: breadth is a first-class goal. LingAfriq should teach **dozens of African languages within two years and keep going**, without ever shipping unvalidated content. Breadth comes from (a) a repeatable content and QA pipeline, (b) the readiness levels below, so a language can go live early at a small but correct scope and deepen over time, and (c) an AI stack chosen for long-tail coverage (`LINGAFRIQ_AI_MODEL_STACK.md` §2 and §4).

Readiness levels (each language moves up independently): - **L0 "Taste":** greetings, numbers, 50–100 words, a culture card; native-recorded audio. - **L1 "Starter":** A1 with native audio for every item and ASR "were you understood?" feedback. - **L2 "Core":** A1–A2 with tone and pronunciation scoring, dynamic TTS, and AI tutor modes. - **L3 "Full":** A1–B1, realtime voice conversation, personas, and checkpoints toward the credential.

Stage 1 (Wave 1, building now): 7 languages. Yoruba, Igbo, Hausa, Nigerian Pidgin (Naijá), Swahili, isiXhosa, isiZulu. Target: all seven at **L1 within the first quarter of the content programme and L2 within two quarters**, with L3 first for whichever languages pass the realtime and persona evals. AI coverage check: - **ASR:** Intron covers all seven (code-switching in six). - **Dynamic TTS:** - Intron covers yo, ig, ha, pcm and sw. - Azure covers isiZulu. - **isiXhosa needs a commissioned custom voice**, because no commercial vendor voice exists. - **Realtime native audio:** Gemini Live covers yo, ha, sw and zu; ig, pcm and xh use the cascade. - **Tone scoring:** in-house. It matters most for yo, ig and ha.

Wave 2 (next, about 14 languages; start L0 once Stage 1 reaches L1): Amharic, Twi (Akan), Somali, Kinyarwanda, Afrikaans, Shona, Wolof, Oromo, Luganda, Lingala, Sesotho, Setswana, Chichewa/Nyanja, Malagasy. They were chosen for speaker numbers, diaspora demand, and existing vendor coverage (most have Intron ASR, Gemini Live or Google support), which makes them the fastest to bring to L1/L2.

Wave 3 (about 25 languages; L0 → L1): Tigrinya, Ewe, Fon, Fula/Fulfulde (Latin + Adlam), Bambara (Latin + N'Ko), Kikongo, Kikuyu, Dholuo, Kirundi, Tamazight (Tifinagh + Latin), Ga, Efik/Ibibio, Tiv, Kanuri, Edo (Bini), Sango, Krio, Cape Verdean Kriolu, Sepedi (Northern Sotho), Xitsonga, Tshivenda, siSwati, isiNdebele, Kamba, Mauritian Creole. Make a deliberate decision on **Egyptian Arabic and Moroccan Darija** (large demand, strong competitors).

Wave 4, the "Living Archive" (50+ languages, community-built L0 starter kits that can grow): Urhobo, Nupe, Igala, Idoma, Ijaw (Izon), Itsekiri, Ebira, Bemba, Tonga, Lozi, Makhuwa, Sena, Umbundu, Kimbundu, Tshiluba, Mende, Temne, Dagbani, Dagaare, Mooré, Dyula, Soninke, Serer, Kabiye, Sidamo, Wolaytta, Afar, Hadiyya, Dinka, Nuer, Acholi, Lugbara, Runyankole, Kalenjin, Maasai, Luhya, Gusii, Chaga, Sukuma, Nyamwezi, Hassaniya, Nobiin, Khoekhoe (Nama), N|uu (critically endangered; only with community consent and benefit-sharing), Ge'ez (liturgical), plus Ajami, N'Ko, Adlam, Tifinagh and Ge'ez script tracks. Partner with Masakhane, Lanfrica/NaijaVoices, African Next Voices, Lelapa, Intron, university linguistics departments (UNILAG, UI Ibadan, UNN, ABU Zaria, Makerere, UDSM, Wits, UCT, Rhodes, Addis Ababa University) and community and diaspora associations.

Running total: 7 → about 21 → about 46 → 95+ languages, each at the readiness level its validated content supports. Dates depend on reviewer capacity; treat these as planning targets, not promises.

How to pick the next language (score 1–5 each, revisit quarterly): speakers (L1+L2) and diaspora demand (waitlist, search data); availability of paid native reviewers and voice talent; AI coverage (ASR/TTS/MT/LLM eval results from the AI stack doc); partner support; mission value (endangerment).

6.4.1 The fast-scaling content and QA pipeline ("language factory")

1. **Language kit (week 0–1):** orthography spec (Unicode, NFC, diacritics, keyboard layout), dialect standard, phone inventory and tone system, a phoneme-to-viseme table, a seed lexicon from licensed corpora, and a native reviewer panel (≥3, paid, contracts with consent and licence terms).
2. **AI bootstrap (week 1–3):** LLM and MT draft the frequency lexicon, glosses, example sentences, dialogues within the lexicon, distractors, cultural-note drafts with citations, and eval sets. Spitch handles tone-mark suggestions for Yoruba, and an LLM with validators handles other languages. Everything is tagged `ai_assisted: true`.
3. **Native validation (week 2–6):** two independent natives plus linguist adjudication for every learner-facing string. **Tone marks, cultural claims and audio are never AI-only.** Disagreements become dialect notes.
4. **Audio (week 3–6):** native recordings of every item (≥2 voices at L0/L1, ≥4 at L2), forced alignment for highlighting, scoring references and lip-sync.
5. **Speech enablement (parallel):** ASR via Intron or Omnilingual, evaluated on a 300-utterance set. TTS via vendor or custom voice, gated on native MOS and tone-error rate. Tone-scoring calibration with native judges.
6. **Pilot and stats (week 6–8):** a beta cohort, item statistics, and the "report a problem" queue. Promote to the next readiness level only when the gates pass.
7. **Throughput target:** one content squad (lead, 2 writers, linguist time, per-language reviewer panels) runs **4–6 languages in parallel**, so 2–3 squads clear a wave of 14 within about two quarters **[planning assumption]**. Tooling (content studio §14, validators §6.5, eval harness) is shared, so each new language costs mainly reviewer and voice time.

What AI may bootstrap vs what natives must validate: the full matrix is in `LINGAFRIQ_AI_MODEL_STACK.md` §7. In short: AI drafts word lists, glosses, examples, dialogues, distractors, explanations and UI translations. Natives must approve every learner-facing string, **all tone and diacritic marks**, cultural notes and persona facts. Curriculum audio is always human-recorded.

Dialect policy: declare the standard taught (Standard Yoruba, Central Igbo + Onitsha/Owerri variants as audio, Kano Hausa, Asante Twi, Kiswahili Sanifu + Kenyan colloquial notes, Sheng as a fun side track) and mark regional variants explicitly. The existing `dialectPreference` model is a good hook.

6.5 Content schema (replace all four bundles)

```
Lexeme:
  id: yo.lex.ọkọ.vehicle   # stable, language-scoped
  lemma: "ọkọ̀"            # NFC-normalised, full diacritics
  orthography: {standard: "ọkọ̀", ascii_fallback: "oko", script: Latn}
  ipa: "/ɔ́kɔ̀/"            # with tone diacritics or Chao numbers
  tone_pattern: [M, L]
  syllables: ["ọ", "kọ̀"]
  senses: [{gloss_en: "vehicle", gloss_fr: "véhicule", domain: transport, register: neutral}]
  minimal_pairs: [yo.lex.ọkọ.husband, yo.lex.ọkọ.hoe]
  audio: [{voice_id: yo_f_lagos_01, speed: natural, uri: audio/yo/9f2c…ogg, aligned_phones: [...]},
          {voice_id: yo_m_ibadan_02, speed: slow, …}]
  image: {uri, alt_text, cultural_review: approved}
  cefr: A1; frequency_rank: 812
  provenance: {author, source, licence, created_at}
  review: {status: approved, reviewers: [native_1, linguist_1], tone_checked: true, audio_checked:
true, version: 3}
Pattern (grammar): id, description_l1, form, examples[], contrasts[], common_errors[], cefr
Dialogue: id, scene, roles, turns[{speaker, text, gloss, audio, tone_pattern}],
comprehension_items[], can_do_ids[]
Item (exercise): id, type, stimulus refs, answer spec (with accepted variants incl. diacritic-
insensitive matching rules), distractor rationale, IRT params (a, b), skill tags
Unit: id, can_do_ids[], input (story/dialogue), patterns[], lexemes[], items[], speaking tasks[],
culture card, checkpoint
```

Rules: file names are **content hashes (ASCII)**, never orthography. All text is NFC. The validator rejects missing tone marks for tonal languages, mismatched audio, unreviewed items, and placeholder patterns (`/word\d|example|sample|lorem/i`) in CI.

6.6 Authoring and review pipeline

1. **Syllabus design** by an applied linguist per language: can-do map, frequency-based lexicon (from corpora such as JW300 alternatives with clear licences, Masakhane datasets, news, radio transcripts, plus community elicitation), pattern sequence.
2. **Drafting:** human writers, with LLMs allowed for *variants and distractors only*, on a grounded prompt with the approved lexicon. Every AI-drafted string is tagged `ai_assisted: true`.
3. **Native review, two independent reviewers** (different regions where dialects differ) plus **linguist adjudication**. Checks: orthography and tone, naturalness, register, cultural appropriateness, translation accuracy. Track inter-reviewer agreement; disagreement becomes dialect notes, not silent edits.
4. **Audio production:** studio or high-quality remote recording with ≥2 voices per language at L0/L1 and ≥4 at L2+ (gender, age, region), natural and slow takes, forced alignment for word and phone timings, QA listen-through. Contributors sign clear licence and consent forms with revenue share or fair payment (the UGC voice contribution feature becomes this pipeline, with consent).
5. **Pilot:** release to a beta cohort; collect item statistics (p-value, discrimination, time, hint use) and learner reports ("this is wrong" button on every item, routed to reviewers).
6. **Versioning:** content packs are semver'd and signed; the app downloads diffs; every item keeps its history.

6.7 How content is displayed across features

- **One renderer** for target-language text (Tone Lens, tap-to-hear, long-press for gloss/IPA/minimal pairs, save to deck) used by lessons, chat, Polie, personas, games, the magazine and the dictionary.
- **Diacritic-aware input:** a custom keyboard bar for each language (ẹ ọ ṣ + tone keys for Yoruba; ị ọ ụ ṅ for Igbo; ɓ ɗ ƙ y for Hausa; ɛ ɔ for Twi), with "lenient" grading that accepts missing tone marks in early levels but always shows the correct form, then tightens with level.
- **Script learning tracks** for Ge'ez/Fidel (Amharic, Tigrinya), N'Ko, Adlam, Tifinagh and Ajami, with stroke-order animation and handwriting recognition practice.

---

7. AI modes and voice: best-in-class architecture

7.1 Current state

- **Three paths to the LLM:** direct client calls to Groq/OpenAI/Hugging Face with compiled keys (P0-3), a backend completion proxy that accepts the client's system prompt (P0-9), and backend persona chat. Mode logic is spread across `ai_chat_provider_groq.dart` (2,516 lines), `polie_workspace_screen.dart` (6,485 lines), `ai/modes/*`, `ai/pedagogy/*`, `ai/persona_cognition/*` (a two-pass "self-critique" orchestrator run *on the phone*).
- **No streaming** on the backend fallback (`ai_chat_provider_groq.dart:1302`), so release users wait for full completions.
- **Client-side post-processing as a safety net:** `DiacriticsEnforcer.enforceWithMetadata` patches model output. Diacritic restoration is a hard NLP problem; a regex pass can introduce errors as easily as it fixes them.
- **Voice:** STT through Groq Whisper and a Hugging Face serverless call (`lingafriq-voice-service/main.py:46-60`) with **no authentication on that service**; TTS through device `flutter_tts` (most Android/iOS devices have no Yoruba/Igbo/Hausa/Twi voice), MMS-TTS and XTTS-v2 (licensing, P0-24). The **Intron proxy** (`routes/intron.route.ts`, `utils/intronLanguageMap.ts`) is well built and **never called** by the app.
- **Pronunciation:** see P0-18/19. Whisper-family models normalise or drop tone diacritics, so "toneScore" computed from them (`tutor_pronunciation_mode_screen.dart:115`) does not measure tone.

7.2 Target architecture (model choices: see `LINGAFRIQ_AI_MODEL_STACK.md`)

```
App (Flutter)
  ├─ Realtime voice session (WebRTC via LiveKit; you already ship livekit_client)
  └─ HTTPS (text modes, grading, content)
        │
AI Gateway (backend service, TypeScript or Python)
  ├─ AuthN/Z, per-user quotas, cost budgets, kill switches, feature flags
  ├─ Prompt registry (versioned templates per mode; no client prompts)
  ├─ Grounding: curriculum KG + learner model + persona source index (RAG)
  ├─ Model router (per language × task × tier), fallbacks, caching
  ├─ Safety: input/output classifiers, age-tier policies, PII scrubbing
  ├─ Structured output (JSON schema validation, retries)
  └─ Telemetry: traces, token cost, latency, eval sampling
        │
Speech stack (self-hosted GPU + vendors)
  ├─ STT: Intron (African-accent STT, already integrated) | Omnilingual ASR (Apache-2.0, 1,600+
langs) fine-tuned | vendor fallbacks
  ├─ TTS: licensed/commissioned voices per language (see 7.5)
```

```
  ├─ Pronunciation & tone scorer (see 7.4)
  └─ Forced aligner (lip-sync + highlighting)
```

7.3 Modes that should exist (and how each is grounded)

Mode	What it does	Grounding and guardrails
Polie Conversation	Free conversation at the learner's level, with recasts, a hint ladder, and "say it better" suggestions	Learner model (known words, target words due), CEFR-controlled vocabulary; output validator flags out-of-level words; every turn tagged with the skills practised
Scenario Roleplay	Market bargaining, taxi, family visit, clinic, job interview, church/mosque etiquette	Scenario scripts with goals and success criteria authored by humans; the model improvises inside them
Tutor: Explain	Grammar and usage questions answered in the learner's L1	RAG over the approved pattern library only; must cite the pattern card; refuses to invent rules
Tutor: Correct my writing	Diacritic and tone restoration, grammar, register	Hybrid: rule and lexicon checks first, model second, with uncertainty shown ("tone uncertain here, listen to native audio")
Story Mode	Graded micro-stories with audio, tap-to-gloss, comprehension questions	Generated from the approved lexicon; **human-approved before publishing** to the shared library; personal stories are labelled as AI-generated
Shadowing / Pronunciation coach	Listen, repeat, see the pitch contour vs the native contour, targeted feedback	Real scorer (7.4)
Translator with teaching	Translate and explain why	Only behind a quality gate per language pair; never the unofficial Google scraper
Live Translate / Point-and-say	Camera or voice to target language	Same; plus a visible confidence indicator

7.4 Pronunciation and tone scoring (the signature capability)

1. **Segmental scoring:** goodness-of-pronunciation (GOP) using a multilingual CTC acoustic model (Omnilingual wav2vec2/CTC, Apache-2.0, fine-tuned per language on native recordings; **not** MMS weights, which are CC-BY-NC) forced-aligned to the expected phone sequence. Per-phone scores, with substitutions mapped to learner-friendly advice ("your *gb* sounded like *b*; close lips **and** the back of the tongue together").
2. **Tone scoring:** extract F0 (pYIN or CREPE), normalise per speaker (semitones relative to the speaker's median, z-scored), segment by aligned syllables, then compare the contour to the expected pattern with a small classifier trained on native recordings, per language. Handle Yoruba H/M/L + glides, Igbo H/L/downstep, Hausa H/L/falling, Twi H/L, Zulu depressor effects. Output per-syllable tone correctness with a confidence value, plus a visual of the learner's contour overlaid on the native one (Tone Lens).
3. **Fluency:** speech rate, pause distribution, self-repairs.
4. **Calibration:** collect human ratings (native raters, 1–5) on a few thousand attempts per language; fit the scorer to agree with them (target Pearson r ≥0.7 with raters, inter-rater as the ceiling). Publish the methodology.
5. **Honesty rules:** when confidence is low (noise, clipping), say so and ask for a retry; never fabricate a number. Run on the server at first; distil a small on-device model for offline tone practice later (the 300M Omnilingual variant is a candidate for offline use **[Unverified on-device performance]**).

7.5 TTS strategy

- **Commission native voice talent** per Stage-1 language (≥2 voices; mandatory for isiXhosa, which has no commercial vendor voice) under a commercial licence, then fine-tune a commercially licensed

TTS architecture on them. Prefer recorded human audio for all curriculum items and use TTS for dynamic AI turns.
- **Vendors to evaluate** (verify licences, quality and tone accuracy per language before committing): Intron (TTS for yo/ig/ha/pcm/sw; STT for all seven), Spitch (yo/ig/ha/pcm), Azure (zu, sw), Google/ElevenLabs/Gemini TTS (sw; ElevenLabs also ha). YarnGPT is CC-BY-NC-SA and so not usable commercially. Full licence and coverage research is in `LINGAFRIQ_AI_MODEL_STACK.md` §3.5. Run blind native MOS tests with a tone-accuracy rubric.
- **Never ship a voice that gets tones wrong.** For tonal languages, a mis-toned TTS teaches errors at scale. Gate each voice per language on a tone-accuracy threshold judged by natives.

7.6 LLM choices for low-resource languages

- Frontier models handle Swahili, Amharic, Hausa and Yoruba unevenly, and diacritics and tones are the usual failure point. Use a **router**: frontier model for reasoning and English explanations; a model that tests best on each target language for generation (evaluate Gemini, GPT and Claude, plus commercially licensed open models such as Gemma 3/AfriqueGemma and Qwen3, fine-tuned with licensed data; Aya models are CC-BY-NC and N-ATLaS is capped at 1,000 users without a licence; see `LINGAFRIQ_AI_MODEL_STACK.md` §2.4); **always constrain generation to the approved lexicon** at lower levels.
- **Evals before launch and on every prompt or model change:** a per-language test set (≥300 items per Stage-1 language, and per language before it reaches L2) scored by natives and automated checks (diacritic completeness, out-of-level vocabulary rate, factuality for culture, refusal correctness, persona fidelity). Block deploys on regression.
- **Cost and latency:** cache deterministic outputs (explanations, story drafts), use small models for classification and grading, stream all chat, set a time-to-first-audio target of under 800 ms for voice turns and under 400 ms for first text token, and set per-user daily budgets by tier.

7.7 Realtime voice conversation

LiveKit Agents (or similar) running a **cascaded pipeline** per turn: VAD → streaming STT (Intron/Omnilingual) → LLM (streaming) → TTS (streaming) → client, with barge-in, end-of-turn detection tuned for each language's prosody, and avatar lip-sync from TTS alignments (§4.3). Use speech-native "realtime" models only for the languages where blind native tests show they pronounce tones correctly. A "push-to-talk" fallback for noisy environments and low bandwidth (Opus at 16–24 kbps; text fallback on 2G/3G).

---

8. Historical Personas: from chatbot to revelation

8.1 Current state

- **Two independent systems.** Client registry: 27 personas (`ai/personas/historical_persona_registry.dart`, 1,455 lines) with eras, core events, voice profiles and scenario templates whose `branches: {}` are empty (e.g., `:270-272`). Backend: seeds in `scripts/extendedHistoricalPersonas/personas-e01…e08.ts` with `primary_sources` fields, but chat uses a plain system prompt with no retrieval, and responses are stored with `sources: []` and a hard-coded `confidence: 0.9` (`historicalPersonality.service.ts:195`).
- **The selection screen uses the backend** (`services/ai/historical_personality_service.dart`), which tries **four URL variants per call** (`:310-313, 366-369, 407-410`), so every 404 multiplies latency.
- **Persona missions use the client Groq key** and fail silently in release builds (P0-13).
- **Safety is a substring blocklist** that blocks the core content of these personas: anti-colonial war (Yaa Asantewaa), apartheid discrimination (Mandela), the Zulu kingdom's military history (Shaka). It also

blocks "skill" and "whatever" (P0-14).
- **Deception:** "Do not mention… being an AI model" (P0-15).
- **Factual slips** show the lack of historian review: Nkrumah's languages `English/Twi` (`personas-e01.ts:20`; he was Nzema from Nkroful); "First sub-Saharan colony to reach independence" (`:39`) needs nuance (Sudan became independent in 1956).

8.2 Vision: "Sit with history"

Each persona is a **short, authored, replayable experience** with three layers: a narrative quest, a language lesson, and a history lesson, all grounded and honest.

Example: "The Golden Stool" with Yaa Asantewaa (Twi, A2–B1, 25 min, 4 chapters). 1. *Kumasi, 1900.* An era-accurate illustrated scene with ambient sound (talking drums, market). The learner is a young messenger. Objective: greet the Queen Mother correctly (honorifics, the Twi greeting sequence). The Tone Lens is on; the avatar kneels if the learner chooses the respectful option. 2. *The demand.* The British governor's demand for the Golden Stool is read in English. Yaa Asantewaa responds in Twi with subtitles toggled by level. The learner must understand three key sentences (comprehension checkpoint). 3. *The council.* The learner argues a position in simplified Twi (scaffolded sentence builders). The persona answers in character, grounded in sources, and marks clearly when something is *inferred* ("Historians are not certain what I said here; the record tells us…"). 4. *Aftermath.* A historian's note (human-written) and a mini-quiz, plus a "Primary source" card (photo of an archival document or museum object with attribution) and a vocabulary set added to the learner's deck.

8.3 Grounding and integrity

- **Source dossier per persona,** curated by a historian: primary sources (speeches, letters, court records, oral histories), reputable secondary sources, contested points, known myths, and respectful-portrayal guidance from descendants or community bodies where they exist. Store it as a retrieval index; every persona turn retrieves from it.
- **Answer contract:** `{speech_text, language_teaching_points[], claims:[{text, support: documented|inferred|legend, source_ids[]}], emotion, next_beat}`, validated server-side. The UI shows a small "Source" chip on documented claims and an "Inferred" label on speculation.
- **Knowledge horizon:** the persona knows nothing after their death. When the learner asks about later events, the persona answers in character with curiosity ("I did not live to see that; tell me…"). A classifier decides this, not keyword lists.
- **Sensitivity rules** written with historians: violence is discussed historically and age-appropriately, never glorified; no slurs reproduced even if in sources (paraphrase with a note); living-memory figures (Mandela, Sankara, Lumumba) are handled with family/estate guidance and possibly limited to narrated "as told by" modes. Some figures should get a narrator rather than first person.
- **Disclosure:** a persistent "AI recreation" badge, and an opening line that explains how the persona was built.

8.4 Craft

- **Voice:** a voice actor per persona speaking the target language with appropriate regional accent, cloned only with consent and a licence (or synthesised voices clearly labelled as such). Avatar busts with era-accurate dress, reviewed.
- **Branching:** fill the empty `branches` with authored beats; let the model improvise *between* beats and keep the story on rails.
- **Classroom mode:** teachers assign a persona chapter; students' turns are graded against can-dos; a debrief sheet is generated for the teacher.

- **Collections:** "Queens of Africa", "Scholars of Timbuktu", "Independence Generation", "Writers & Thinkers" (e.g., Thomas Mofolo, Ahmed Baba, Cheikh Anta Diop), each pairing figures with the languages they spoke.
- **Evals:** historian-rated fidelity on 50 scripted probes per persona, an adversarial suite (jailbreaks, anachronism traps, attempts to make the persona endorse atrocities), and language-quality checks by natives.

---

9. Community: tribes, villages, classrooms, chat

9.1 Current state

- **Familiar surfaces that aren't yet learning-native.** WhatsApp-style chat (`wa-*`: status, starred, media gallery), Snapchat-style (`snap-*`: camera, stories, streaks), X-style feed (`x-*`: compose, explore, lists, notifications), social audio rooms, tribes, villages, classrooms, micro-mentors, friend quests. Familiar patterns lower the learning curve, which is valuable; but each one is shallow today and none uses the target language, and together they triple the moderation surface. All are kept and elevated (§18.6).
- **Security:** P0-4/5/6 (room joins, identity spoofing, global broadcasts). The live socket server's default origin list includes `http://localhost:5500` and `http://localhost:3000` in production unless `CORS_ORIGINS` is set (`socket.service.ts:141-147`); a second, unused socket server with `'*'` CORS sits in `socket/handlers.ts` (dead code that will confuse the next developer). Tokens are accepted in the query string (`socket.service.ts:199`), where they end up in proxy logs. Tribe, village and competition channels can be joined by anyone through `join_room`, and the intended subscription handlers were never wired up.
- **Scale:** presence and the socket map are in process memory (`onlineUsers`), PM2 runs a single fork (`ecosystem.config.js:11-12`), and there is no Socket.IO Redis adapter, so it cannot scale past one node. Every connect broadcasts the full online list to everyone ($O(N^2)$).
- **Safety:** no age tiers, no block/mute UX audit, cosmetic toxicity filter, soft-deleted "disappearing" snaps, external `media_url` allowed (tracking pixels, malicious content), a public feed by default, and no CSAM hash matching on uploads.
- **Fake social proof:** mock co-learners in Village flashcards (`village/flashcard_focus_screen.dart:26-34`).

9.2 Target: one "Village" social layer built for learning

Principle: every social feature must create *language practice with real people* or *belonging*. Copy what made WhatsApp, TikTok and Snapchat work (speed, voice, ephemerality, creativity), but point it at learning.

Surface	What it is	Why it is better than a clone
Circles (tribes renamed; villages and friends kept as their own places)	Small groups (5–50): family circle, study circle, class, language village	Small groups have higher engagement and are easier to moderate than giant rooms
Chat	1:1 and group, text + **voice notes as the primary medium**, reactions, replies, read state (contacts only)	Voice notes are how much of Africa already communicates; LingAfriq adds **learning overlays**: Tone Lens on messages, tap any word to gloss/save, "check my message" before sending, and optional AI transcription of voice notes with gloss
Practice Rooms (alongside Village Radio audio rooms)	Scheduled or drop-in voice rooms with a host, topic and level; the AI can act as a facilitator that suggests prompts and vocabulary	Structured speaking, not open mic

Surface	What it is	Why it is better than a clone
Clips (the Snap camera and stories, made learning-native; the X-style feed stays as the Square)	15–60 s vertical video or audio clips: "phrase of the day", proverb explainers, tongue twisters, duets ("say it after me" split-screen like TikTok duets), stitch-style corrections from native speakers	Short video drives discovery; duets turn content into practice
Native Corrections	Learners post a phrase or voice note; verified natives correct it (HiNative-style), with reputation	Peer correction at scale, which also feeds content QA
Classrooms	Teacher dashboard (web), roster import, assignments (lessons, persona chapters, speaking tasks), auto-graded speaking with teacher override, progress by can-do, live sessions (LiveKit) with recording consent	Schools, diaspora Saturday schools and churches/mosques running language classes are a real B2B market
Events	Seasonal festivals (Ọdún Ọ̀ṣun, Eid, Timkat, Durbar, Homowo, Umhlanga), live cultural sessions	Culture as a reason to come back

9.3 Private messaging done properly

- **E2EE for 1:1 and small groups** using MLS (RFC 9420, designed for group messaging) or the Signal protocol (libsignal). Server stores ciphertext only. Learning overlays (gloss, Tone Lens) run **on device**; anything needing a server model is opt-in per message.
- **Under-18 accounts:** no E2EE DMs with unknown adults. Messaging is allowed only within guardian-approved circles and classrooms, where messages are not E2EE so they can be moderated; this trade-off is explained to families. Adult accounts get E2EE by default.
- **Delivery:** durable server-side message log with per-device cursors, idempotent client message ids (the backend already has `messageIdempotency.guard.ts`, which is good), offline queue, push via FCM/APNs with data-only payloads (no message content in pushes for E2EE chats).
- **Media:** signed upload URLs, server-side transcoding (Opus for voice, HEVC/AV1 or H.264 for video), size limits, EXIF stripping, perceptual hashing and known-CSAM matching (PhotoDNA or equivalent via a vendor) on unencrypted surfaces, hard deletion on expiry.

9.4 Moderation and child safety stack

1. **Prevention:** age tiers, default-private profiles for minors, no location sharing, no contact discovery for minors, rate limits on new accounts, friction for first DMs.
2. **Detection:** multilingual toxicity classifiers evaluated on African languages and code-switching (Pidgin/English, Sheng, Swahili/English); word-boundary-aware lexicons curated by native moderators per language (fix the *ode*/"code" problem); grooming-pattern heuristics; image/video classifiers; hash matching.
3. **Response:** block, mute, report on every message and profile in at most two taps; a moderator console with queues, evidence, and SLAs; escalation to law enforcement and the relevant hotline (e.g., NCMEC CyberTipline reporting for US-linked CSAM, plus local equivalents) handled by a trained safety lead.
4. **Transparency:** community guidelines written in plain language and in the supported UI languages, appeals, and a periodic transparency report.

9.5 Realtime infrastructure

Socket.IO with the Redis adapter (or a managed service such as Ably/Pusher/Stream Chat if speed matters more than control), horizontally scaled with sticky sessions; presence in Redis with TTL; per-room

authorisation middleware; event schemas versioned and validated (zod) on both sides; backpressure and per-socket rate limits; load-tested to 10k concurrent connections per node before launch events.

---

10. Games: fun, educational, and honest

10.1 Current state

About 45 game screens (`screens/games/*`, `games/*`) built on a shared "gamekit". Problems: (a) **fake scoring** in the speaking games (P0-18); (b) **content** comes from a 85–118-word pool per language with no audio and with tone errors; (c) **tone trainer** applies one H/M/L model with one correct tone per *word* for all languages (`tone_trainer_game.dart:47-66`); (d) games are a separate destination rather than part of spaced practice, so playing them does not reliably update the learner model; (e) finished game code is unreachable (Tone Forge, the mastery results screen, and a GameKit rewrite of Proverb Unlocker), and all of it is revived in §17.5.

10.2 Design rules for every game

- **Every game is a retrieval or perception exercise in disguise,** reads and writes the learner knowledge graph (§6.3), and uses due items first.
- **Sessions of 60–180 s,** one core verb, instant feedback ≤160 ms, juicy but brief celebration, a "why was I wrong" explanation with audio.
- **Difficulty adapts** (target 80–85% success). Multiplayer is matched by level.
- **No dark patterns:** no energy gates on core learning, no loot boxes, no fake opponents presented as real people (label bots as bots).

10.3 Ten flagship games (headliners; every existing game stays in the Arcade)

Game	Core loop	Skill trained	Notes
Talking Drum (tone perception)	A phrase is "drummed" (pitch contour played on a talking-drum timbre); the learner picks the phrase whose tones match, then speaks it and sees their contour vs the drum	Tone perception and production	A culturally rooted mechanic (Yoruba *dùndún*, Akan *atumpan* really do "speak" tone); per-language tone inventories
Minimal Pair Duel	Hear one of 2–4 near-identical words (ọkọ/ọkọ̀/ọkọ́), tap the right picture; head-to-head async with friends	Tone and vowel discrimination	Uses the minimal_pairs field; leagues by accuracy, not speed only
Market Haggle	Voice roleplay with a market trader (AI, scripted goals): get the price down without being rude; numbers, politeness and bargaining phrases	Speaking, numbers, pragmatics	Scored on goal achieved + politeness + correctness; replayable traders with personalities
Danfo / Matatu Rush	Listen to rapid conductor calls and stop names; tap to board the right bus before it leaves	Listening at native speed	Real recorded audio with accents; difficulty by speed
Proverb Weaver	Assemble a proverb from fragments, hear it, then pick the situation where it applies	Advanced vocabulary, culture	B1+; proverbs reviewed by elders/linguists
Griot's Story Chain	Collaborative story: each player records a sentence continuing the story; AI checks grammar and suggests a fix; the result plays as an illustrated audio story	Speaking, creativity	Great for classrooms and circles
Word Forge (morphology)	Build Swahili/Zulu words from noun-class prefixes and verb slots (a-na-ku-penda); see the meaning change live	Bantu morphology, agglutination	Language-specific; Yoruba variant builds serial verbs

Game	Core loop	Skill trained	Notes
Fidel Flight / Script Sprint	Trace and recognise Ge'ez/N'Ko/Adlam/Tifinagh characters in a flying game	Script literacy	Handwriting recognition on device
Greeting Gauntlet	Fast scenarios: time of day × person (elder, peer, child) × setting; respond with the right greeting and gesture	Pragmatics	Avatar performs the gesture (kneel, bow, handshake style)
Echo Karaoke (rebuilt)	Sing or chant short songs, children's rhymes and praise phrases with a live pitch line	Rhythm, tone, fluency	Real F0 tracking; licensed or commissioned music

These ten are the **headliners** of the Arcade. Several are elevations of existing games (Talking Drum ← Drum Word and Drum Rhythm; Market Haggle ← Market Bargaining; Danfo/Matatu Rush ← Taxi Survival; Proverb Weaver ← Proverb Unlocker; Greeting Gauntlet ← Greeting Diplomacy; Echo Karaoke ← Pronunciation Karaoke). **Every other existing game stays** on its skill shelf, with its own elevation in §18.5, and Tone Forge is revived as a featured tone game.

10.4 Meta-game

Weekly **Village Festival** themes (food, family, travel) that rotate content across all games; a **Mastery Map** showing which skills each game improved; **co-op goals** with circles ("our circle learned 300 words this week") rather than pure competition.

---

11. Tutorials and quizzes

11.1 Current state

- "Tutorials" are mostly **feature tours** (`features_guide`, `game_onboarding_overlay.dart`, four unreachable `*_preloader.dart` help screens, revived as contextual first-use tips in §18) rather than skill tutorials.
- Quizzes exist in four disconnected systems (language quiz, history quiz, random quiz, unit quizzes) with separate backend models (`languageQuiz`, `historyQuiz`, `randomQuiz`, curriculum `unit_quiz`). Placeholder bundles contain `"Sample MCQ"` with answer `"a"`. There is no item bank, no difficulty calibration, no distractor rationale and no link to the learner model.

11.2 Tutorials that teach (micro-lessons with a "learn it, see it, try it" shape)

- **Sound tutorials per language,** 60–90 s each with native video of the mouth: Yoruba tones and the subdot vowels, *gb/kp*; Igbo vowel harmony and the dotted vowels; Hausa implosives ɓ/ɗ and ejective ƙ; Zulu/Xhosa clicks c/q/x with the slow-motion mouth view; Amharic ejectives and gemination; Swahili noun classes (as a visual "class wheel"). Each ends with a 30-second perception check and a speaking try.
- **Orthography tutorials:** typing diacritics with the LingAfriq keyboard bar; why tone marks matter (the minimal-pair demo).
- **Culture tutorials:** greeting choreography (who greets first, kneeling/prostrating, handshake styles), naming customs, taboo topics.
- **Just-in-time coaching:** tips appear the first time a learner hits a pattern, not in a tour at install.

11.3 Quizzes that measure

- **One item bank** (schema in §6.5) with IRT parameters calibrated from pilot data; adaptive placement and checkpoint tests (CAT) that end in 8–15 items with a reported confidence.

- **Item types:** listen-and-choose, tone identification, picture matching, dictation (diacritic-aware grading), sentence building, speaking prompts scored by the real scorer, short conversation tasks scored against can-dos.
- **Feedback design:** after every item, show the correct form with audio and the Tone Lens, and for wrong answers one sentence on *why* (distractor rationale written by the content team).
- **Checkpoint per unit** maps to can-do statements; passing updates the credential evidence (§12.4).
- **Integrity for credentials:** proctoring only where stakes justify it (the Passport), with clear privacy terms; never use random or opaque scores for credentials.

---

12. Gamification and retention without dark patterns

12.1 Current state

XP, three currencies (`ngwenya`, `cowries`, `ancestral_beads`), leagues, hearts (`models/userHearts.model.ts`), badges, streaks, magic items, ancestral tree, heritage milestones, competitions, gifting, launch events. **The integrity layer is broken** (P0-7/8): currency and points can be self-minted. Push reminders spam and guilt-trip (P0-23: "Your N-day streak is at risk! … 🔥"). Three currencies with unclear sinks inflate and confuse.

12.2 Principles

- **Reward learning, not tapping.** XP is computed server-side from *evidence of learning* (items recalled at spaced intervals, speaking tasks passed, can-dos demonstrated), with diminishing returns for grinding easy items.
- **One soft currency** (keep *Cowries*: culturally resonant and historically real as money) with clear sinks: cosmetic avatar items, streak repair, persona chapters early access, donating to a community language-preservation fund (visible impact).
- **Streaks that respect life:** "weekly streak" option, freeze tokens earned by practice, a forgiving repair window, and no loss-aversion copy at night. Reminders at most once a day, within the learner's chosen window and timezone, and quiet hours by default. Celebrate milestones rather than threatening losses.
- **Hearts:** keep the feature, but move it out of core lessons into an opt-in **Challenge Mode** (lessons and games played "on hearts" for bragging rights and a badge). Mistakes are how people learn; punishing them in core lessons is a Duolingo pattern that drew criticism.
- **Leagues:** opt-in, matched by time zone and activity level, ranked on learning XP; minors see only circle or class leagues.
- **Collaborative goals:** circle and classroom quests, family challenges ("teach grandma's phrase of the week back to her"), a "Heritage Tree" that grows as family members learn.
- **Meaningful status:** "Can-do badges" verified by checkpoints ("Can bargain in a market, Yoruba A2") are shareable and valuable, unlike generic badges.

12.3 Ethical guardrails

No loot boxes. No real-money purchases by minors. No fake scarcity timers. No fake "friends are waiting" pushes. No mock participants presented as real: replace the hard-coded co-learners in `flashcard_focus_screen.dart:26-34` with real matched learners or clearly labelled practice bots (the group-study feature stays). Every monetised item states exactly what it does.

12.4 Credentials (Passport)

Rebuild the Passport only after content and scoring are real: evidence-based (a portfolio of passed can-do tasks plus a proctored CAT), aligned to CEFR descriptors with a published methodology, signed and

verifiable (W3C Verifiable Credentials or Open Badges 3.0), revocable.

---

13. Security, privacy, reliability, performance, offline, accessibility, i18n, testing

13.1 Security hardening (beyond the P0 list)

- **Auth:** short-lived access tokens (10–15 min) plus rotating refresh tokens stored server-side (revocable, reuse-detected); move from HS256 shared secrets to asymmetric keys (EdDSA/RS256) with `kid` rotation; passkeys first, then Google/Apple sign-in, then email magic link; drop the legacy Django password hash on next login (rehash with Argon2id); enforce at least 8 characters with a breached-password check (login accepts anything of 6+ characters, `auth.controller.ts:107`; the sign-up rule is unverified). The controller trims `JWT_SECRET` (`auth.controller.ts:23`) but the middleware and socket do not (`middleware/auth.middleware.ts:20-27`); a trailing newline in the env var would make every token fail verification in middleware.
- **API:** zod/Joi validation on every route (some exist); object-level authorisation tests for every `/:id` route (IDOR suite); remove client-supplied identity fields everywhere; request size limits per route (the global 10 MB JSON limit, `app.ts:162`, is too generous).
- **Logging:** stop logging the first 40 characters of the Authorization header (`middleware/requestLogger.middleware.ts:26`) and the password length at login (`auth.controller.ts:108-113`).
- **Mobile:** certificate pinning is passed as a dart-define but the Android network config leaves pinning commented out with `YOUR_CERTIFICATE_PIN_HERE` placeholders (`android/app/src/main/res/xml/network_security_config.xml:17-26`); decide and implement consistently. Use authenticated encryption (AES-GCM) for cached data (`lib/services/offline/cache_encryption.dart:28` uses `Encrypter(AES(key))`, the `encrypt` package's default SIC/CTR mode, with no MAC). Play Integrity / App Attest on the AI gateway and reward endpoints to cut bot abuse. Obfuscate release builds (`--obfuscate --split-debug-info`; the release workflow has split-debug-info, add obfuscate).
- **Secrets hygiene:** gitleaks in pre-commit and CI; a secrets manager (Doppler, 1Password, GCP Secret Manager); a quarterly rotation calendar. `scripts/secrets-hygiene-check.mjs` exists; wire it into CI as a blocking step.
- **Supply chain:** Dependabot/Renovate, `npm audit --omit=dev` blocking on high, `flutter pub outdated` monthly, SBOM generation, pinned GitHub Actions by SHA. The admin panel runs React 18 + Vite 4 (2023-era); upgrade.
- **Pen test** by a third party before any school or child-focused launch.

13.2 Privacy and compliance

- **Nigeria:** NDPA 2023 (it superseded the NDPR as the primary law) and the NDPC's implementation directive; register with the NDPC as a data controller of major importance if thresholds are met **[verify thresholds with counsel]**; DPIA for voice data and minors. **Kenya** Data Protection Act 2019, **South Africa** POPIA, **Ghana** Data Protection Act 2012, **EU/UK** GDPR (diaspora users), **US** COPPA if any US children use it, plus the UK Age Appropriate Design Code if marketing there.
- **Voice data is biometric-adjacent:** explicit opt-in to store recordings, default deletion after scoring, separate consent for contributing recordings to model training (with payment or benefit-sharing for contributors), and data-subject export and deletion (the backend has `userExport.ts` and `deleteUserAccount`; expose both in the app).
- **AI transparency:** label AI-generated content and personas (EU AI Act Art. 50).
- **Data map and retention schedule** per collection (107 Mongo models today); purge jobs, logged.

13.3 Minors and families

- Age assurance at sign-up; **guardian account** that links children, approves circles and classrooms, sets time limits and sees progress; school accounts provisioned by teachers with school consent.
- Child experience: no public profile, no DMs outside approved circles, no snaps or public feed, no ads, no purchases, AI conversations with a stricter policy and no persistent memory of personal details.
- A named safety lead, documented incident response, and moderator training on grooming patterns.

13.4 Reliability and observability

- **Fail loud, not silent.** Remove every "return a fake success" path: pronunciation zeros with HTTP 200, the persona "I hear you", the offline queue stub. Replace with typed errors and visible states.
- **Startup:** parallelise init, defer non-critical services until after the first frame, start Sentry first, and target <2 s cold start to first meaningful frame on a mid-range Android. Stop the 1.2 s artificial delay.
- **Backend:** one deployable API + separate worker service on a job queue (BullMQ with Redis, which is already partly present) using leader election for cron; health and readiness probes; graceful shutdown; at least two API instances behind the load balancer; MongoDB Atlas with backups and point-in-time restore tested quarterly.
- **Observability:** OpenTelemetry traces across app → API → AI gateway → speech services; RED metrics per route; SLOs (API p95 <300 ms for reads, chat delivery p95 <500 ms, AI first token <400 ms); cost dashboards per AI feature; an alert on the "fallback path rate" (how often a fallback ran).
- **API contract:** one versioned OpenAPI spec generated from code, with a generated Dart client; delete the client's multi-path guessing (`lib/services/ai/historical_personality_service.dart:310-314`, which tries five URL variants) and `legacyBackendUrlPrefixes` of raw `http://` IPs (`lib/services/env_config.dart:35-38`).

13.5 Offline-first, for real

- **One local database** (Drift/SQLite or Isar) and **one durable outbox** with idempotency keys, exponential backoff and a visible sync state. Consolidate the five parallel sync implementations (`offline_handler.dart`, `background_sync_service.dart`, `sync_v2_service.dart`, `persisted_outbox_service.dart`, `backend_sync_provider.dart`) into one, built on the best of them. This is infrastructure only: every offline feature (Downloads, Point & Say queue, offline review) keeps working on top of it.
- **Downloadable units** (content pack + audio, typically 10–30 MB per unit with Opus audio), with storage management and Wi-Fi-only by default. Offline speaking practice with on-device tone feedback (distilled model) as a later milestone.
- **Low-bandwidth mode:** text-first AI, compressed audio, image-light UI; respect Android Data Saver.

13.6 Performance

- APK/AAB size: remove the duplicated curriculum JSONs (about 34 MB of identical files plus duplicated bundles, Python and Dart files shipped as assets), the unused fonts, and the 3 developer example files (move to `dev/`); older duplicate screens shrink the build as they are consolidated into their surviving feature (§17.5). Ship large content as downloadable language packs. Target under 40 MB download.
- Split the god files (`polie_workspace_screen.dart` 6,485 lines, `ai_chat_provider_groq.dart` 2,516, `api_provider.dart` 2,468, `game_provider.dart` 2,139) into feature modules with clear boundaries.
- Image and Rive budgets; `const` widgets; list virtualisation; avoid rebuild storms from large Riverpod providers.

13.7 Accessibility

- Screen-reader labels on every interactive element (471 `Semantics(` already exist; audit for coverage and meaningful labels), large-text support up to 200% without clipping, WCAG 2.2 AA contrast, captions and transcripts for all audio and video, haptic and visual alternatives to sound-only cues, Tone Lens not colour-only, switch/keyboard navigation on tablets and web, reduced motion, dyslexia-friendly option (Andika is a good base), and hearing-impaired paths (visual pitch feedback is a gift here).

13.8 Internationalisation

- Move all 1,104 hard-coded UI strings to ARB; professional translation of the UI into French, Portuguese, Arabic, Swahili, Amharic, Hausa, Yoruba and Igbo, **with full diacritics reviewed by natives**; pseudo-localisation in CI to catch overflow; RTL layouts for Arabic/Ajami.
- Explanations are delivered in the learner's L1 (French speakers learning Wolof, for example), not English only.

13.9 Testing and CI/CD

- **Gates:** `flutter analyze` fatal on warnings for new code; unit tests for the learner model, FSRS, graders and validators; golden tests for the design system and Rive chips; integration tests for onboarding → first lesson → sync; backend contract tests from the OpenAPI spec; an authorisation test suite; content validator (§6.5) as a blocking job; AI eval suite (§7.6, §8) blocking for prompt and model changes.
- **Coverage target:** 70% on domain logic, not vanity totals. Today there are 64 Flutter test files for 306k lines and 52 backend tests.
- **Release:** trunk-based with feature flags (a `featureFlag.model.ts` already exists); staged rollouts (1% → 10% → 50% → 100%) with crash-free and ANR gates; Shorebird or similar for Dart patching if policy allows; remove the auto-increment commit-and-push step in `build-and-release.yml:59-66` (CI committing to the main branch races with developers).
- **Repo hygiene:** move the roughly 470 status Markdown files into `/docs/archive`, write one honest `README`, one `ARCHITECTURE.md` and one `STATUS.md` that is generated from tests and feature flags rather than prose.

---

14. Admin and content operations

14.1 Current state

`lingafriq-admin` has 33 pages (lesson builder, quiz builder, lexicon manager, media library, telemetry, subscriptions, tribes/villages, culture magazine…) but **no review workflow, no roles beyond admin, no content versioning, no audit trail UI, and 0 tests**. Lesson drafts autosave to `localStorage` (`src/pages/LessonBuilder.tsx:258`). The app-side native verification tools are dead code.

14.2 Target: a content studio built around review

- **Roles:** Owner, Content Lead (per language), Writer, Native Reviewer, Linguist, Audio Producer, Historian (personas), Moderator, Support, Analyst. Least privilege enforced on the server.
- **Workflow:** Draft → AI-assist (optional, flagged) → Native review ×2 → Linguist approve → Audio recorded and aligned → QA → Scheduled publish → Monitor item stats → Revise. Every state change is audited.

- **Editors that understand the content:** a tone-aware text field (diacritic palette, NFC enforcement, live Tone Lens preview), audio waveform with alignment view, minimal-pair linker, dialogue editor with per-turn audio, can-do mapper, and an item preview that renders exactly like the app.
- **Quality dashboards:** item difficulty and discrimination, "report a problem" queue, reviewer agreement, audio coverage per language and level, placeholder detector, untranslated UI string count.
- **Moderation console:** report queues with evidence, user history, actions (warn, mute, suspend, ban), appeals, law-enforcement escalation workflow, CSAM handling restricted to trained staff.
- **Persona console:** source dossiers, claims audit, eval results per persona, kill switch.
- **AI ops:** prompt registry with versions and diffs, eval runs, cost per feature, model routing table, incident switches.
- **Tech:** upgrade to current React and Vite, typed API client generated from OpenAPI, SSO for staff with mandatory 2FA, server-side sessions or httpOnly cookies instead of `sessionStorage` tokens, Playwright tests on critical flows.

---

15. Analytics, experimentation and copy

- **North-star metric:** *weekly learners who demonstrated a new can-do* (learning outcome), with guardrails on D1/D7/D30 retention, minutes practised, speaking minutes, and crash-free sessions. Do not optimise on streak length alone.
- **Event taxonomy** (one schema, versioned): `session_started`, `item_answered{item_id, correct, latency, hints}`, `speech_attempt_scored{phones_ok, tone_ok, confidence}`, `cando_demonstrated`, `ai_turn{mode, tokens, latency, fallback_used}`, `content_reported`. Send on-device aggregates where possible (privacy).
- **Experimentation:** feature-flag-based A/B with pre-registered hypotheses and minimum detectable effects; never experiment on minors' safety settings.
- **Learning-outcome research:** an external efficacy study (pre/post oral proficiency interview with blinded raters) for two Stage-1 languages (one tonal West African language such as Yoruba, one Bantu language such as isiZulu) within a year. That result is your best marketing asset, and it must be real (no invented stats; Playbook §9).
- **Copy:** warm, clear, specific; grade-7 reading level; verb-first buttons ("Start lesson", "Hear it again"); name continuity ("Publish" → "Published"); no em dashes in UI strings; no "World-class/Elite/Seamless" in product copy (Playbook §9 and §15.4). The village voice lives in characters, not in every label. Error copy follows "what happened, why, what next" (e.g., "Couldn't score that recording. It was too quiet. Try again closer to the mic.").

---

16. Prioritized roadmap

Effort key: **S** ≤ 3 engineer-days · **M** 1–2 weeks · **L** 3–6 weeks · **XL** a quarter or more. Estimates assume a small team (3–4 engineers, 1 designer, a content lead, and paid native reviewers). They are judgement calls and should be re-estimated with the team.

16.1 P0, week 1–2 ("stop the bleeding")

#	Action	Fixes	Effort
1	Revoke and rotate the Firebase service-account key (`firebase-adminsdk-us33e`, plus the … `bc5f413f35` key file loaded by `messagingService.ts:29` and any copies that sat in `node-backend/dist/services` on the Mac), the Apple `.p8` key, and every AI key ever shipped in an app build. Purge them from git history (`git filter-repo`) and from every machine and backup. Add gitleaks to CI.	P0-1/2/3	S
2	Stand up a server-side **AI gateway** (auth, quotas, server-owned prompts, cost caps) and move every client Groq/HF/Stability/Replicate/Leonardo call behind it. Ship an app update that removes the keys.	P0-3/9/13	M
3	Socket lockdown: identity only from the verified JWT, server-authorised `join_room`, room-scoped emits instead of `io.emit`, friend-scoped presence, token from `auth` not the query string.	P0-4/5/6	M
4	Make reward endpoints server-only; compute XP from verified events; reconcile balances and remove fraudulent ones.	P0-7/8	S–M
5	Turn off DMs, snaps, calls and the public feed for under-18s (and for unknown ages) until §13.3 ships; add an age gate.	P0-10	S
6	Replace the toxicity filter with a hosted moderation API plus report/block flows; hard-delete expired snaps with server-uploaded media only.	P0-11/12	M
7	Persona fixes: AI disclosure, remove "don't mention being an AI", replace the substring safety filter, show an honest error instead of the canned fallback, correct the Nkrumah data.	P0-13/14/15	S
8	Remove fake scoring (random scores, zero-on-failure HTTP 200s) and keep those games visible in an honest **practice mode** (no score shown, "scoring coming soon") until a real scorer exists; remove the false tone claims and the placeholder bundles from the build.	P0-16/17/18/19	S
9	Delete the semver logout, fix onboarding persistence, replace the stub offline queue with the existing persisted outbox (pick one) and stop marking unsent ops complete.	P0-20/21/22	M
10	Notification dedupe (one per user per day, local timezone, quiet hours) and workers run only in the worker process.	P0-23	S
11	Swap out non-commercial models (MMS-TTS, XTTS-v2, NLLB) and the unofficial Translate scraper; get legal sign-off on every model and dataset licence.	P0-24/26	M
12	Generic login errors, per-account lockout and backoff, shorter tokens with revocable refresh.	P0-25	S
13	RevenueCat webhooks and server-side entitlement checks.	P0-27	S–M

16.2 P1, months 1–3 ("make it true")

Action	Section	Effort
Content studio and language factory (§6.4.1, §14): schema, validators, two-native review, audio recording and alignment. **All seven Stage-1 languages (Yoruba, Igbo, Hausa, Pidgin, Swahili, isiXhosa, isiZulu) at L1**, with native audio for every item; start Wave 2 at L0	§6, §14	XL (content), L (tooling)
Learner model + FSRS-6 scheduling + one item bank; adaptive placement	§6.3, §11.3	L
New IA (Today / Learn / Speak / Play / Village + Me, Search, Lens, Word Card, All-features directory) and one design system with tokens; re-render every feature on it; revive the 25 unreachable features and consolidate duplicate implementations into the surviving feature; retire the six extra **design systems** (code, not features)	§3, §17, §18	L
Onboarding v2: speak in 60 seconds, account later, guardian path	§5, §13.3	M
Real pronunciation and tone scoring v1 (GOP + F0 contour against native references) for Yoruba, Igbo and Hausa first, then the other Stage-1 languages, with confidence and an "unsure" state	§7.4	L
Avatar v1 (first build; avatars don't exist yet): character bible, then Polie plus one host on the View Model contract, following §4.6 steps 1–6, with lip-sync from aligned audio, all tutor states, verified on device	§4	L

Action	Section	Effort
Personas v1: five rigorously sourced personas with dossiers, citations, historian review, eval suite	§8	L
Village v1 for adults: circles and classrooms with moderation, report/block, server-authorised rooms	§9	L
Observability, SLOs, OpenAPI contract with a generated Dart client, CI gates, staged rollouts	§13.4, §13.9	M
Privacy programme: DPIA, data map, retention, export/delete in-app, NDPC registration check, privacy policy rewrite	§13.2	M
UI string extraction to ARB and native-reviewed translations with diacritics	§13.8	M
Gamification rebalance: one currency, ethical streaks, hearts moved to opt-in Challenge Mode	§12	M

16.3 P2, months 4–9 ("make it loved")

Action	Effort
Stage 1 to L2 (L3 where evals pass). **Wave 2 (14 languages: Amharic, Twi, Somali, Kinyarwanda, Afrikaans, Shona, Wolof, Oromo, Luganda, Lingala, Sesotho, Setswana, Chichewa, Malagasy) L0 → L1**. Wave 3 L0 kits begin. Community contributor programme with paid, credited native speakers	XL
Custom TTS voices for languages without vendor voices (isiXhosa first, then Wave 2 gaps), trained on commissioned recordings	L
Realtime voice tutor (speech-to-speech with barge-in) on the gateway, with tone-aware feedback	L
Flagship games from §10 on real scoring (Tone Lens duels, Market Haggle, Talking Drum)	L
Guardian and school accounts, teacher dashboards, curriculum alignment for schools	L
E2EE for private messages (MLS), group calls, Stories for circles	L
Offline-first content packs with downloadable units and offline review	M–L
Passport credential v1 with verifiable credentials, published methodology	M
Efficacy study design and recruitment	M
Accessibility audit with disabled users and fixes to WCAG 2.2 AA	M

16.4 Moonshots (9–24 months)

Moonshot	Why it matters	Effort
95+ languages via the Living Archive (Wave 4 community starter kits growing to L1)	Makes LingAfriq the home of African languages, not just the big ten	XL
Open tone corpus and on-device tone model for 10+ tonal African languages, built with paid contributors and released under an open licence	Becomes the industry reference for African tonal-language scoring; a real moat and a public good	XL
"Grandma Mode": family members record phrases in their own dialect and the app turns them into lessons for the grandchildren in the diaspora	Emotional hook no competitor has; generates authentic dialect content with consent	L
Living history: personas grounded in archives (with museum and university partners) and AR scenes at heritage sites	Turns history into an experience; partnerships open institutional funding	XL
Ajami, Nsibidi, Adinkra and Ge'ez script journeys	Celebrates African writing systems, which other apps ignore	L
Heritage Language Fund: a share of revenue funds documentation of endangered languages, with progress shown in-app	Mission credibility, press, grant funding	M (programme)
Teacher marketplace: verified native tutors for live sessions, with fair revenue share	Human connection at scale and a second revenue line	L

16.5 What to stop doing

- Stop starting **new** features until the existing 151 are finished and connected. Every one of them is kept (§18); the next 9 months are about taking each to its potential and wiring it into the learning loop. Put new breadth into **languages**.
- Stop writing status Markdown files that claim completion; let tests and flags speak.
- Stop shipping AI-generated language content without native review.
- Stop treating the WhatsApp-, Snapchat- and X-style screens as copies; keep their familiar patterns and make each one learning-native (Language Status, Clips, the Square with Tone Lens; §18.6), so the Village becomes something only LingAfriq can offer.

---

17. Information architecture: every feature has a home

Tope's rule: **no feature is lost or cut.** Today the app has about 125 named routes (`my_app.dart:278-627`), a 35-item drawer (`app_drawer_material3.dart`), 5 tabs (Home, Courses, AI, Social, Profile) and 88 built-but-unreachable screens. The problem is not too many features. The problem is that nothing tells a learner **which feature to use when**. The new IA gives every feature a home, a reason to appear, and a level at which it opens.

17.1 Navigation map

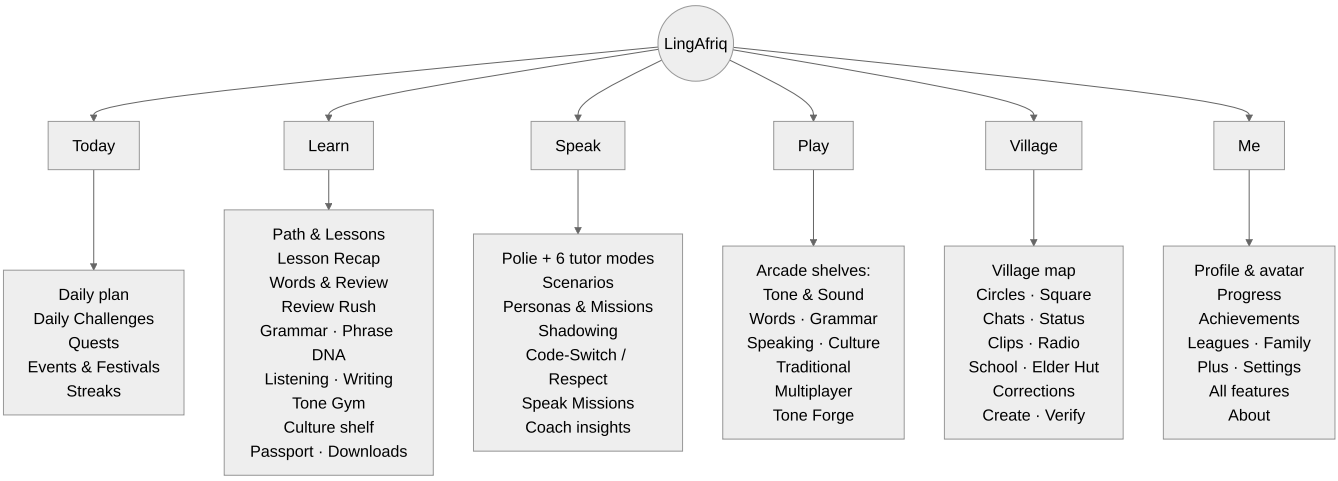
Five tabs + Me + three global tools.

Surface	What lives there	Brings together (today's entry points)
Today (tab 1)	One 10-minute plan: due reviews, next lesson, a speaking moment, a challenge, events, streak	Home tab, dashboard, Daily Goals, Quests, Daily Challenges (revived), Seasonal Events
Learn (tab 2)	The Path (units by CEFR level), lessons, Words & Review, Grammar, Listening, Writing, Tone Gym, Culture shelf (Magazine, Heritage, History, Mannerisms, Cultural Hub), Passport, Downloads	Courses tab, Lessons, Authentic Path, Learning Path, Grammar hub, Vocabulary Builder, Phrase DNA, Tone & Rhythm, Culture Magazine, Heritage, Passport
Speak (tab 3)	Polie (all tutor modes as chips), Scenarios, Personas & Missions, Shadowing, Code-Switch and Respect coaches, Speak Missions, Coach insights	AI tab, AI Chat, Polie modes, tutor dashboard, Historical Personas, Persona Missions, Speak Engine Lab, conversation scenarios
Play (tab 4)	The Arcade: every game on skill shelves (Tone & Sound, Words, Grammar, Speaking, Culture, Traditional, Multiplayer)	Games catalogues ×3, cultural games, traditional games, Tone Forge (revived)
Village (tab 5)	Map of places, Circles (tribes), Square (feed, rooms), Chats, Status, Clips, Radio (audio rooms), School (classroom), Elder Hut (mentors), Corrections, Create, Contribute	Social tab, Global/Private/Tribe chat, wa-*, snap-*, x-*, villages, tribes, classroom, social audio, micro-mentors, UGC, voice contribution
Me (avatar, top-left)	Profile, avatar wardrobe, Progress, Achievements, Leagues, Family, Plus, Settings, **All features**, About	Profile tab, settings, badges, leaderboards, magic items, features guide, Stitch hub
Search (top bar, global)	Words (→ Word Card / Living Dictionary), people, circles, lessons, games, **features by name**	search_languages_page, community search, people search
Lens (top bar camera button, global)	Point & Say, Live Translate (Conversation Bridge), Import Media	Drawer items Point & Say, Live translate, Import Media

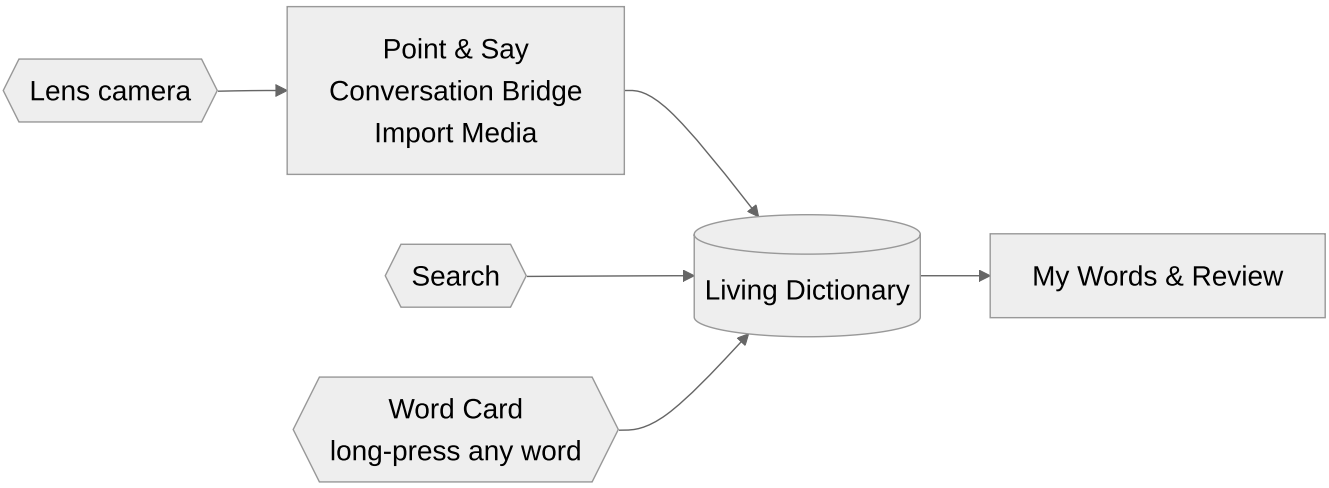
Surface	What lives there	Brings together (today's entry points)
Word Card (long-press any target-language word, anywhere)	Meaning, audio (dialects), Tone Lens, examples, "save", "practise"	Living Dictionary, My Words

Nothing is lost when the drawer goes. The drawer is replaced by **Me → All features**: an A–Z, searchable directory of every feature in §18, grouped by hub, with locked ones showing "unlocks at..." and a preview. Global Search also finds features by name ("ludo", "status", "proverb").

17.2 Sitemap



Global tools, available on every tab:



17.3 Contextual surfacing: features find the learner

Moment	Surfaces
Learner finishes a lesson	Lesson Recap → "say it" → a matching game ("Tone twins in Play")
A word appears anywhere (lesson, chat, feed, status, game)	Long-press → Word Card → Living Dictionary, save, practise
Camera button or a lesson about objects	Lens → Point & Say; the lesson offers "find 3 of these at home"
Learner struggles with tones	Tone Gym drill in Today; Tone Forge and Talking Drum in Play
Learner reaches a scenario unit (market)	Scenario Theatre → Market Haggle game → Village Market place
Weekend	Circle Clash, Village Radio show, Festival event

Moment	Surfaces
Learner is A2+ and active	"Try Native Corrections", "Post a Status in Yoruba"
Inactivity	One honest reminder with a 2-minute session (not 12 pushes a day, P0-23)

17.4 Progressive unlocking

Adults are never hard-blocked: a locked feature shows **"unlocks at..."** with a **Preview** and an **"Open anyway"** option. The exception is safety gates for minors (§13), which are enforced on the server.

Stage	Opens
Day 1 (no account needed)	Today, first unit, Polie basic, Lens (Point & Say), 3 games, Word Card, read-only Magazine and Events
Account created	Streaks, all games, Circles (join), Square (read), Global handle
Week 1 / A1 unit 3	Personas, Scenarios, Chats (adults), Status, Practice Rooms
A2	Clips, Native Corrections (post), Passport, Code-Switch and Respect coaches, Radio (speak)
B1	Create (UGC), Contribute/Verify (native speakers at any level), host rooms, mentoring
Under 18	Circles only with guardian-approved contacts, no public feed posting, no DMs with adults, no calls; School and games fully open

17.5 Re-audit of unreachable code: revive, consolidate, never cut

The earlier "183 unused files" figure counted Dart files that **no file imports**. A stricter reachability scan from `lib/main.dart` finds **330 files unreachable, 88 of them screens** (an unreachable screen can still be imported by another unreachable file). Re-auditing all 88 screens:

- **Revive (25 screen-level features, wired into the new IA):** Tone Forge; Conversation Analytics; Review Dashboard and vocabulary dashboard (Words & Review); Roleplay Completion Summary; Roleplay Progress Dashboard; Chat Search; Contact Info (block/report); Moderation Tools; Chat Welcome & Guidelines; Lesson Item Verification; Native Speaker Contribution; Game Mastery Results; Teacher Classroom Dashboard; Daily Challenges; help preloaders ×4 → first-use tips; Lesson Recap; Offline Downloads; Global Handle edit; Review Rush (gamified review); Social Audio Following; Scheduled Sessions; Family/Guardian Dashboard; Subscription (no live paywall exists today); About Us; Shadowing.
- **Consolidate (an older or parallel version of a feature that survives; best parts harvested):** 5 AI chat variants → Polie workspace; 4 classroom variants → live classroom; 6 onboarding variants and placement quiz → unified onboarding (Kijiji's village story kept); 7 game templates, universal loading/game screens, start guide and victory results → GameKit; `games/proverb_unlocker` GameKit rewrite **replaces** the live random-logic version; older chat, feed, profile, settings, curriculum, translation, practice and UGC "enhanced" variants → their live counterparts (full list in the per-feature "Code" lines of §18).
- **Developer artefacts (not features, so nothing user-facing is lost):** `examples/errorhandler_integration_example.dart`, `examples/performance_utilities_example.dart`, `ai_chat/polie_tsx_flutter_mapping.dart`. Move them to a `dev/` folder outside the app build.

17.6 Exemplar deep-dive: Point & Say

What it is. Point the camera at an object or a sign, learn the word in your language, say it, get feedback. It is the most "magical" feature in the app and the best hook for the first session.

Current state (evidence). `screens/ar/point_and_say_screen.dart` (982 lines), route `point-and-say` (`my_app.dart:500-506`) and a drawer item.

- The photo is downscaled (`:176`). Object labels come from on-device `google_mlkit_object_detection`, best effort (`:195-219`). Text comes from ML Kit's Latin `TextRecognizer` via `media_import_service.dart:89-93`.
- English labels are translated through `hybrid_polie/translation_service.dart`. The fallback chain is backend → Hugging Face NLLB-200 (`:32`, CC-BY-NC, not licensed for commercial use, P0-24) → the free MyMemory API (`:280-290`).
- The learner records audio. The flow is a voice-service health check → `OfflineFallbackPolicy` → `PointAndSayQueueService` (153 lines; an offline queue with optional tone evaluation via learning v2).

Defects.

1. The target language is typed as free text ("Enter a language (e.g., yoruba)", `:427-430`), so typos break it.
2. ML Kit's generic labels are English, coarse ("Food", "Home good") and culturally blind: no *ọ̀gẹ̀dẹ̀* vs *plantain* distinction, no *agbada*, no *jollof*.
3. Translation is word-level with a non-commercial model and an anonymous free API, so quality and licence are both unsafe.
4. There's no native audio for the word, so the learner imitates nothing.
5. The scoring path depends on a placeholder tone endpoint (§1 P0-18/19).
6. Latin-only OCR mangles stacked diacritics **[Unverified on device]** and cannot read Ge'ez or Ajami.

Elevation: "Lens".

1. **Recognise.** On-device detection for instant boxes, then a server vision model (multimodal Gemini 3 Flash; AI stack §3.1, with §3.13 for OCR) for **culturally specific** labels with a confidence score. A curated African object set (foods, clothing, household items, market goods) is grounded in the Living Dictionary.
2. **Teach.** Each tag becomes a **Word Card**: word with tones, native audio (from the verified lexicon, or labelled TTS), example sentence, dialect variants, and a plural or classifier where relevant (Zulu noun class prefixes, Swahili m-/mi-).
3. **Say.** Hold to speak. Real GOP plus tone scoring (§7.4), with the native contour overlaid on yours.
4. **Collect.** Words go to a personal **Lens collection** and into FSRS review. Weekly **"Home Safari"** quests: find 5 kitchen items.
5. **Share.** Optionally turn the photo into a Clip with the phrase overlay (Village → Clips). Faces are blurred before sharing.
6. **Signs and menus.** OCR mode translates signs, menus and labels and offers "read it aloud". A diacritic-restoring pass handles stacked tone marks.
7. **Offline.** An offline pack of the top 300 objects per language with cached audio. The existing queue service syncs scores when back online (after the P0-22 fix).
8. **Privacy.** Photos are processed then deleted unless saved. Minors' photos never leave the device without consent.

UX and render. Full-screen camera with soft rounded boxes that "breathe". Tapping a box flips it into a Word Card with the target word at 28 sp, a tone contour line under the syllables and a large mic button. The language comes from the learner profile (no text box), with a chip to switch. Haptics on a successful pronunciation; the avatar appears in the corner to celebrate. A reduced-motion variant is included.

Lives in. The **Lens button** in the top bar on every tab; Learn (object lessons: "find these at home"); Today (Home Safari challenge); Play (Flashcard Safari uses the same pipeline); Word Card → "see it in the wild".

17.7 Exemplar deep-dive: Living Dictionary

What it is. The app's word store, which learners can add to.

Current state (evidence).

- **Client:** `screens/vocabulary/living_dictionary_screen.dart` (620 lines) with `services/learning/living_dictionary_service.dart`. It does GET/POST/DELETE on `ApiContract.learningV2.livingDictionaryEntries`, caches in SharedPreferences, and shows an offline banner (`:390`) and an "Add to living dictionary" action (`:244`).
- **Backend:** `learningV2.route.ts:150-165` (requireSignin) and `controllers/learningV2.controller.ts:834-930`. `listLivingDictionary` queries `{userId}` and creates `LivingLexemeModel` entries with that `userId`, so **it is a per-user personal word list, not a shared dictionary**.
- **Admin:** a separate CRUD/import exists (`routes/admin/livingDictionary.admin.route.ts:13-18`, requireAdmin). User entries have no review status.

Defects.

1. The name promises a living, shared lexicon; the product is a private notes list.
2. Learner-entered words have no verification, so if they were shared they would spread errors.
3. There are no audio, tone, dialect or example fields in the learner flow.
4. It is not linked from lessons, chat or games, so words have to be retyped.
5. The admin lexicon and user entries are separate stores with no promotion path.

Elevation: a two-layer living lexicon.

1. **Verified Lexicon (shared).** Every entry has a lemma with full tone marks, IPA, part of speech, noun class, plural, dialect variants, 2–3 native audio recordings (age and gender spread), example sentences from real corpora, cultural notes, register (formal/elder/slang), etymology and loanword origin, and related proverbs. Each entry carries a status: AI-drafted → native-verified → linguist-approved, shown as a badge. This is the engine of the language factory (§6.4.1): one schema for all 7 Stage-1 languages and then every wave.
2. **My Words (personal).** What the current endpoint already does, kept: saved words, personal notes and the learner's own recordings. Each saved word links to its verified entry when one exists.
3. **Living loop.** A learner suggests a word or meaning → it goes to the Verify queue (§18 Village → Contribute) → native speakers confirm → it is promoted into the Verified Lexicon with credit to the suggester ("Added by @ada, verified by 3 speakers"). Corrections from Native Corrections and the Square ("How do you say…?") feed the same queue.
4. **Everywhere.** Long-press any word in a lesson, chat, feed, status, game or Lens result to open the **Word Card**. Global Search starts with words. A daily "word of the day" with its story appears in Today.
5. **Pan-African view.** "Same idea in other languages": water = omi / mmiri / ruwa / maji / amanzi, linking languages and celebrating cognates. It powers Multilingual Relay.
6. **Offline.** A per-language lexicon pack (top 5k entries plus audio) syncs the personal layer.

UX and render. The Word Card is a bottom sheet with a large lemma (tone marks never clipped; Noto Sans with diacritic-safe line height), a Tone Lens contour, play buttons for each speaker, dialect chips, examples with audio, and Save and Practise buttons. The full entry page adds proverbs, culture and the "same idea in other languages" strip. Verification badges are honest ("AI draft, awaiting native check").

Lives in. Global Search (first result type); long-press on any word; Learn → Words (My Words and Review); Lens results; Village → Contribute → Verify.

18. Feature catalogue and deep-dives (every feature kept)

151 user-facing features, each with its own deep-dive below, grouped by hub. Where several screens implement one feature (old and new versions), they are one row and the best version wins (consolidation);

no feature is dropped. **21 catalogue entries revive previously unrouted screens (25 screen-level features: some entries revive two or more related screens together; see §17.5).** Evidence lines are generated from a static scan of each file (line counts, direct Groq calls, `Random`, mock markers, swallowed catches, hard-coded `Text(` vs l10n) plus the routing graph; they are signals, not proof of behaviour, and on-device behaviour is **[Unverified]** unless a `file:line` is cited.

State legend: Works = reachable and functional in code; Works (client AI) = works only while a paid key is compiled into the app (P0-3); Partial = reachable but thin, duplicated or cheatable; Simulated / Faked = shows fake activity or scores; Unrouted → revive = built but unreachable today.

18.1 Catalogue summary

#	Feature	Hub	State	Elevation in one line	Lives in
1	Home / Today session	Today	Works	Make it a single adaptive "Today" session (10–15 min), a playlist built by the learner model: 3 due reviews → 1 new input (story or dialogue) → 1 spea...	Today tab (root).
2	Daily Goals	Today	Works (client AI)	Goals become intent-based ("speak for 5 minutes", "learn 10 words", "one conversation") rather than XP.	Today → goal ring; Me → Settings → Goals.
3	Daily Challenges	Today	Unrouted → **revive**	Revive as the Challenge board behind the Today widget.	Today → Challenges strip → board.
4	Quests	Today	Works (client AI)	Story quests set in the village narrative ("Help Mama Ngozi prepare for the naming ceremony": learn kinship words, greet the elder, haggle for yams in...	Today → Quest card; Play → Quests.
5	Seasonal Events / Village Festival	Today	Works	A cultural calendar: Ọdún Ọ̀ṣun, Eid, Timkat, Durbar, Homowo, Umhlanga, Kwanzaa and Heritage Day.	Today → Event banner; Village → Events.
6	Streaks & reminders	Today	Works	Ethical streaks (§12): a weekly-streak option, earned freezes and repair windows.	Today header; Me → Progress.
7	Lessons map / Courses	Learn	Works	One Path map per language: CEFR bands as regions of a village-to-city journey, units as stops, each showing its can-do ("I can greet elders correctly"...	Learn tab (root).
8	Authentic Path (A1–C1)	Learn	Works	Becomes the "Real-world track": authentic materials (radio clips, songs, market recordings, Nollywood/Bongo scenes, licensed) with scaffolding, unlock...	Learn → Path → "Real-world" lane (A2+).
9	Lesson Flow	Learn	Works	An adaptive lesson player: input → noticing (Tone Lens) → guided practice → speaking → mini-check, with stages skipped when mastered.	Opened from Path and Today.
10	Lesson detail & Level certificate	Learn	Works	Lesson detail shows can-dos, words and the culture note, with a preview audio.	Learn → Path → lesson; Me → Credentials.
11	Learning Path (personal plan)	Learn	Works	Make it the "My plan" view of Path: the learner's goal (heritage, travel, exam) mapped to units and weekly targets, with an ETA to the next can-do.	Learn → Path → "My plan".
12	Placement test	Learn	Works	An adaptive 3-minute CAT placement with listening, reading and one speaking prompt.	Onboarding; Learn → Path → "Retake placement".

#	Feature	Hub	State	Elevation in one line	Lives in
13	Grammar Hub	Learn	Works	A Pattern Library: each pattern card has a plain-language explanation in the learner's language, 5 audio examples, contrasts and common errors.	Learn → Patterns.
14	Vocabulary Builder	Learn	Works	Themed word packs (market, family, body, church/mosque, football) with native audio, Tone Lens, pictures reviewed for cultural accuracy, and "add all ...	Learn → Words → Packs.
15	My Words & Review (SRS)	Learn	Unrouted → **revive**	One FSRS-6 review fed by every feature (save a word anywhere and it appears).	Learn → Words → Review / Insights; Today pulls due items.
16	Review Rush (gamified review)	Learn	Unrouted → **revive**	Revive as Review Rush: a 90-second sprint over due cards with combo multipliers that only reward accuracy.	Learn → Words → Review → "Rush"; Play → Quick games.
17	Listening Practice	Learn	Works	Ear training: graded listening at native speed with accents (Lagos, Ibadan, Kano, Nairobi, Durban), a slow-down slider, dictation with diacritic-aware...	Learn → Skills → Listening.
18	Writing Practice	Learn	Works (client AI)	Write & check: a diacritic keyboard bar, tone-aware grading that is lenient early and strict later, and AI feedback through the gateway that shows unc...	Learn → Skills → Writing.
19	Phrase DNA	Learn	Works	Turn it into sentence frames you can speak: "I want to buy ___" with slot fillers from the learner's own words.	Learn → Skills → Phrase DNA; Speak → Warm-ups.
20	Tone & Rhythm Trainer	Learn	Works	The Tone Gym, LingAfriq's signature: perception drills (minimal pairs), production with a live pitch contour against the native one, per-language tone...	Speak → Tone Gym (also Learn → Skills).
21	Tonal Lesson (Village)	Learn	Works	Becomes the story-led tone lesson (an elder teaches greetings with tone), linked to the Tone Gym for practice and placed on the Path where tone is int...	Learn → Path (tone units); Speak → Tone Gym → Lessons.
22	Language Quiz	Learn	Works	Merge into the item bank: unit checkpoints and adaptive quizzes with rationales and audio (§11.3), while keeping the familiar "take a quiz" entry.	Learn → Path → Checkpoints; Today → "Quick quiz".
23	History & History Quiz	Learn	Works	Living History: short illustrated stories with sources, linked to Historical Personas ("Meet her" button) and quizzes that cite the story.	Learn → Culture → History; Speak → Personas cross-link.
24	Mannerisms & etiquette	Learn	Works	Etiquette cards with practice: kneeling and prostration greetings, handshakes, eating etiquette, taboo topics.	Learn → Culture → Etiquette.
25	Cultural Hub	Learn	Works	Becomes the Culture landing inside Learn, organising Magazine, History, Etiquette, Heritage archive, Proverbs and Music in one place.	Learn → Culture (root).
26	Culture Magazine	Learn	Works (client AI)	A bilingual magazine: articles at the reader's level with tap-to-gloss, audio narration by native voices, "words from this article" to save, and autho...	Learn → Culture → Magazine; Today → culture card.

#	Feature	Hub	State	Elevation in one line	Lives in
27	Heritage Milestones	Learn	Works	Family milestones for heritage learners: "Greeted grandma in Yoruba", "Understood a family WhatsApp voice note", "Said a prayer at a naming ceremony".	Me → Heritage; Today celebrates milestones.
28	FLB Heritage Archive	Learn	Works	The Living Archive (Wave 4 languages): folktales, praise poetry (oríkì), songs and proverbs recorded with community consent, each with transcript, tra…	Learn → Culture → Archive.
29	Dialect Preference	Learn	Works	Real dialect lanes: audio variants per item (Ọ̀yọ́ vs Ìjẹ̀bú Yoruba; Owerri vs Onitsha Igbo; Kenyan vs Tanzanian Swahili), with notes where words diff…	Me → Settings → Language & dialect; Word Card toggle.
30	Living Dictionary	Learn	Works	See exemplar deep-dive	Global: search bar + long-press on any word; Learn → Words.
31	Offline Downloads	Learn	Unrouted → **revive**	Revive as Downloads: per-unit packs (text + Opus audio), storage meter, Wi-Fi-only toggle, auto-download of the next unit, and an offline Tone Gym onc…	Me → Downloads; Path shows a download icon per unit.
32	Language detail & Suggest a language	Learn	Works	Language pages show speakers, regions, readiness level (L0–L3) and a sample phrase.	Learn → language switcher → "All languages".
33	Passport (proctored test & credential)	Learn	Works	The LingAfriq Passport: evidence portfolio plus an adaptive proctored test, mapped to CEFR descriptors with a published methodology, issued as a verif…	Me → Credentials (unlocks at A2).
34	Lesson Recap	Learn	Unrouted → **revive**	Revive as the 60-second recap after every lesson: three retrieval questions, the new words with audio and Tone Lens, one sentence to say aloud (scored…	End of every lesson (Learn → lesson → recap).
35	Polie AI Tutor workspace	Speak	Works (client AI)	Polie becomes a voice-first companion: hold-to-talk with live transcript and Tone Lens on Polie's replies, recasts ("You said X; natives say Y"), a hi…	Speak tab (root) → Polie.
36	Tutor: Dialogue mode	Speak	Works (client AI)	Scripted-goal dialogues ("order food", "ask directions") where Polie improvises inside human-authored scenarios and scores goal completion, politeness…	Speak → Polie → mode chips.
37	Tutor: Grammar mode	Speak	Works (client AI)	Answers come only from the Pattern Library (retrieval), cite the pattern card, and spawn a 3-item micro-drill.	Speak → Polie → mode chips.
38	Tutor: Story mode	Speak	Works (client AI)	Choose-your-path stories at the learner's level using known plus due words, narrated by native TTS, with comprehension checks.	Speak → Polie → mode chips.
39	Tutor: Translation mode	Speak	Works (client AI)	Translate & teach: shows the translation, word-by-word alignment, register options (formal/informal/elder) and why.	Speak → Polie → mode chips.
40	Tutor: Pronunciation mode	Speak	Works (client AI)	Real scoring (§7.4): per-syllable tone and phone feedback with the native contour overlay.	Speak → Polie → mode chips.
41	Tutor: Assess mode	Speak	Works (client AI)	An oral proficiency interview lite: a 5-minute adaptive conversation rated against can-do descriptors, with native-rater calibration.	Speak → Polie → mode chips.

#	Feature	Hub	State	Elevation in one line	Lives in
42	Tutor dashboard	Speak	Works	Coach view: Polie recommends the next mode based on weak skills; the progress tab shows speaking minutes, tone accuracy trend and conversation goals a...	Speak → Polie → "Coach".
43	Shadowing	Speak	Unrouted → **revive**	Revive as Shadow Me: a native clip plays and the learner speaks along a fraction of a second behind, with karaoke-style word highlighting from forced ...	Speak → Practice → Shadowing; Today slot.
44	Conversation Scenarios & Roleplay	Speak	Unrouted → **revive**	Scenario Theatre: a library of real-life scenes with goals and a cast.	Speak → Scenarios.
45	Conversation Analytics	Speak	Unrouted → **revive**	Revive as "How I speak": words used vs avoided, average utterance length, code-switch rate, tone accuracy over time and the most frequent corrections,...	Speak → Coach → Insights; Me → Progress.
46	Speak Engine Lab (code-switch & register coach)	Speak	Works	Split into two named features.	Speak → Practice → Code-Switch Coach / Respect Coach.
47	Speak Missions	Speak	Works (client AI)	Weekly speaking missions in the real world ("Greet three people in Yoruba today and record one"), with a rubric-based evaluation and an optional share...	Speak → Missions; Today → Challenges.
48	Historical Personas	Speak	Works	See exemplar deep-dive	Speak → Personas.
49	Persona Missions	Speak	Works (client AI)	The story-chapter format from §8: era scene, objective (greet the Queen Mother correctly), branching dialogue, cultural debrief with sources, and a re...	Speak → Personas → Missions.
50	LingChat AI conversation	Speak	Works	Becomes "Practice with Polie" inside chat: open a practice DM with Polie or a persona from the chat list, so AI practice sits where conversations alre...	Village → Chats → pinned "Polie (AI)".
51	Synthetic Voice Styles	Speak	Works	Voice picker with commercial voices per language (Intron, Spitch, Azure, custom; AI stack §3.5): preview, gender, region, speed.	Me → Settings → Voice; Speak → Polie settings.
52	Point & Say	Speak	Works	See exemplar deep-dive	Global Lens button; Learn/Speak contextual.
53	Live Translate	Speak	Works	Conversation Bridge: two people, two languages, one phone.	Global Lens → "Bridge"; Speak → Practice.
54	Import Media (OCR & clips)	Speak	Works	Learn from anything: import a song, a WhatsApp voice note, a sermon clip or a sign photo, and LingAfriq turns it into a personal lesson (transcript, g...	Global Lens → "Import"; share-sheet target from other apps.
55	Games Arcade (hub & catalogue)	Play	Works	One Arcade with shelves: "For you today" (games that train the learner's weakest skills), Tone & Sound, Words, Grammar, Speaking, Culture, Traditional...	Play tab (root).
56	Game shell, guides & results	Play	Unrouted → **revive**	One GameKit frame.	Inside every game.
57	Word Match	Play	Partial (thin content)	Due-word driven matching with audio on every tap and a "tone twin" round where two words differ only in tone.	Play → Words shelf.

#	Feature	Hub	State	Elevation in one line	Lives in
58	Word Match (audio)	Play	Partial (thin content)	Listening-first: hear it, match it.	Play → Tone & Sound shelf.
59	Fill in the Blank	Play	Partial (thin content)	Cloze from real dialogues with diacritic-aware typing and a hint that plays the audio.	Play → Grammar shelf.
60	Picture–Word Association	Play	Partial (thin content)	Culturally reviewed illustrations (local foods, clothing, places) and a "say it" bonus that awards points only for real scored speech.	Play → Words shelf.
61	Memory Map	Play	Partial (thin content)	Audio-card variant (flip to hear) and family-themed decks, so heritage learners memorise kinship terms.	Play → Words shelf.
62	Speed Challenge	Play	Partial (thin content)	Adaptive speed that targets 80–85% success, with a "calm mode" without a timer for accessibility.	Play → Words shelf.
63	Speed Round	Play	Partial (thin content)	Merge mechanics with Speed Challenge as two modes of one game ("Sprint" and "Rounds"); both survive as modes.	Play → Words shelf.
64	Story Builder	Play	Partial (thin content)	Builds illustrated audio stories from the learner's level; finished stories can be shared to circles.	Play → Grammar shelf.
65	Grammar Detective	Play	Partial (thin content)	Cases framed as village mysteries; each solved case unlocks a Pattern Library card.	Play → Grammar shelf.
66	Grammar Jam	Play	Partial (thin content)	Rhythm-paced sentence building over Afrobeats/amapiano-style beats (commissioned), with rhythm tied to syllables.	Play → Grammar shelf.
67	Greeting Diplomacy	Play	Partial (thin content)	Becomes Greeting Gauntlet (§10.3): time × person × setting, with avatar gestures and spoken answers.	Play → Culture shelf.
68	Market Bargaining	Play	Partial (thin content)	Becomes Market Haggle: voice roleplay with trader personalities, real numbers practice, and politeness scoring.	Play → Speaking shelf.
69	Quiz Chef	Play	Partial (thin content)	Recipes as lessons: cook jollof, egusi, ugali or chakalaka by following spoken instructions in the target language.	Play → Culture shelf.
70	Roleplay Adventure	Play	Partial (thin content)	Shares the Scenario Theatre engine; adventures become multi-scene trips (Lagos → Ibadan → Ọ̀ṣogbo).	Play → Speaking shelf.
71	Taxi Survival	Play	Partial (thin content)	Becomes Danfo/Matatu Rush: real recorded conductor calls at native speed; board the right bus.	Play → Tone & Sound shelf.
72	Tone Trainer	Play	Partial (thin content)	Per-language tone inventories with minimal pairs and real audio; feeds the Tone Gym ladder.	Play → Tone & Sound shelf.
73	Pronunciation Duel	Play	Faked → fix	Real recording and scoring; async duels with friends on minimal pairs; honest "practice mode" (no score) until the scorer ships.	Play → Speaking shelf.
74	Pronunciation Game	Play	Partial (thin content)	Merge its rounds into the Tone Gym ladder as a game mode; real scoring.	Play → Speaking shelf.
75	Pronunciation Karaoke	Play	Faked → fix	Becomes Echo Karaoke: real F0 line, licensed songs, children's rhymes and praise phrases.	Play → Speaking shelf.
76	Emoji Translator	Play	Partial (thin content)	Emoji stories → sentences, then send the sentence as a chat sticker made from your avatar.	Play → Words shelf.

#	Feature	Hub	State	Elevation in one line	Lives in
77	Elders' Blessings	Play	Partial (thin content)	An elder character (avatar) gives blessings for life events (birth, marriage, travel); the learner responds correctly.	Play → Culture shelf.
78	Folktale Reconstruction	Play	Partial (thin content)	Àlọ́ / hadithi / iintsomi storytelling: rebuild the tale, then narrate it with scoring.	Play → Culture shelf.
79	Clan Lineage Story Builder	Play	Partial (thin content)	Teaches kinship vocabulary through a family tree you build; links to the Ancestral Tree feature.	Play → Culture shelf.
80	Conversation Relay	Play	Partial (thin content)	Classroom and circle relay: each player adds a turn by voice; AI checks; the chain plays back as an audio drama.	Play → Speaking shelf.
81	Board Word Games	Play	Partial (thin content)	Diacritic-aware word puzzles (Wordle-like with tone marks as a mechanic).	Play → Words shelf.
82	Cultural Games collection	Play	Partial (thin content)	Becomes the Arcade's "Culture" shelf (the screen remains as a themed shelf).	Play → Culture shelf.
83	Accent Puzzle	Play	Partial (thin content)	Decode the same sentence spoken in different regional accents; builds real-world listening.	Play → Tone & Sound shelf.
84	Call & Response	Play	Partial (thin content)	Musical call-and-response with real singers; responses scored on rhythm and tone.	Play → Speaking shelf.
85	Cultural Etiquette	Play	Partial (thin content)	Linked to Etiquette cards; consequences shown kindly ("Mama is surprised you didn't kneel").	Play → Culture shelf.
86	Drum Word	Play	Partial (thin content)	Becomes Talking Drum (§10.3): drummed tone contours → words, then speak it.	Play → Tone & Sound shelf.
87	Flashcard Safari	Play	Partial (thin content)	Shares the Point & Say camera pipeline: a scavenger hunt ("find something red, say it").	Play → Words shelf.
88	Food Quest	Play	Partial (thin content)	Market-to-pot quests with regional dishes; ties to Quiz Chef recipes.	Play → Culture shelf.
89	Liar Liar	Play	Partial (thin content)	Social bluffing: friends submit sentences, some wrong; others spot the lie.	Play → Grammar shelf.
90	Multilingual Relay	Play	Partial (thin content)	Pan-African relay: one word passed through several languages (Yoruba → Swahili → Zulu), celebrating cognates and loanwords.	Play → Words shelf.
91	Phrase Sniper	Play	Partial (thin content)	Listening reflex game with native audio; accessibility mode without time pressure.	Play → Tone & Sound shelf.
92	Proverb Unlocker	Play	Partial (thin content)	Becomes Proverb Weaver (§10.3).	Play → Culture shelf.
93	Rhythm Typing	Play	Partial (thin content)	Diacritic typing trainer to rhythm; builds tone-mark keyboard fluency.	Play → Words shelf.
94	Tongue Twister Race	Play	Partial (thin content)	Real scoring with fluency metrics; native tongue twisters per language; shareable duet clips.	Play → Speaking shelf.
95	Village Quest	Play	Partial (thin content)	Becomes the playable layer of the Village map: talk to NPCs in each place (market, café, elder hut).	Play → Speaking shelf.
96	Ayo (mancala)	Play	Partial (thin content)	Keep the authentic game; questions become due-word reviews; online play with friends; a cultural history card.	Play → Traditional shelf.

#	Feature	Hub	State	Elevation in one line	Lives in
97	Snakes & Ladders (cultural)	Play	Partial (thin content)	Family mode for heritage households (pass-and-play), with questions at each player's own level.	Play → Traditional shelf.
98	Language Ludo	Play	Partial (thin content)	Multiplayer with friends and circles; speaking challenges on special squares.	Play → Traditional shelf.
99	Suwe (hopscotch)	Play	Partial (thin content)	Kids-friendly counting and body-movement vocabulary; a guardian-safe offline mode.	Play → Traditional shelf.
100	Whot!	Play	Partial (thin content)	Card names and calls in the target language ("Check up!", "Last card!"); online play with friends.	Play → Traditional shelf.
101	Drum Rhythm	Play	Partial (thin content)	Merge with Talking Drum as its rhythm mode; rhythm mapped to syllable timing.	Play → Tone & Sound shelf.
102	Tone Forge	Play	Unrouted → **revive**	Revive as a headline Tone & Sound game: forge words by shaping their pitch contour on the Tone Lens (drag syllables up and down), hear the result and ...	Play → Tone & Sound (featured).
103	Villages hub & Language Villages	Village	Works	The Village is the community home: an illustrated map per language where each place is a real activity.	Village tab (root).
104	Swahili Village Map	Village	Works	The template for every language: Yoruba compound, Igbo obi, Hausa birni, Zulu umuzi, Xhosa village, a Lagos street for Pidgin.	Village → map (per language).
105	Village Market	Village	Works	Live Market Haggle stall, price and number drills, and the Magic Items shop (cosmetics only, no paid power).	Village → map → Market.
106	Village Café	Village	Works	Doorway to Practice Rooms and audio rooms: "3 tables open in Yoruba A2 right now".	Village → map → Café.
107	Elder Hut	Village	Works	Home of Micro-Mentors and proverbs: book a 10-minute session with a verified native speaker or elder, hear a proverb of the day with its story, and as...	Village → map → Elder Hut.
108	Practice Rooms & Sessions	Village	Simulated → make real	Real small-group rooms (2–6 learners at the same level) with a structured script: warm-up, topic, roleplay and a recap.	Village → Café → Practice Rooms; Today when a room matches.
109	Flashcard Focus (group)	Village	Simulated → make real	Replace the mocks with real matched learners or clearly labelled practice bots; study together with a shared deck and a race to recall.	Village → Café → Study together.
110	Matching Pairs (group)	Village	Works	Becomes the multiplayer mode of Memory Map: turn-based with friends, audio cards.	Village → Café; Play → Words.
111	Tribes	Village	Partial (P0/dupes)	Keep the feature, rename to Circles (study teams of 5–30) with optional language or city themes.	Village → Circles.
112	Tribe vs Tribe, Tribal Duel & Inter-tribe Leaderboard	Village	Partial (P0/dupes)	Circle Clash: weekend team events where points come only from verified learning (reviews done, lessons passed, scored speaking).	Village → Circles → Clash; Today → Events.
113	Global Chat	Village	Works	Language Squares: public rooms per language and level with a "target-language only" mode, Tone Lens on messages, tap-to-translate, and one-tap "correc...	Village → Square → Rooms.

#	Feature	Hub	State	Elevation in one line	Lives in
114	Private Chat (DMs)	Village	Unrouted → **revive**	Language-exchange DMs: each message can be written in the target language with an inline "check my message" (AI suggestion before sending) and a partn...	Village → Chats.
115	Call History	Village	Works	Each call entry shows minutes spoken in the target language and words learned (opt-in transcript), and offers "save phrases from this call".	Village → Chats → Calls.
116	Chat Welcome & Community Guidelines	Village	Unrouted → **revive**	Revive as the first-entry pledge: 3 friendly rules (be kind, help learners, no harassment) with an illustrated agreement, shown once before the first ...	Village first entry; Me → Safety.
117	Moderation Tools	Village	Unrouted → **revive**	Revive as Community Stewards tools for circle leaders and trusted volunteers: reported queue, mute/timeout, slow mode, pinned rules, with escalation t...	Village → Circles → Steward tools (role-gated).
118	Social Hub & Connections	Village	Works	Language partners: suggested partners who speak the language you learn and learn the one you speak (tandem matching), with friend profiles showing lan...	Village → People.
119	Friend Quests & Challenge a Friend	Village	Works	Buddy quests: shared weekly goals (both practise five days), async challenges on real skills (tone duel, review race), and nudges that don't nag.	Village → People → Quests; Today.
120	Share Progress	Village	Works	Beautiful share cards: the learner's avatar saying a phrase they learned (short clip), Passport stamps and streak milestones in the target language.	Me → Progress → Share; after milestones.
121	Social Gifting	Village	Partial (P0/dupes)	Gift a lesson / streak freeze / cosmetic using server-validated currency; send a voice blessing with the gift ("Ẹ ku odun!").	Village → People → Gift.
122	Community Corrections	Village	Works	Native Corrections: learners post short writing or audio; verified natives correct with inline edits and audio; reputation for helpful correctors; bes...	Village → Corrections.
123	Ancestral Tree	Village	Works	Family Tree in your language: heritage learners build their family tree with kinship terms (ìyá àgbà, nne nne, ugogo), record relatives' names with co...	Me → Heritage; Learn → Culture.
124	Micro-Mentors	Village	Works	Book a native for 10–20 minute focused sessions (pronunciation clinic, heritage conversation, exam prep), with verified mentors, ratings, safeguarding...	Village → Elder Hut → Mentors.
125	Live Classroom	Village	Works	School in the Village: scheduled live classes by level with a teacher, raise-hand speaker queue, shared whiteboard with Tone Lens, auto-notes (transcr...	Village → School.
126	Teacher Classroom Dashboard	Village	Unrouted → **revive**	Revive as LingAfriq for Schools: class roster, assign units, see weak skills per student, run a live game (Conversation Relay, Liar Liar) on a project...	Me → Teaching (role-gated); web.

#	Feature	Hub	State	Elevation in one line	Lives in
127	Social Audio Rooms	Village	Unrouted → **revive**	Village Radio: scheduled audio shows hosted by natives (folktale night, Pidgin news, Zulu music talk), learner rooms by level, and revived following a...	Village → Radio.
128	Status (WhatsApp-style)	Village	Works	Language Status: post a daily status in the target language (text, photo or voice) with a word-of-the-day sticker; friends react with corrections or "...	Village → Status.
129	Starred Messages & Media Gallery	Village	Works	Starred = Saved phrases: starring a message adds it to My Words with audio; the media gallery doubles as a personal phrasebook of voice notes.	Village → Chats → Saved.
130	Snaps & Streaks (Snapchat-style)	Village	Partial (P0/dupes)	Clips: send a photo or 10-second video with a phrase overlay in the target language (camera labels from Point & Say suggest words); streaks count only...	Village → Clips; Lens camera.
131	Community Feed (X-style)	Village	Works	The Square feed: posts in the target language with Tone Lens and tap-to-translate; native "verified speaker" badges; threads for questions ("How do yo...	Village → Square.
132	Explore, Search & Community Profile	Village	Works	Explore becomes the Square's discover page (trending phrases, events, top correctors); search folds into the global Search; community profiles show la...	Village → Square → Discover; global Search.
133	Create Content (UGC)	Village	Works	Creator Studio: templates for mini-lessons, quizzes and stories; AI checks for diacritics and accuracy; native-review queue; published content credite...	Village → Create (unlocks at B1).
134	Voice Contribution	Village	Works	Lend your voice: natives record prompted sentences with explicit consent and licence choice (AI stack §5), get credited, and earn currency or cash.	Village → Contribute.
135	Native Speaker Contribution & Lesson Item Verification	Village	Unrouted → **revive**	Revive as the Verify queue: native speakers swipe through AI-drafted lesson items (correct / fix / reject), add audio, and earn reputation.	Village → Contribute → Verify (verified natives).
136	Native Content Review	Village	Works	Becomes the reviewer dashboard above the Verify queue: stats, disagreements and escalations to linguists.	Village → Contribute → Review.
137	Profile	Me	Partial (P0/dupes)	My LingAfriq identity: avatar (§4), languages and levels, heritage story (optional), Passport stamps, contributions and circles.	Me (avatar, top-left).
138	Global Handle (@id)	Me	Unrouted → **revive**	Revive in onboarding (after account creation) and profile: choose an @handle, with suggestions in the target language (e.g.	Me → Profile → Handle; onboarding.
139	Progress Dashboard	Me	Works	Can-do progress: CEFR can-do statements ticked with evidence, skill radar (listen, speak, read, write, tone), words known, speaking minutes and foreca...	Me → Progress.
140	Achievements & Badges	Me	Works	Cultural badges earned by real milestones: first proverb used correctly, 100 tones nailed, first native conversation.	Me → Achievements.

#	Feature	Hub	State	Elevation in one line	Lives in
141	Leaderboards	Me	Works	Leagues of 30 similar learners with weekly promotion, an opt-out, and circle and country boards; points only from verified learning.	Me → Leagues; Village → Circles.
142	Magic Items	Me	Works	Cosmetic and helper shop: avatar outfits from regional designers, streak freezes and hint tokens; server-validated currency; nothing pay-to-win.	Me → Wardrobe & Shop; Village → Market.
143	Settings, Privacy & Appearance	Me	Partial (P0/dupes)	Grouped settings: Learning (daily goal, reminders, voice), Privacy & Safety (who can message me, discoverability, data export/delete per NDPA), Appear...	Me → Settings.
144	Subscription & Plans	Me	Unrouted → **revive**	Revive as LingAfriq Plus: free core learning forever; Plus adds unlimited Polie voice, offline packs, mentor credits and family sharing.	Me → Plus.
145	Family / Guardian Dashboard	Me	Unrouted → **revive**	Revive as the Guardian hub: link child profiles, see progress, set safe-social rules (circles only, no DMs), screen time and a weekly report.	Me → Family (guardians).
146	About & Policy	Me	Unrouted → **revive**	Revive About as Our Story: the mission, the native-speaker community, language-credit roll (contributors) and AI transparency (which models, how data ...	Me → About.
147	Sign in, Sign up & Email Verification	Me	Partial (P0/dupes)	Try-before-signup (first lesson without an account), then passkey, Google/Apple, phone OTP (key for Nigeria/Kenya) and email; generic error messages; ...	Before the app; Me → Account.
148	Onboarding & Placement	Me	Partial (P0/dupes)	See exemplar deep-dive	First run.
149	Splash	Me	Works	A 1-second greeting in the learner's target language, with the avatar waving (Rive), cached for offline.	Launch.
150	Features Guide, Navigation Hub & First-use Tips	Me	Unrouted → **revive**	Becomes the All features directory (A–Z, searchable, with locked items showing "unlocks at..." and a preview), which replaces the drawer so nothing is...	Me → All features; contextual first use.
151	Staff & Micro-mentor Reports	Me	Works	Keep in-app for staff on the go, behind server-checked roles, with a fuller version in the admin console (§14).	Me → Staff (role-gated).

18.2 Today hub (6 features)

Home / Today session

- **What it is.** The landing screen: continue-learning card, streak, stats and shortcuts.
- **Code.** `screens/tabs_view/home/home_tab.dart`, `screens/dashboard/dashboard_screen_material3.dart`, `screens/dashboard/modern_dashboard_screen.dart`. Scan: 3 file(s), 2,515 lines; 7 mock/fake marker(s); 40 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `modern_dashboard_screen.dart`.
- **Current state.** Works (routed as tab 1). `modern_dashboard_screen.dart` is an unrouted earlier version.
- **Defects.** Two dashboards compete. "Continue learning" falls back to a placeholder card when no lesson is resolved (`dashboard_screen_material3.dart:472,616`). Shortcuts are a grid of features rather than a plan for today.

- **Elevation.** Make it a **single adaptive "Today" session** (10–15 min), a playlist built by the learner model: 3 due reviews → 1 new input (story or dialogue) → 1 speaking moment (Polie or a persona line) → 1 culture card → 1 game round. One big "Start" button and a ring that fills. Underneath sit a "Because you..." shelf (contextual surfacing of other features) and the streak/quest strip. Port the best widgets from `modern_dashboard_screen.dart` into the new layout (consolidation; the feature survives).
- **UX and render.** A hero card with an avatar greeting in the target language (Tone Lens on the greeting). Skeletons matching the layout while loading. Empty state on day 1: "Your first session takes 6 minutes." Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Today tab (root).

Daily Goals

- **What it is.** Set a daily XP or minutes target and track it.
- **Code.** `screens/goals/daily_goals_screen.dart`. Scan: 1 file(s), 486 lines; 1 direct Groq-client call site(s) (P0-3 dependency); 12 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `daily_goals`). Calls Groq from the client (see evidence).
- **Defects.** Goals are XP-based, and XP can be self-minted (P0-8). The AI goal suggestion depends on the client Groq key (P0-3).
- **Elevation.** Goals become **intent-based** ("speak for 5 minutes", "learn 10 words", "one conversation") rather than XP. The weekly goal flexes with real life (travel mode, exam mode). The AI suggestion goes through the gateway and explains itself ("You are strong at listening; let's push speaking").
- **UX and render.** Goal ring on Today. Edit sheet with three presets and a custom option; no guilt copy. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Today → goal ring; Me → Settings → Goals.

Daily Challenges *(revived)*

- **What it is.** Three rotating micro-challenges a day (for example "use 5 new words in chat").
- **Code.** `screens/goals/daily_challenges_screen.dart`, `widgets/gamification/daily_challenges_widget.dart`. Scan: 2 file(s), 910 lines; 7 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `daily_challenges_screen.dart`.
- **Current state. Unrouted screen** (REVIVE). The widget and provider are live (`providers/daily_challenges_provider.dart`), but the full screen is not reachable.
- **Defects.** There is no detail view, so challenges can only be seen, not explored. Challenge completion must be verified server-side (P0-7/8).
- **Elevation.** Revive as the **Challenge board** behind the Today widget. Challenges are generated from the learner's weak skills and cross-link features (one challenge says "Point & Say three things in your kitchen"). Each challenge deep-links into the feature that completes it, which is the main contextual-discovery engine for lesser-known features.
- **UX and render.** Three cards with progress chips and a "Do it" deep-link. Completion uses a modest animation (rare tier). Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Today → Challenges strip → board.

Quests

- **What it is.** Multi-step quests with rewards.
- **Code.** `screens/gamification/quest_screen.dart`. Scan: 1 file(s), 677 lines; 2 direct Groq-client call site(s) (P0-3 dependency); 18 `Text(` widgets vs 0 l10n lookups.

- **Current state.** Works (route `quest`). Client Groq calls detected.
- **Defects.** Quest generation relies on client AI (P0-3). Rewards can be minted client-side (P0-7).
- **Elevation.** Story quests set in the **village narrative** ("Help Mama Ngozi prepare for the naming ceremony": learn kinship words, greet the elder, haggle for yams in Market Haggle). Chapters unlock weekly and tie together Learn, Speak and Play.
- **UX and render.** Quest map with illustrated stops; each stop is a deep-link. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Today → Quest card; Play → Quests.

Seasonal Events / Village Festival

- **What it is.** Time-limited cultural events.
- **Code.** `screens/gamification/seasonal_events_screen.dart`. Scan: 1 file(s), 543 lines; 2 mock/fake marker(s); 10 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `events`) and falls back to hard-coded events when the API fails (`seasonal_events_screen.dart:49-54`).
- **Defects.** The hard-coded fallback can show stale or wrong-date events. Event content isn't reviewed for cultural accuracy.
- **Elevation.** A **cultural calendar**: Ọdún Ọ̀ṣun, Eid, Timkat, Durbar, Homowo, Umhlanga, Kwanzaa and Heritage Day. Each event bundles a story, 20 words, a persona appearance, a themed game rotation and a circle challenge. Events are localised by region, and there's an honest "no events right now" state instead of fake events.
- **UX and render.** Event banner on Today; event page with a countdown, a culture explainer (reviewed) and activities. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Today → Event banner; Village → Events.

Streaks & reminders

- **What it is.** The daily streak counter and push reminders.
- **Code.** `screens/snap/snap_streaks_screen.dart`. Scan: 1 file(s), 103 lines; 4 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works but harmful: reminders every 2 h with no dedupe (P0-23). Snap streak counts are client-supplied (`socket.service.ts:888-905`).
- **Defects.** Loss-aversion copy at night, duplicate pushes, and streaks that can be faked.
- **Elevation.** Ethical streaks (§12): a weekly-streak option, earned freezes and repair windows. **One** reminder in the learner's chosen window, written by their favourite character in the target language ("Ẹ kú alẹ́! Two minutes with me?"). Celebrate milestones.
- **UX and render.** Streak flame becomes a calabash that fills; tap for history; no red alarms. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Today header; Me → Progress.

18.3 Learn hub (28 features)

Lessons map / Courses

- **What it is.** The course map of units and lessons per language.
- **Code.** `screens/lessons/lessons_map_entry_screen.dart`, `screens/tabs_view/courses/courses_tab.dart`, `lessons/screens/lessons_list_screen.dart`, `screens/curriculum/curriculum_screen_material3.dart`, `screens/curriculum/curriculum_screen.dart`, `screens/tutor/curriculum_viewer_screen.dart`. Scan: 6 file(s), 1,998 lines; 5 mock/fake marker(s); 29

`Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `curriculum_screen.dart`, `curriculum_viewer_screen.dart`.

- **Current state.** Works (routes `curriculum`, Courses tab). `curriculum_screen.dart` and `tutor/curriculum_viewer_screen.dart` are unrouted variants (CONSOLIDATE).
- **Defects.** The map draws from placeholder or template curriculum (P0-16). Four entry points to the same thing. No can-do framing.
- **Elevation.** One **Path** map per language: CEFR bands as regions of a village-to-city journey, units as stops, each showing its can-do ("I can greet elders correctly"). Downloadable per unit. Checkpoints sit at region borders. Keep the useful views of the variants as modes of the same map (list view, curriculum outline for teachers).
- **UX and render.** Vertical journey with region art; a progress ring per unit; the current unit pinned; offline badge. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Learn tab (root).

Authentic Path (A1–C1)

- **What it is.** A curated "authentic" A1–C2 track.
- **Code.** `screens/curriculum/authentic_curriculum_entry_screen.dart`, `screens/lesson/authentic_lesson_flow_screen.dart`. Scan: 2 file(s), 518 lines; 1 direct Groq-client call site(s) (P0-3 dependency); 8 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `authentic-path`). Content is template-repetitive (126 lessons/language, 85–111 unique words, grammar lines repeated 1,546×; §6).
- **Defects.** Content quality (P0-16). The name implies the main course is not authentic.
- **Elevation.** Becomes the **"Real-world track"**: authentic materials (radio clips, songs, market recordings, Nollywood/Bongo scenes, licensed) with scaffolding, unlocked from A2. It sits alongside the core Path rather than competing with it.
- **UX and render.** Media-first lesson cards with a transcript toggle, Tone Lens and tap-to-gloss. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Learn → Path → "Real-world" lane (A2+).

Lesson Flow

- **What it is.** The 10-stage lesson player.
- **Code.** `screens/lesson/lesson_flow_screen.dart`, `screens/lesson/lesson_flow_stages.dart`. Scan: 2 file(s), 661 lines; 1 `Random` use(s); 3 silently swallowed catch(es); 6 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `lesson-flow`). No audio in content (§6).
- **Defects.** Template content, no audio, and stages that don't adapt.
- **Elevation.** An adaptive lesson player: input → noticing (Tone Lens) → guided practice → speaking → mini-check, with stages skipped when mastered. Every item has native audio and a "Why?" sheet.
- **UX and render.** Full-screen, one task per screen, haptic feedback within 160 ms; progress bar segmented by stage. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Opened from Path and Today.

Lesson detail & Level certificate

- **What it is.** Lesson overview and a certificate on finishing a level.
- **Code.** `screens/curriculum/lesson_detail_screen.dart`, `screens/curriculum/level_certificate_screen.dart`. Scan: 2 file(s), 1,009 lines; 21 `Text(` widgets vs 0 l10n lookups.

- **Current state.** Works (pushed from curriculum).
- **Defects.** Certificates aren't evidence-based or verifiable.
- **Elevation.** Lesson detail shows can-dos, words and the culture note, with a preview audio. Certificates become **verifiable can-do badges** (Open Badges 3.0) that are shareable and link to the Passport.
- **UX and render.** Certificate rendered with the learner's avatar and a QR verification link. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Learn → Path → lesson; Me → Credentials.

Learning Path (personal plan)

- **What it is.** A personalised learning-path screen.
- **Code.** `screens/learning/learning_path_screen.dart`. Scan: 1 file(s), 514 lines; 10 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `learning-path`).
- **Defects.** Duplicates the course map conceptually.
- **Elevation.** Make it the **"My plan"** view of Path: the learner's goal (heritage, travel, exam) mapped to units and weekly targets, with an ETA to the next can-do.
- **UX and render.** Timeline with milestones; editable goal. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Learn → Path → "My plan".

Placement test

- **What it is.** Placement during onboarding.
- **Code.** `screens/onboarding/placement_test_screen.dart`, `screens/onboarding/placement_quiz_screen.dart`. Scan: 2 file(s), 1,021 lines; 18 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `placement_quiz_screen.dart`.
- **Current state.** Works (`placement_test_screen.dart`); `placement_quiz_screen.dart` is an unrouted variant (CONSOLIDATE).
- **Defects.** Not adaptive; no speaking.
- **Elevation.** An adaptive 3-minute **CAT placement** with listening, reading and one speaking prompt. Results place the learner on the Path and are retakeable.
- **UX and render.** Calm, one item per screen, with a clear "skip, I'm a beginner" option. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Onboarding; Learn → Path → "Retake placement".

Grammar Hub

- **What it is.** Grammar topics, lessons and exercises.
- **Code.** `screens/grammar/grammar_hub_screen.dart`, `screens/grammar/grammar_lesson_screen.dart`, `screens/grammar/grammar_exercise_screen.dart`. Scan: 3 file(s), 2,067 lines; 37 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (routes `grammar-hub`, `grammar-lesson`, `grammar-exercise`).
- **Defects.** Grammar content is the repeated template lines (§6).
- **Elevation.** A **Pattern Library**: each pattern card has a plain-language explanation in the learner's language, 5 audio examples, contrasts and common errors. Interactive pattern builders (Word Forge for Bantu noun classes, serial verbs for Yoruba). Linked from every lesson and from Polie's explanations.
- **UX and render.** Cards with colour-coded morphology (not colour-only); swipe through examples. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.

- **Lives in.** Learn → Patterns.

Vocabulary Builder

- **What it is.** Themed vocabulary sets with images.
- **Code.** `screens/content/vocabulary_builder_screen.dart`. Scan: 1 file(s), 1,209 lines; 3 mock/fake marker(s); 7 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `vocabulary-builder`).
- **Defects.** Images are stock or remote with spinners; words have no audio or tone data.
- **Elevation.** Themed **word packs** (market, family, body, church/mosque, football) with native audio, Tone Lens, pictures reviewed for cultural accuracy, and "add all to review". Each word opens the Living Dictionary card.
- **UX and render.** Grid of picture cards; tap to hear; long-press for the Word Card. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Learn → Words → Packs.

My Words & Review (SRS) *(revived)*

- **What it is.** Saved words and flashcard review.
- **Code.** `screens/vocabulary/vocabulary_screen.dart`, `screens/vocabulary/flashcard_review_screen.dart`, `screens/vocabulary/vocabulary_flashcard_screen.dart`, `screens/ai_chat/review_dashboard_screen.dart`, `screens/ai_chat/vocabulary_dashboard_screen.dart`. Scan: 5 file(s), 2,781 lines; 57 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `vocabulary_flashcard_screen.dart`, `review_dashboard_screen.dart`, `vocabulary_dashboard_screen.dart`.
- **Current state.** Works (`my-vocabulary`, `flashcard-review`). `vocabulary_flashcard_screen.dart` is an unrouted variant (CONSOLIDATE). **`review_dashboard_screen.dart` and `vocabulary_dashboard_screen.dart` are unrouted analytics screens (REVIVE as tabs).**
- **Defects.** Four separate SRS implementations (§6.3). No FSRS. Saved words in chat, personas and games don't flow here.
- **Elevation.** One **FSRS-6 review** fed by every feature (save a word anywhere and it appears). The revived dashboards become the "Insights" tab: forecast of due cards, retention curve, hardest words, and tone-error patterns.
- **UX and render.** Review card with audio first, then meaning, then a speak-it button; swipe grading; dashboard charts generated from real data. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Learn → Words → Review / Insights; Today pulls due items.

Review Rush (gamified review) *(revived)*

- **What it is.** Timed, gamified review of saved words.
- **Code.** `screens/review/gamified_review_screen.dart`. Scan: 1 file(s), 367 lines; 10 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `gamified_review_screen.dart`.
- **Current state. Unrouted** (REVIVE). Uses `apiProvider`, `gamificationProvider` and `userProvider`, so it is wired to real data.
- **Defects.** Not reachable from any menu.
- **Elevation.** Revive as **Review Rush**: a 90-second sprint over due cards with combo multipliers that only reward accuracy. Reachable from Review and from the Play shelf. Results update FSRS like normal review.
- **UX and render.** Arcade skin over the same card component; reduced-motion version. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.

- **Lives in.** Learn → Words → Review → "Rush"; Play → Quick games.

Listening Practice

- **What it is.** Listening exercises and a listening quiz.
- **Code.** `screens/content/listening_practice_screen.dart`, `screens/tutor/listening_quiz_screen.dart`. Scan: 2 file(s), 1,241 lines; 16 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `listening_quiz_screen.dart`.
- **Current state.** Works (route `listening-practice`). `tutor/listening_quiz_screen.dart` is unrouted (CONSOLIDATE as a mode).
- **Defects.** Content lacks real audio (§6).
- **Elevation. Ear training**: graded listening at native speed with accents (Lagos, Ibadan, Kano, Nairobi, Durban), a slow-down slider, dictation with diacritic-aware grading, and the quiz mode merged in. Danfo/Matatu Rush (new game) reuses the same audio.
- **UX and render.** Waveform with tap-to-replay segments; transcript reveal after answering. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Learn → Skills → Listening.

Writing Practice

- **What it is.** Writing exercises with AI feedback.
- **Code.** `screens/content/writing_practice_screen.dart`. Scan: 1 file(s), 920 lines; 2 direct Groq-client call site(s) (P0-3 dependency); 11 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `writing-practice`); AI feedback calls Groq from the client.
- **Defects.** Client key dependency (P0-3). A regex diacritic "enforcer" can introduce errors (§7.1).
- **Elevation. Write & check**: a diacritic keyboard bar, tone-aware grading that is lenient early and strict later, and AI feedback through the gateway that shows uncertainty ("tone unclear here, listen"). Handwriting mode for Ge'ez, N'Ko and Adlam scripts.
- **UX and render.** Inline corrections with an explanation popover; never auto-replace text. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Learn → Skills → Writing.

Phrase DNA

- **What it is.** Phrase templates that learners recombine (sentence frames).
- **Code.** `screens/learning/phrase_dna/phrase_dna_templates_screen.dart`, `screens/learning/phrase_dna/phrase_dna_session_screen.dart`. Scan: 2 file(s), 460 lines; 6 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (routes `phrase-dna-templates` plus the session).
- **Defects.** Templates come from the same thin content.
- **Elevation.** Turn it into **sentence frames you can speak**: "I want to buy ___" with slot fillers from the learner's own words. The learner sees how the tone changes in context (tone sandhi where relevant) and says the full sentence with scoring. A strong bridge from words to speaking.
- **UX and render.** Slot chips that snap into place; Tone Lens updates live. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Learn → Skills → Phrase DNA; Speak → Warm-ups.

Tone & Rhythm Trainer

- **What it is.** Tone and rhythm practice.

- **Code.** `screens/learning/tone_rhythm/tone_rhythm_trainer_screen.dart`, `screens/practice/tone_drill_screen.dart`. Scan: 2 file(s), 1,009 lines; 4 silently swallowed catch(es); 20 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `tone_drill_screen.dart`.
- **Current state.** Works (route `tone-rhythm-trainer`). `practice/tone_drill_screen.dart` is unrouted (CONSOLIDATE as "Drills").
- **Defects.** Tone models are simplified (one H/M/L scheme for all languages; §10.1).
- **Elevation.** The **Tone Gym**, LingAfriq's signature: perception drills (minimal pairs), production with a live pitch contour against the native one, per-language tone inventories (Yoruba H/M/L + glides, Igbo downstep, Hausa falling) and a progression ladder. Fed by the real scorer (§7.4).
- **UX and render.** Large pitch canvas; the native contour in a calm colour, the learner's overlaid; shape plus label, never colour-only. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Speak → Tone Gym (also Learn → Skills).

Tonal Lesson (Village)

- **What it is.** A tone-focused lesson inside the Village.
- **Code.** `screens/village/tonal_lesson_screen.dart`. Scan: 1 file(s), 324 lines; 2 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `tonal-lesson`).
- **Defects.** Sits in the Village with no link to the Tone Gym or Path.
- **Elevation.** Becomes the **story-led tone lesson** (an elder teaches greetings with tone), linked to the Tone Gym for practice and placed on the Path where tone is introduced.
- **UX and render.** Character-led scene with the Tone Lens on every line. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Learn → Path (tone units); Speak → Tone Gym → Lessons.

Language Quiz

- **What it is.** Section quizzes per language.
- **Code.** `language_quiz/screens/language_quiz_sections_screen.dart`, `screens/tabs_view/home/take_quiz_screen.dart`. Scan: 2 file(s), 740 lines; 3 `Random` use(s); 1 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (pushed from home). Placeholder MCQs exist in bundles ("Sample MCQ"; §11).
- **Defects.** Placeholder items; no calibration.
- **Elevation.** Merge into the **item bank**: unit checkpoints and adaptive quizzes with rationales and audio (§11.3), while keeping the familiar "take a quiz" entry.
- **UX and render.** Clean MCQ/listening item templates; result screen with a can-do update. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Learn → Path → Checkpoints; Today → "Quick quiz".

History & History Quiz

- **What it is.** History reading list and history quizzes.
- **Code.** `history/screens/history_list_screen.dart`, `history_quiz/screens/history_quiz_sections_screen.dart`. Scan: 2 file(s), 527 lines; 1 `Random` use(s).
- **Current state.** Works (pushed from home/drawer).
- **Defects.** Facts aren't sourced or reviewed (see persona data errors, §8).
- **Elevation. Living History**: short illustrated stories with sources, linked to Historical Personas ("Meet her" button) and quizzes that cite the story. Reviewed by historians.

- **UX and render.** Story cards with a "Sources" footer; persona portrait chip. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Learn → Culture → History; Speak → Personas cross-link.

Mannerisms & etiquette

- **What it is.** Cultural mannerisms list.
- **Code.** `mannerisms/screens/mannerism_list_screen.dart`. Scan: 1 file(s), 185 lines.
- **Current state.** Works (pushed from home).
- **Defects.** Text list only; no practice.
- **Elevation. Etiquette cards with practice**: kneeling and prostration greetings, handshakes, eating etiquette, taboo topics. Each card has a short video (commissioned), "why", and a Greeting Gauntlet round; the avatar demonstrates.
- **UX and render.** Card stack with gesture illustrations; regional variants flagged. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Learn → Culture → Etiquette.

Cultural Hub

- **What it is.** Hub for cultural content.
- **Code.** `screens/content/cultural_hub_screen.dart`. Scan: 1 file(s), 898 lines; 16 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `cultural-hub`).
- **Defects.** One of several culture entry points (magazine, heritage, history, mannerisms).
- **Elevation.** Becomes the **Culture** landing inside Learn, organising Magazine, History, Etiquette, Heritage archive, Proverbs and Music in one place.
- **UX and render.** Editorial layout with one featured story and shelves. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Learn → Culture (root).

Culture Magazine

- **What it is.** Editorial cultural articles.
- **Code.** `screens/magazine/culture_magazine_screen_enhanced.dart`, `screens/magazine/culture_magazine_screen.dart`, `screens/magazine/culture_magazine_enhanced_features.dart`. Scan: 3 file(s), 2,256 lines; 15 direct Groq-client call site(s) (P0-3 dependency); 2 mock/fake marker(s); 2 silently swallowed catch(es); 22 `Text(` widgets vs 2 l10n lookups; not reachable from `main.dart`: `culture_magazine_screen.dart`.
- **Current state.** Works (route `magazine` → enhanced). `culture_magazine_screen.dart` is an unrouted earlier version (CONSOLIDATE). AI features call Groq from the client (12 call sites in `culture_magazine_enhanced_features.dart`).
- **Defects.** Client key dependency (P0-3). Backend `cultureArticleGenerator.service.ts` uses gpt-4o to generate articles, which is unreviewed AI content.
- **Elevation.** A **bilingual magazine**: articles at the reader's level with tap-to-gloss, audio narration by native voices, "words from this article" to save, and author credits for African writers (commissioned, paid). AI-assisted drafts are labelled and reviewed.
- **UX and render.** Typography-led layout with Andika/Noto; reading progress; offline save. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Learn → Culture → Magazine; Today → culture card.

Heritage Milestones

- **What it is.** Heritage-learner milestone tracker.
- **Code.** `screens/learning/heritage_milestones_screen.dart`. Scan: 1 file(s), 137 lines; 4 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `heritage-milestones`).
- **Defects.** Milestones are generic.
- **Elevation. Family milestones** for heritage learners: "Greeted grandma in Yoruba", "Understood a family WhatsApp voice note", "Said a prayer at a naming ceremony". Self-reported with an optional audio memory saved to a private family album.
- **UX and render.** Timeline with photos and voice memories; private by default. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me → Heritage; Today celebrates milestones.

FLB Heritage Archive

- **What it is.** An archive of heritage items (folklore, legends, background) with detail pages.
- **Code.** `screens/heritage/flb_heritage_archive_screen.dart`, `screens/heritage/flb_heritage_detail_screen.dart`. Scan: 2 file(s), 468 lines; 2 mock/fake marker(s).
- **Current state.** Works (routes `flb-heritage-archive`, `flb-heritage-detail`).
- **Defects.** Provenance and licensing of items are unclear.
- **Elevation.** The **Living Archive** (Wave 4 languages): folktales, praise poetry (oríkì), songs and proverbs recorded with community consent, each with transcript, translation, Tone Lens, contributor credit and licence. It is also how endangered-language starter kits are presented.
- **UX and render.** Museum-like detail pages with audio first, then text. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Learn → Culture → Archive.

Dialect Preference

- **What it is.** Choose the dialect variant.
- **Code.** `screens/learning/dialect_preference_screen.dart`. Scan: 1 file(s), 30 lines; 1 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `dialect-preference`).
- **Defects.** Dialect choice doesn't change content (no dialect-tagged content yet).
- **Elevation.** Real **dialect lanes**: audio variants per item (Ọ̀yọ́ vs Ìjẹ̀bú Yoruba; Owerri vs Onitsha Igbo; Kenyan vs Tanzanian Swahili), with notes where words differ and a "hear other dialects" toggle on every Word Card.
- **UX and render.** Map-based picker with sample audio per dialect. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me → Settings → Language & dialect; Word Card toggle.

Living Dictionary

- **What it is.** A searchable dictionary that learners can add to.
- **Code.** `screens/vocabulary/living_dictionary_screen.dart`, `services/learning/living_dictionary_service.dart`. Scan: 2 file(s), 750 lines; 21 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `living-dictionary`). See the deep-dive in §17.7.
- **Defects.** Per-user word list only (the backend filters by `userId`); no shared verified lexicon.
- **Elevation.** See §17.7 (exemplar deep-dive).
- **UX and render.** See §17.7. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Global: search bar + long-press on any word; Learn → Words.

Offline Downloads *(revived)*

- **What it is.** Manage downloaded lessons and content for offline use.
- **Code.** `screens/offline/offline_content_screen.dart`. Scan: 1 file(s), 375 lines; 17 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `offline_content_screen.dart`.
- **Current state. Unrouted** (REVIVE). Wired to `offlineContentProvider`, `offlineDownloadProvider` and `subscriptionProvider`.
- **Defects.** Offline is a key need in the target markets, but this screen is unreachable.
- **Elevation.** Revive as **Downloads**: per-unit packs (text + Opus audio), storage meter, Wi-Fi-only toggle, auto-download of the next unit, and an offline Tone Gym once the on-device model ships. Ties into the single outbox (§13.5).
- **UX and render.** List with size per pack, progress and a clear sync status. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me → Downloads; Path shows a download icon per unit.

Language detail & Suggest a language

- **What it is.** Per-language info page and a form to request new languages.
- **Code.** `screens/tabs_view/home/language_detail_screen.dart`, `screens/tabs_view/profile/suggest_language_screen.dart`. Scan: 2 file(s), 508 lines; 3 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (pushed from home and profile).
- **Defects.** Suggestions go nowhere visible.
- **Elevation.** Language pages show speakers, regions, readiness level (L0–L3) and a sample phrase. **Suggest a language** becomes the public waitlist for Waves 2–4, with a "sign up as a native reviewer" path. It directly feeds the language-selection score (§6.4).
- **UX and render.** Language page with a map, audio sample and readiness badge; waitlist counter. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Learn → language switcher → "All languages".

Passport (proctored test & credential)

- **What it is.** A proctored proficiency test and credential.
- **Code.** `screens/passport/passport_proctored_session_screen.dart`, `screens/passport/passport_credential_screen.dart`. Scan: 2 file(s), 354 lines; 11 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works as UI (routes `passport-proctored`, `passport-credential`).
- **Defects.** Scoring and content aren't valid yet (§12.4); proctoring privacy terms are unclear.
- **Elevation.** The **LingAfriq Passport**: evidence portfolio plus an adaptive proctored test, mapped to CEFR descriptors with a published methodology, issued as a verifiable credential. Launch only after real scoring.
- **UX and render.** Formal, calm design; clear privacy explanation before the camera turns on. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me → Credentials (unlocks at A2).

Lesson Recap *(revived)*

- **What it is.** End-of-lesson recap screen (328 lines).
- **Code.** `screens/learning/lesson_recap_screen.dart`. Scan: 1 file(s), 328 lines; 6 `Random` use(s); 6 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `lesson_recap_screen.dart`.

- **Current state. Unrouted** (REVIVE).
- **Defects.** Lessons end without consolidation, so there is no retrieval moment.
- **Elevation.** Revive as the **60-second recap** after every lesson: three retrieval questions, the new words with audio and Tone Lens, one sentence to say aloud (scored), and "what's next". New words enter FSRS review automatically.
- **UX and render.** Card stack with large type; one celebratory moment, not confetti spam. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** End of every lesson (Learn → lesson → recap).

18.4 Speak hub (20 features)

Polie AI Tutor workspace

- **What it is.** Polie, the AI tutor: free chat plus mode selection (conversation, explain, correct and others).
- **Code.** `screens/ai_chat/polie_workspace_screen.dart`, `screens/ai_chat/ai_chat_select_screen.dart`, `screens/ai_chat/ai_mode_selection_screen.dart`, `screens/ai_chat/polie_mode_selection_screen.dart`, `screens/ai_chat/ai_chat_language_setup_screen.dart`, `screens/ai_chat/ai_chat_screen.dart` (+4 more). Scan: 10 file(s), 10,869 lines; 37 direct Groq-client call site(s) (P0-3 dependency); 2 `Random` use(s); 4 mock/fake marker(s); 9 silently swallowed catch(es); 93 `Text(` widgets vs 7 l10n lookups; not reachable from `main.dart`: `ai_chat_screen.dart`, `ai_chat_screen_new.dart`, `ai_chat_screen_with_tracking.dart`, `conversation_enhanced_screen.dart`, `ai_language_selection_screen.dart`.
- **Current state.** Works (AI tab; routes `ai_chat_select`, `polie_mode_selection`). Client-side Groq calls throughout (11 in the workspace alone). Five unrouted earlier versions (CONSOLIDATE; harvest the tracking and "enhanced conversation" features).
- **Defects.** 6,485-line god screen. Client keys (P0-3). No streaming on the backend fallback (`ai_chat_provider_groq.dart:1302`). Silent fallbacks.
- **Elevation.** Polie becomes a **voice-first companion**: hold-to-talk with live transcript and Tone Lens on Polie's replies, recasts ("You said X; natives say Y"), a hint ladder, and "save this phrase". Modes are chips at the top: Chat, Explain, Correct me, Roleplay, Story and Translate. Polie knows the learner model, so it uses due words naturally. All through the AI gateway with streaming. Avatar lip-sync (§4).
- **UX and render.** Chat surface with large target-language text (18 sp+), a gloss on tap, and the avatar bust in the header reacting (listening/thinking/speaking). Honest "AI unavailable" state. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Speak tab (root) → Polie.

Tutor: Dialogue mode

- **What it is.** Guided dialogues with the tutor.
- **Code.** `screens/tutor/tutor_dialogue_mode_screen.dart`. Scan: 1 file(s), 799 lines; 11 direct Groq-client call site(s) (P0-3 dependency); 5 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works as UI (pushed from the tutor dashboard); the AI calls Groq directly from the client (see evidence).
- **Defects.** Client key dependency (P0-3): in release builds these modes fail and fall back silently. No grounding in approved content.
- **Elevation.** Scripted-goal dialogues ("order food", "ask directions") where Polie improvises inside human-authored scenarios and scores goal completion, politeness and accuracy. Learners can replay with a different character (elder, child, trader).
- **UX and render.** Shared tutor shell: header with avatar and mode chip, transcript, mic button, and a feedback drawer. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48

dp targets, dark mode and l10n.
- **Lives in.** Speak → Polie → mode chips.

Tutor: Grammar mode

- **What it is.** Ask grammar questions, get explanations and drills.
- **Code.** `screens/tutor/tutor_grammar_mode_screen.dart`. Scan: 1 file(s), 795 lines; 12 direct Groq-client call site(s) (P0-3 dependency); 1 mock/fake marker(s); 17 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works as UI (pushed from the tutor dashboard); the AI calls Groq directly from the client (see evidence).
- **Defects.** Client key dependency (P0-3): in release builds these modes fail and fall back silently. No grounding in approved content.
- **Elevation.** Answers come **only from the Pattern Library** (retrieval), cite the pattern card, and spawn a 3-item micro-drill. Explanations in the learner's L1 (English, French, Portuguese, Arabic).
- **UX and render.** Shared tutor shell: header with avatar and mode chip, transcript, mic button, and a feedback drawer. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Speak → Polie → mode chips.

Tutor: Story mode

- **What it is.** AI-generated graded stories.
- **Code.** `screens/tutor/tutor_story_mode_screen.dart`. Scan: 1 file(s), 1,008 lines; 11 direct Groq-client call site(s) (P0-3 dependency); 3 silently swallowed catch(es); 11 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works as UI (pushed from the tutor dashboard); the AI calls Groq directly from the client (see evidence).
- **Defects.** Client key dependency (P0-3): in release builds these modes fail and fall back silently. No grounding in approved content.
- **Elevation. Choose-your-path stories** at the learner's level using known plus due words, narrated by native TTS, with comprehension checks. Labelled AI-generated; the best stories go to human review and then the shared library.
- **UX and render.** Shared tutor shell: header with avatar and mode chip, transcript, mic button, and a feedback drawer. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Speak → Polie → mode chips.

Tutor: Translation mode

- **What it is.** Translate with explanation.
- **Code.** `screens/tutor/tutor_translation_mode_screen.dart`. Scan: 1 file(s), 730 lines; 14 direct Groq-client call site(s) (P0-3 dependency); 14 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works as UI (pushed from the tutor dashboard); the AI calls Groq directly from the client (see evidence).
- **Defects.** Client key dependency (P0-3): in release builds these modes fail and fall back silently. No grounding in approved content.
- **Elevation. Translate & teach**: shows the translation, word-by-word alignment, register options (formal/informal/elder) and why. Uses the commercial MT stack (AI stack §3.7); confidence badge per language pair. Merges the unrouted `enhanced_translation_screen.dart` features (CONSOLIDATE).
- **UX and render.** Shared tutor shell: header with avatar and mode chip, transcript, mic button, and a feedback drawer. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Speak → Polie → mode chips.

Tutor: Pronunciation mode

- **What it is.** Record and get pronunciation feedback.
- **Code.** `screens/tutor/tutor_pronunciation_mode_screen.dart`. Scan: 1 file(s), 701 lines; 6 direct Groq-client call site(s) (P0-3 dependency); 4 mock/fake marker(s); 1 silently swallowed catch(es); 16 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works as UI (pushed from the tutor dashboard); the AI calls Groq directly from the client (see evidence).
- **Defects.** Client key dependency (P0-3): in release builds these modes fail and fall back silently. No grounding in approved content.
- **Elevation.** Real scoring (§7.4): per-syllable tone and phone feedback with the native contour overlay. Replaces the fake rhythm score (`score*0.9`, P0-18). Merges unrouted `practice/pronunciation_practice_screen.dart` (CONSOLIDATE).
- **UX and render.** Shared tutor shell: header with avatar and mode chip, transcript, mic button, and a feedback drawer. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Speak → Polie → mode chips.

Tutor: Assess mode

- **What it is.** AI-run assessment conversation.
- **Code.** `screens/tutor/tutor_assess_mode_screen.dart`. Scan: 1 file(s), 586 lines; 11 direct Groq-client call site(s) (P0-3 dependency); 1 silently swallowed catch(es); 9 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works as UI (pushed from the tutor dashboard); the AI calls Groq directly from the client (see evidence).
- **Defects.** Client key dependency (P0-3): in release builds these modes fail and fall back silently. No grounding in approved content.
- **Elevation.** An **oral proficiency interview lite**: a 5-minute adaptive conversation rated against can-do descriptors, with native-rater calibration. Feeds the Passport evidence portfolio.
- **UX and render.** Shared tutor shell: header with avatar and mode chip, transcript, mic button, and a feedback drawer. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Speak → Polie → mode chips.

Tutor dashboard

- **What it is.** Entry to the tutor modes and tutor progress.
- **Code.** `screens/tutor/tutor_dashboard_screen.dart`, `screens/ai_chat/tutor_progress_dashboard_screen.dart`. Scan: 2 file(s), 860 lines; 15 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `tutor_progress_dashboard_screen.dart`.
- **Current state.** Works. `ai_chat/tutor_progress_dashboard_screen.dart` is an unrouted progress view (CONSOLIDATE as the dashboard's "Progress" tab).
- **Defects.** Modes are listed rather than recommended.
- **Elevation. Coach view**: Polie recommends the next mode based on weak skills; the progress tab shows speaking minutes, tone accuracy trend and conversation goals achieved.
- **UX and render.** Cards with "Recommended" ribbon; small charts. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Speak → Polie → "Coach".

Shadowing *(revived)*

- **What it is.** Listen-and-repeat shadowing exercise.

- **Code.** `screens/tutor/shadowing_exercise_screen.dart`. Scan: 1 file(s), 385 lines; 2 direct Groq-client call site(s) (P0-3 dependency); 8 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `shadowing_exercise_screen.dart`.
- **Current state. Unrouted** (REVIVE). Client Groq calls detected.
- **Defects.** Unreachable; client key dependency.
- **Elevation.** Revive as **Shadow Me**: a native clip plays and the learner speaks along a fraction of a second behind, with karaoke-style word highlighting from forced alignment and a tone contour comparison. Daily 2-minute shadowing appears in Today.
- **UX and render.** Lyrics-style scrolling transcript; dual waveform; "slow" toggle. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Speak → Practice → Shadowing; Today slot.

Conversation Scenarios & Roleplay *(revived)*

- **What it is.** Scenario-based roleplay (market, taxi, clinic) with summaries and progress.
- **Code.** `screens/content/conversation_scenarios_screen.dart`, `screens/ai_chat/roleplay_scenario_selection_screen.dart`, `screens/ai_chat/roleplay_completion_summary_screen.dart`, `screens/ai_chat/roleplay_progress_dashboard_screen.dart`. Scan: 4 file(s), 2,651 lines; 3 direct Groq-client call site(s) (P0-3 dependency); 1 mock/fake marker(s); 29 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `roleplay_scenario_selection_screen.dart`, `roleplay_completion_summary_screen.dart`, `roleplay_progress_dashboard_screen.dart`.
- **Current state.** Scenarios work (route `conversation-scenarios`). **Scenario selection (CONSOLIDATE), completion summary (REVIVE) and progress dashboard (REVIVE) are unrouted.**
- **Defects.** Without summary and progress, roleplay has no closure or progression. Client AI dependency.
- **Elevation. Scenario Theatre**: a library of real-life scenes with goals and a cast. The revived completion summary shows goals met, best phrases, corrected mistakes and words to save. The revived progress dashboard becomes a "scene passport" collecting stamps per scenario and level. Market Haggle and Greeting Gauntlet share the engine.
- **UX and render.** Scene cards with illustrated settings; summary screen with replay of key moments. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Speak → Scenarios.

Conversation Analytics *(revived)*

- **What it is.** Analytics of the learner's AI conversations.
- **Code.** `screens/ai_chat/conversation_analytics_screen.dart`. Scan: 1 file(s), 615 lines; 10 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `conversation_analytics_screen.dart`.
- **Current state. Unrouted** (REVIVE). Wired to `conversationAnalyticsServiceProvider`.
- **Defects.** Insight data is collected but never shown.
- **Elevation.** Revive as **"How I speak"**: words used vs avoided, average utterance length, code-switch rate, tone accuracy over time and the most frequent corrections, with one actionable suggestion per week.
- **UX and render.** Simple, honest charts with an explanation line under each. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Speak → Coach → Insights; Me → Progress.

Speak Engine Lab (code-switch & register coach)

- **What it is.** Experimental lab: code-switching sessions and a register (formality) coach.

- **Code.** `screens/learning/speak_engine_lab_screen.dart`. Scan: 1 file(s), 189 lines; 12 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `speak-engine-lab`), backed by `code_switch_session_service.dart` and `register_coach_service.dart`.
- **Defects.** "Lab" framing hides two genuinely novel features.
- **Elevation.** Split into two named features. **Code-Switch Coach**: practise natural Yoruba-English or Swahili-English mixing the way people really talk, scored for naturalness. **Respect Coach**: rewrite a sentence for an elder, peer or child, with honorifics (Yoruba ẹ/ọ, Igbo titles, Zulu hlonipha). Both are rare in the market.
- **UX and render.** Side-by-side "casual vs respectful" cards with audio. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Speak → Practice → Code-Switch Coach / Respect Coach.

Speak Missions

- **What it is.** Speaking missions evaluated by AI.
- **Code.** `screens/content/speak_mission_evaluate_screen.dart`. Scan: 1 file(s), 233 lines; 4 direct Groq-client call site(s) (P0-3 dependency); 2 silently swallowed catch(es); 5 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works as UI; evaluation calls Groq from the client and records audio.
- **Defects.** Client AI dependency; evaluation not grounded in a rubric.
- **Elevation. Weekly speaking missions** in the real world ("Greet three people in Yoruba today and record one"), with a rubric-based evaluation and an optional share to the circle. Ties to Daily Challenges.
- **UX and render.** Mission card with checklist; recording sheet; rubric feedback. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Speak → Missions; Today → Challenges.

Historical Personas

- **What it is.** Chat with historical African figures.
- **Code.** `screens/personalities/personality_selection_screen.dart`, `screens/personalities/personality_chat_screen.dart`. Scan: 2 file(s), 776 lines; 16 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works as UI (route `historical-personas`). Backend persona service flagged: told not to mention being AI (P0-15); confidence 0.9 with empty sources; data errors (Nkrumah listed as Twi).
- **Defects.** P0-13/14/15; ungrounded answers.
- **Elevation.** See §8: grounded, cited, AI-disclosed personas with illustrated story chapters (Yaa Asantewaa and the Golden Stool), voice in the target language, and historian review. Collections: Queens & Warriors, Builders of Nations, Writers & Thinkers, Scientists & Healers.
- **UX and render.** Portrait busts (Rive), a sources drawer on every answer, and an "AI recreation" label. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Speak → Personas.

Persona Missions

- **What it is.** Goal-based missions with personas.
- **Code.** `screens/persona_missions/persona_missions_hub_screen.dart`, `screens/persona_missions/persona_mission_setup_screen.dart`, `screens/persona_missions/persona_mission_session_screen.dart`. Scan: 3 file(s), 427 lines; 2 direct Groq-client call site(s) (P0-3 dependency); 10 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works as UI; session calls Groq from the client (silently fails in release, P0-13).

- **Defects.** P0-13 canned fallback "I hear you. Let us continue."
- **Elevation.** The **story-chapter format** from §8: era scene, objective (greet the Queen Mother correctly), branching dialogue, cultural debrief with sources, and a reward that ties to the persona's history.
- **UX and render.** Illustrated scene header, ambient audio, objective tracker. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Speak → Personas → Missions.

LingChat AI conversation

- **What it is.** An AI conversation partner inside the chat section.
- **Code.** `screens/chat/lingchat_ai_conversation_screen.dart`. Scan: 1 file(s), 45 lines; 1 direct Groq-client call site(s) (P0-3 dependency).
- **Current state.** Works (route `lingchat_ai`); a thin 45-line wrapper over Polie modes.
- **Defects.** Duplicates Polie in a different place.
- **Elevation.** Becomes **"Practice with Polie"** inside chat: open a practice DM with Polie or a persona from the chat list, so AI practice sits where conversations already happen. It reuses Polie (consolidated engine) but keeps this entry point.
- **UX and render.** Appears as a pinned contact in Chats with an "AI" label. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Chats → pinned "Polie (AI)".

Synthetic Voice Styles

- **What it is.** Choose the synthetic voice style used for TTS.
- **Code.** `screens/settings/synthetic_voice_styles_screen.dart`. Scan: 1 file(s), 196 lines; 4 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `synthetic-voice-styles`).
- **Defects.** Voices come from non-commercial models (MMS-TTS, XTTS-v2; P0-24).
- **Elevation. Voice picker** with commercial voices per language (Intron, Spitch, Azure, custom; AI stack §3.5): preview, gender, region, speed. The chosen voice also becomes Polie's voice. Clearly labelled synthetic.
- **UX and render.** Voice cards with a play sample and region tag. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me → Settings → Voice; Speak → Polie settings.

Point & Say

- **What it is.** Point the camera at an object or text, learn the word and say it.
- **Code.** `screens/ar/point_and_say_screen.dart`, `services/offline/point_and_say_queue_service.dart`. Scan: 2 file(s), 1,135 lines; 7 silently swallowed catch(es); 22 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `point-and-say`). See the deep-dive in §17.6.
- **Defects.** See §17.6.
- **Elevation.** See §17.6 (exemplar deep-dive).
- **UX and render.** See §17.6. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Global Lens button; Learn/Speak contextual.

Live Translate

- **What it is.** Real-time speech translation session (on-device STT streamed over Socket.IO).

- **Code.** `screens/live_translate/live_translate_screen.dart`, `services/live_translate/live_translate_client.dart`. Scan: 2 file(s), 632 lines; 2 mock/fake marker(s); 3 silently swallowed catch(es).
- **Current state.** Works as UI (drawer item). It uses on-device speech recognition streamed through `live_translate_event` (`live_translate_client.dart:6,35,75`).
- **Defects.** On-device recognisers rarely support Yoruba, Igbo, Hausa, Xhosa or Pidgin, so the source side fails for those languages. The socket inherits P0-4/5 weaknesses.
- **Elevation. Conversation Bridge**: two people, two languages, one phone. Each speaks in turn; server-side ASR (Intron) handles African languages and the commercial MT stack translates. Every turn shows Tone Lens and "learn this phrase". A learning feature, not just a utility: the transcript becomes a saved mini-lesson.
- **UX and render.** Split-screen facing layout (one half upside-down for the other person); big mic buttons; confidence indicator. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Global Lens → "Bridge"; Speak → Practice.

Import Media (OCR & clips)

- **What it is.** Import images, audio or video to extract text, transcribe and generate learning material.
- **Code.** `screens/media/import_media_screen.dart`, `screens/media/media_clip_player_screen.dart`. Scan: 2 file(s), 163 lines.
- **Current state.** Works (route `import_media`). ML Kit Latin OCR (`media_import_service.dart:93`); transcription and generation through the backend.
- **Defects.** ML Kit Latin OCR mangles stacked tone marks [verify on device]. Generated content isn't reviewed.
- **Elevation. Learn from anything**: import a song, a WhatsApp voice note, a sermon clip or a sign photo, and LingAfriq turns it into a personal lesson (transcript, glosses, Tone Lens, 10 flashcards, a shadowing clip). Server OCR with a diacritic-aware second pass (AI stack §3.13). Private by default; copyright notice for sharing.
- **UX and render.** Import sheet (camera/files/share-sheet target); processing steps shown honestly. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Global Lens → "Import"; share-sheet target from other apps.

18.5 Play hub (48 features)

Games Arcade (hub & catalogue)

- **What it is.** The catalogue of all games.
- **Code.** `screens/games/games_screen_material3.dart`, `screens/games/games_screen.dart`, `screens/games/games_screen_enhanced_with_all_games.dart`, `screens/games/game_catalog.dart`, `screens/games/game_router.dart`, `screens/games/language_games_screen.dart` (+2 more). Scan: 8 file(s), 2,993 lines; 3 silently swallowed catch(es); 22 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `language_games_screen.dart`, `language_games_screen_components.dart`, `lazy_game_list.dart`.
- **Current state.** Works (routes `games`, `games_api_languages`, `games_enhanced_catalog`: three routed catalogues). `language_games_screen*.dart` and `lazy_game_list.dart` are unrouted variants (CONSOLIDATE).
- **Defects.** Three competing catalogues; games are a separate destination and don't update the learner model reliably.
- **Elevation.** One **Arcade** with shelves: "For you today" (games that train the learner's weakest skills), Tone & Sound, Words, Grammar, Speaking, Culture, Traditional Games, Multiplayer and Classroom.

Each game card shows the skill it trains and the learner's mastery. Lazy loading from `lazy_game_list.dart` is kept as the loading strategy.
- **UX and render.** Shelf layout with game art (one consistent illustration style); skill chips; "New" and "Event" ribbons. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play tab (root).

Game shell, guides & results *(revived)*

- **What it is.** Shared game frame: how-to-play guide, loading and results.
- **Code.** `screens/games/templates/game_template_shell.dart`, `screens/games/game_onboarding_overlay.dart`, `screens/games/game_start_guide_screen.dart`, `screens/games/game_mastery_results_screen.dart`, `screens/games/victory_results_screen.dart`, `screens/games/universal_loading_screen.dart` (+2 more). Scan: 8 file(s), 1,943 lines; 9 `Random` use(s); 27 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `game_start_guide_screen.dart`, `game_mastery_results_screen.dart`, `victory_results_screen.dart`, `universal_loading_screen.dart`, `universal_game_screen.dart`, `template_timed_quiz.dart`.
- **Current state.** Overlay and shell work. **`game_mastery_results_screen.dart` (REVIVE as the standard results screen)**. Start guide, victory results, universal loading, universal game screen and six templates are unrouted (CONSOLIDATE into GameKit).
- **Defects.** Inconsistent results screens; no mastery feedback after play.
- **Elevation.** One **GameKit** frame. A 10-second how-to (from the start guide) on first play; the revived **mastery results** screen shows which words and skills improved, mistakes with audio, "add to review", and a replay button. Every game reports to the learner model.
- **UX and render.** Consistent header (timer, score, quit), results with skill bars; confetti only on personal bests. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Inside every game.

Word Match

- **What it is.** Match words to meanings.
- **Code.** `screens/games/word_match_game.dart`. Scan: 1 file(s), 455 lines; 4 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Due-word driven matching with audio on every tap and a "tone twin" round where two words differ only in tone.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Words shelf.

Word Match (audio)

- **What it is.** Match heard words to meanings.
- **Code.** `screens/games/word_match_audio_game.dart`. Scan: 1 file(s), 527 lines; 2 `Random` use(s); 3 `Text(` widgets vs 1 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI

turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Listening-first: hear it, match it. Speakers rotate across regions to build accent tolerance.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Tone & Sound shelf.

Fill in the Blank

- **What it is.** Complete sentences.
- **Code.** `screens/games/fill_in_the_blank_game.dart`. Scan: 1 file(s), 528 lines; 2 `Random` use(s); 11 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Cloze from real dialogues with diacritic-aware typing and a hint that plays the audio.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Grammar shelf.

Picture–Word Association

- **What it is.** Match a word to an image.
- **Code.** `screens/games/picture_word_association_screen.dart`. Scan: 1 file(s), 399 lines; 1 `Random` use(s); 1 mock/fake marker(s); 2 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Culturally reviewed illustrations (local foods, clothing, places) and a "say it" bonus that awards points only for real scored speech.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Words shelf.

Memory Map

- **What it is.** 3×4 card flip memory of word↔meaning pairs.
- **Code.** `screens/games/memory_map_screen.dart`. Scan: 1 file(s), 419 lines; 2 `Random` use(s).
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Audio-card variant (flip to hear) and family-themed decks, so heritage learners memorise kinship terms.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.

- **Lives in.** Play → Words shelf.

Speed Challenge

- **What it is.** Timed vocabulary challenge.
- **Code.** `screens/games/speed_challenge_game.dart`. Scan: 1 file(s), 620 lines; 15 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Adaptive speed that targets 80–85% success, with a "calm mode" without a timer for accessibility.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Words shelf.

Speed Round

- **What it is.** Rapid-fire rounds.
- **Code.** `screens/games/speed_round_game.dart`. Scan: 1 file(s), 424 lines; 4 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Merge mechanics with Speed Challenge as two modes of one game ("Sprint" and "Rounds"); both survive as modes.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Words shelf.

Story Builder

- **What it is.** Arrange sentences into a story.
- **Code.** `screens/games/story_builder_game.dart`. Scan: 1 file(s), 457 lines; 4 `Random` use(s); 1 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Builds illustrated audio stories from the learner's level; finished stories can be shared to circles.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Grammar shelf.

Grammar Detective

- **What it is.** Find grammar errors.
- **Code.** `screens/games/grammar_detective_game.dart`. Scan: 1 file(s), 471 lines; 4 `Random` use(s); 6 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Cases framed as village mysteries; each solved case unlocks a Pattern Library card.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Grammar shelf.

Grammar Jam

- **What it is.** Music-paced grammar game.
- **Code.** `screens/games/grammar_jam_screen.dart`. Scan: 1 file(s), 506 lines; 4 `Random` use(s); 2 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Rhythm-paced sentence building over Afrobeats/amapiano-style beats (commissioned), with rhythm tied to syllables.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Grammar shelf.

Greeting Diplomacy

- **What it is.** Choose the right greeting for the situation.
- **Code.** `screens/games/greeting_diplomacy_screen.dart`. Scan: 1 file(s), 384 lines; 5 `Random` use(s); 3 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Becomes **Greeting Gauntlet** (§10.3): time × person × setting, with avatar gestures and spoken answers.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Culture shelf.

Market Bargaining

- **What it is.** Haggle with a trader (random asking price, `market_bargaining_screen.dart:84`).
- **Code.** `screens/games/market_bargaining_screen.dart`. Scan: 1 file(s), 392 lines; 3 `Random` use(s); 8 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI

turns and effects; any scoring use is called out in the description.
- **Defects.** Asking price is random (`:84`), which is fine as a mechanic; there is no politeness or language scoring. Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Becomes **Market Haggle**: voice roleplay with trader personalities, real numbers practice, and politeness scoring.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Speaking shelf.

Quiz Chef

- **What it is.** Cooking-themed quiz.
- **Code.** `screens/games/quiz_chef_screen.dart`. Scan: 1 file(s), 771 lines; 1 `Random` use(s); 3 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Recipes as lessons: cook jollof, egusi, ugali or chakalaka by following spoken instructions in the target language.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Culture shelf.

Roleplay Adventure

- **What it is.** Branching roleplay adventure.
- **Code.** `screens/games/roleplay_adventure_game.dart`. Scan: 1 file(s), 406 lines; 2 `Random` use(s); 4 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Shares the Scenario Theatre engine; adventures become multi-scene trips (Lagos → Ibadan → Ọ̀ṣogbo).
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Speaking shelf.

Taxi Survival

- **What it is.** Navigate a taxi ride using the language.
- **Code.** `screens/games/taxi_survival_screen.dart`. Scan: 1 file(s), 426 lines; 2 `Random` use(s); 1 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.

- **Elevation.** Becomes **Danfo/Matatu Rush**: real recorded conductor calls at native speed; board the right bus.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Tone & Sound shelf.

Tone Trainer

- **What it is.** Identify tones (one H/M/L scheme for all languages, `tone_trainer_game.dart:47-66`).
- **Code.** `screens/games/tone_trainer_game.dart`. Scan: 1 file(s), 552 lines; 3 `Random` use(s); 5 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Applies one High/Mid/Low scheme to every language (`tone_trainer_game.dart:47-66`), which is false for Swahili (no lexical tone) and incomplete for Igbo downstep (P0-17). Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Per-language tone inventories with minimal pairs and real audio; feeds the Tone Gym ladder.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Tone & Sound shelf.

Pronunciation Duel

- **What it is.** Head-to-head pronunciation (never records; random 75–95 score, `:105-129`, P0-18).
- **Code.** `screens/games/pronunciation_duel_game.dart`. Scan: 1 file(s), 367 lines; 6 `Random` use(s); 5 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Never opens the microphone; random 75–95% score (`pronunciation_duel_game.dart:105-129`, P0-18). Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Real recording and scoring; async duels with friends on minimal pairs; honest "practice mode" (no score) until the scorer ships.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Speaking shelf.

Pronunciation Game

- **What it is.** Pronunciation practice game.
- **Code.** `screens/games/pronunciation_game.dart`. Scan: 1 file(s), 606 lines; 15 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** No verified scoring path [Unverified whether it records]. Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.

- **Elevation.** Merge its rounds into the Tone Gym ladder as a game mode; real scoring.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Speaking shelf.

Pronunciation Karaoke

- **What it is.** Follow advancing words and hold the mic to "sing" (pitch score `60 + rng.nextInt(35)`, `:155`).
- **Code.** `screens/games/pronunciation_karaoke_screen.dart`. Scan: 1 file(s), 469 lines; 3 `Random` use(s); 2 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Pitch score is `60 + rng.nextInt(35)` (`pronunciation_karaoke_screen.dart:155`, P0-18). Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Becomes **Echo Karaoke**: real F0 line, licensed songs, children's rhymes and praise phrases.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Speaking shelf.

Emoji Translator

- **What it is.** Translate emoji sequences into words.
- **Code.** `screens/games/emoji_translator_screen.dart`. Scan: 1 file(s), 414 lines; 1 `Random` use(s); 2 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Emoji stories → sentences, then send the sentence as a chat sticker made from your avatar.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Words shelf.

Elders' Blessings

- **What it is.** Learn blessings and proverbs from elders.
- **Code.** `screens/games/elders_blessings_screen.dart`. Scan: 1 file(s), 380 lines; 2 `Random` use(s); 1 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** An elder character (avatar) gives blessings for life events (birth, marriage, travel); the learner responds correctly. Content reviewed by cultural consultants.

- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Culture shelf.

Folktale Reconstruction

- **What it is.** Reorder folktale scenes.
- **Code.** `screens/games/folktale_reconstruction_screen.dart`. Scan: 1 file(s), 501 lines; 2 `Random` use(s); 3 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Àlọ́ / hadithi / iintsomi storytelling: rebuild the tale, then narrate it with scoring. Tales come from the Living Archive.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Culture shelf.

Clan Lineage Story Builder

- **What it is.** Build a lineage story (clan/kinship).
- **Code.** `screens/games/clan_lineage_story_builder_screen.dart`. Scan: 1 file(s), 554 lines; 7 `Random` use(s); 1 mock/fake marker(s); 2 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Teaches kinship vocabulary through a family tree you build; links to the Ancestral Tree feature.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Culture shelf.

Conversation Relay

- **What it is.** Multi-turn chain using bundled scenarios plus Polie.
- **Code.** `screens/games/conversation_relay_screen.dart`. Scan: 1 file(s), 215 lines; 2 direct Groq-client call site(s) (P0-3 dependency); 1 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Classroom and circle relay: each player adds a turn by voice; AI checks; the chain plays back as an audio drama.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Speaking shelf.

Board Word Games

- **What it is.** Word seed puzzles with spaced letters and a category hint.
- **Code.** `screens/games/board_word_games.dart`. Scan: 1 file(s), 540 lines; 8 `Random` use(s); 9 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Diacritic-aware word puzzles (Wordle-like with tone marks as a mechanic).
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Words shelf.

Cultural Games collection

- **What it is.** Launcher for the cultural games set.
- **Code.** `screens/games/cultural_games.dart`. Scan: 1 file(s), 15 lines.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Becomes the Arcade's "Culture" shelf (the screen remains as a themed shelf).
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Culture shelf.

Accent Puzzle

- **What it is.** Accent decoding puzzle.
- **Code.** `screens/games/cultural/accent_puzzle_game.dart`. Scan: 1 file(s), 375 lines; 2 `Random` use(s); 7 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Decode the same sentence spoken in different regional accents; builds real-world listening.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Tone & Sound shelf.

Call & Response

- **What it is.** Respond to calls (traditional call-and-response).
- **Code.** `screens/games/cultural/call_response_game.dart`. Scan: 1 file(s), 411 lines; 3 `Random` use(s); 7 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI

turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Musical call-and-response with real singers; responses scored on rhythm and tone.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Speaking shelf.

Cultural Etiquette

- **What it is.** Etiquette scenarios.
- **Code.** `screens/games/cultural/cultural_etiquette_game.dart`. Scan: 1 file(s), 360 lines; 2 `Random` use(s); 7 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Linked to Etiquette cards; consequences shown kindly ("Mama is surprised you didn't kneel").
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Culture shelf.

Drum Word

- **What it is.** Match drum patterns to words.
- **Code.** `screens/games/cultural/drum_word_game.dart`. Scan: 1 file(s), 363 lines; 3 `Random` use(s); 8 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Becomes **Talking Drum** (§10.3): drummed tone contours → words, then speak it.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Tone & Sound shelf.

Flashcard Safari

- **What it is.** AR-style vocabulary scanning.
- **Code.** `screens/games/cultural/flashcard_safari_game.dart`. Scan: 1 file(s), 351 lines; 4 `Random` use(s); 7 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Shares the Point & Say camera pipeline: a scavenger hunt ("find something red, say it").
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.

- **Lives in.** Play → Words shelf.

Food Quest

- **What it is.** Food vocabulary quest.
- **Code.** `screens/games/cultural/food_quest_game.dart`. Scan: 1 file(s), 391 lines; 3 `Random` use(s); 7 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Market-to-pot quests with regional dishes; ties to Quiz Chef recipes.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Culture shelf.

Liar Liar

- **What it is.** Detect grammatical errors.
- **Code.** `screens/games/cultural/liar_liar_game.dart`. Scan: 1 file(s), 377 lines; 4 `Random` use(s); 8 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Social bluffing: friends submit sentences, some wrong; others spot the lie. A multiplayer grammar game.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Grammar shelf.

Multilingual Relay

- **What it is.** Relay race across languages.
- **Code.** `screens/games/cultural/multilingual_relay_game.dart`. Scan: 1 file(s), 405 lines; 2 `Random` use(s); 10 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Pan-African relay: one word passed through several languages (Yoruba → Swahili → Zulu), celebrating cognates and loanwords.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Words shelf.

Phrase Sniper

- **What it is.** Hit the right phrase quickly.

- **Code.** `screens/games/cultural/phrase_sniper_game.dart`. Scan: 1 file(s), 392 lines; 3 `Random` use(s); 7 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Listening reflex game with native audio; accessibility mode without time pressure.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Tone & Sound shelf.

Proverb Unlocker

- **What it is.** Fill-in-the-blank ancestral tablet.
- **Code.** `screens/games/cultural/proverb_unlocker_game.dart`. Scan: 1 file(s), 440 lines; 3 `Random` use(s); 1 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Becomes **Proverb Weaver** (§10.3). CONSOLIDATE with the unrouted GameKit rewrite `games/proverb_unlocker/*`, which its header says replaces the old random logic, so the better implementation becomes the live one.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Culture shelf.

Rhythm Typing

- **What it is.** Type to a rhythm.
- **Code.** `screens/games/cultural/rhythm_typing_game.dart`. Scan: 1 file(s), 337 lines; 8 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Diacritic typing trainer to rhythm; builds tone-mark keyboard fluency.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Words shelf.

Tongue Twister Race

- **What it is.** Rapid tongue-twister race.
- **Code.** `screens/games/cultural/tongue_twister_game.dart`. Scan: 1 file(s), 346 lines; 9 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.

- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Real scoring with fluency metrics; native tongue twisters per language; shareable duet clips.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Speaking shelf.

Village Quest

- **What it is.** NPC conversation game.
- **Code.** `screens/games/cultural/village_quest_game.dart`. Scan: 1 file(s), 406 lines; 3 `Random` use(s); 6 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Becomes the playable layer of the Village map: talk to NPCs in each place (market, café, elder hut).
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Speaking shelf.

Ayo (mancala)

- **What it is.** Ayo with a gloss MCQ when capturing.
- **Code.** `screens/games/traditional/ayo_mancala_screen.dart`. Scan: 1 file(s), 552 lines; 3 `Random` use(s); 5 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Keep the authentic game; questions become due-word reviews; online play with friends; a cultural history card.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Traditional shelf.

Snakes & Ladders (cultural)

- **What it is.** Snakes and ladders with language questions.
- **Code.** `screens/games/traditional/cultural_snakes_ladders_screen.dart`. Scan: 1 file(s), 470 lines; 7 `Random` use(s); 1 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Family mode for heritage households (pass-and-play), with questions at each player's own level.

- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Traditional shelf.

Language Ludo

- **What it is.** Ludo with language challenges.
- **Code.** `screens/games/traditional/language_ludo_screen.dart`. Scan: 1 file(s), 552 lines; 5 `Random` use(s); 4 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Multiplayer with friends and circles; speaking challenges on special squares.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Traditional shelf.

Suwe (hopscotch)

- **What it is.** Hopscotch with gloss challenges.
- **Code.** `screens/games/traditional/suwe_hopscotch_screen.dart`. Scan: 1 file(s), 445 lines; 3 `Random` use(s); 3 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Kids-friendly counting and body-movement vocabulary; a guardian-safe offline mode.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Traditional shelf.

Whot!

- **What it is.** The Nigerian card game Whot with language rules.
- **Code.** `screens/games/traditional/whot_card_screen.dart`. Scan: 1 file(s), 543 lines; 4 `Random` use(s); 10 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Card names and calls in the target language ("Check up!", "Last card!"); online play with friends.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Traditional shelf.

Drum Rhythm

- **What it is.** Drum rhythm game (GameKit).
- **Code.** `games/drum_rhythm/drum_rhythm_screen.dart`. Scan: 1 file(s), 318 lines; 7 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (launched from the Arcade via `game_router.dart`). Content comes from the shared 85–118-word pool with no audio (§10.1). `Random` is used (see evidence), mostly for shuffling, AI turns and effects; any scoring use is called out in the description.
- **Defects.** Thin content; no learner-model updates; inconsistent results screen; hard-coded strings.
- **Elevation.** Merge with Talking Drum as its rhythm mode; rhythm mapped to syllable timing.
- **UX and render.** GameKit frame; one illustration style; audio on every target-language item; reduced-motion variant. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Tone & Sound shelf.

Tone Forge *(revived)*

- **What it is.** A tone-crafting game; its header calls it the "flagship implementation" on GameKit.
- **Code.** `games/tone_forge/tone_forge_screen.dart`. Scan: 1 file(s), 399 lines; 2 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `tone_forge_screen.dart`.
- **Current state. Unrouted** (REVIVE). 5 files under `games/tone_forge/`, never imported from the app.
- **Defects.** A flagship tone game that nobody can reach.
- **Elevation.** Revive as a headline Tone & Sound game: forge words by shaping their pitch contour on the Tone Lens (drag syllables up and down), hear the result and its meaning change (ọkọ/ọkọ̀/ọkọ́), then speak it. Pairs with the Tone Gym.
- **UX and render.** Tactile contour editor; blacksmith-forge art direction (a nod to Ogun / the West African smithing tradition), reviewed. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Play → Tone & Sound (featured).

18.6 Village hub (34 features)

Villages hub & Language Villages

- **What it is.** Themed language villages (one per language) with places to visit.
- **Code.** `screens/village/villages_hub_screen.dart`, `screens/social/language_villages_screen.dart`, `screens/village/language_village_place_screen.dart`. Scan: 3 file(s), 894 lines; 1 direct Groq-client call site(s) (P0-3 dependency); 11 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (routes `villages`, `language-village`, `language-village-place`).
- **Defects.** Places are static screens; little living activity; hard-coded strings.
- **Elevation. The Village is the community home**: an illustrated map per language where each place is a real activity. Market = haggling games and buy/sell vocabulary; Café = live practice rooms; Elder Hut = mentors and proverbs; Square = feed and events; School = live classroom; Story Tree = folktales. Live presence dots show friends practising now (privacy-safe, opt-in).
- **UX and render.** Hand-illustrated map per region (not one generic village); tap-to-enter places; ambient soundscape (optional). Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village tab (root).

Swahili Village Map

- **What it is.** A Swahili-specific village map.
- **Code.** `screens/village/swahili_village_map_screen.dart`. Scan: 1 file(s), 367 lines; 4 `Text(` widgets vs 0 l10n lookups.

- **Current state.** Works (route `swahili-village-map`).
- **Defects.** The only language with a map; others are generic.
- **Elevation.** The **template** for every language: Yoruba compound, Igbo obi, Hausa birni, Zulu umuzi, Xhosa village, a Lagos street for Pidgin. Each map is reviewed by cultural consultants.
- **UX and render.** Same map engine, regional art direction. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → map (per language).

Village Market

- **What it is.** Market place in the village.
- **Code.** `screens/village/village_market_screen.dart`. Scan: 1 file(s), 507 lines; 15 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `village-market`).
- **Defects.** Mostly static.
- **Elevation.** Live **Market Haggle** stall, price and number drills, and the Magic Items shop (cosmetics only, no paid power). Weekly market-day event.
- **UX and render.** Stalls as cards; trader NPC avatar. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → map → Market.

Village Café

- **What it is.** Social café space.
- **Code.** `screens/village/village_cafe_screen.dart`. Scan: 1 file(s), 523 lines; 10 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `village-cafe`).
- **Defects.** Little actual activity.
- **Elevation.** Doorway to **Practice Rooms** and audio rooms: "3 tables open in Yoruba A2 right now". Small-talk prompts on each table.
- **UX and render.** Tables showing level and topic; join button. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → map → Café.

Elder Hut

- **What it is.** Elder wisdom space.
- **Code.** `screens/village/elder_hut_screen.dart`. Scan: 1 file(s), 582 lines; 8 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `elder-hut`).
- **Defects.** Static content.
- **Elevation.** Home of **Micro-Mentors** and proverbs: book a 10-minute session with a verified native speaker or elder, hear a proverb of the day with its story, and ask cultural questions (answered by humans, labelled).
- **UX and render.** Warm interior art; mentor cards with verified badge. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → map → Elder Hut.

Practice Rooms & Sessions

- **What it is.** Set up and join group practice sessions, with a collaborative room and summary.

- **Code.** `screens/village/practice_room_setup_screen.dart`, `screens/village/practice_session_screen.dart`, `screens/village/practice_room_collaborative_screen.dart`, `screens/village/session_summary_screen.dart`. Scan: 4 file(s), 2,303 lines; 8 `Random` use(s); 1 silently swallowed catch(es); 27 `Text(` widgets vs 0 l10n lookups.
- **Current state.** UI works (routes `practice-room-*`, `practice-session`, `session-summary`). **The active speaker is simulated with random numbers** (`practice_session_screen.dart:112`).
- **Defects.** Fake activity; no real matching; no safety controls.
- **Elevation. Real small-group rooms** (2–6 learners at the same level) with a structured script: warm-up, topic, roleplay and a recap. Polie acts as a facilitator (turn prompts, vocabulary hints) and the session summary lists words used and corrections. Adult-only, or teacher-supervised for minors.
- **UX and render.** Video/audio tiles; speaking-time meter to encourage balance; report and block in one tap. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Café → Practice Rooms; Today when a room matches.

Flashcard Focus (group)

- **What it is.** Group flashcard session.
- **Code.** `screens/village/flashcard_focus_screen.dart`. Scan: 1 file(s), 451 lines; 3 `Text(` widgets vs 0 l10n lookups.
- **Current state.** UI works (route `flashcard-focus`); participants are hard-coded mocks (`flashcard_focus_screen.dart:26-34`).
- **Defects.** Fake participants.
- **Elevation.** Replace the mocks with **real matched learners or clearly labelled practice bots**; study together with a shared deck and a race to recall.
- **UX and render.** Participant avatars with "bot" labels where applicable. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Café → Study together.

Matching Pairs (group)

- **What it is.** Matching pairs in the village.
- **Code.** `screens/village/matching_pairs_screen.dart`. Scan: 1 file(s), 302 lines; 3 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `matching-pairs`).
- **Defects.** Duplicates Memory Map mechanics.
- **Elevation.** Becomes the **multiplayer mode** of Memory Map: turn-based with friends, audio cards. Both entry points stay.
- **UX and render.** Shared board; turn indicator. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Café; Play → Words.

Tribes

- **What it is.** Join a tribe (team), discover tribes, tribe hub and tribe chat.
- **Code.** `screens/gamification/tribe_selection_screen.dart`, `screens/village/tribe_hub_screen.dart`, `screens/village/tribe_discovery_screen.dart`, `screens/village/my_tribe_screen.dart`, `screens/chat/tribe_chat_screen.dart`, `screens/chat/tribe_chat_screen_material3.dart`. Scan: 6 file(s), 2,391 lines; 1 silently swallowed catch(es); 51 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (routes `tribe`, `tribe-selection`, `tribe-hub`, `tribe-discovery`, `my-tribe`). Duplicate routes for the same screens.

- **Defects.** "Tribe" naming can read as ethnic grouping in an African context; duplicate routes; tribe chat inherits the socket flaws (P0-4/5/6).
- **Elevation.** Keep the feature, rename to **Circles** (study teams of 5–30) with optional language or city themes. Weekly team goals and a team streak; circle chat with moderation; leaders get tools. The "tribe" word remains available for playful naming by users.
- **UX and render.** Circle crest builder (patterns from adire, kente, kanga, Ndebele — reviewed); team progress ring. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Circles.

Tribe vs Tribe, Tribal Duel & Inter-tribe Leaderboard

- **What it is.** Team competitions and leaderboards between tribes.
- **Code.** `screens/social/tribe_vs_tribe_screen.dart`, `screens/village/tribal_duel_screen.dart`, `screens/village/inter_tribe_leaderboard_screen.dart`. Scan: 3 file(s), 1,441 lines; 6 mock/fake marker(s); 33 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works as UI (routes `tribe_vs_tribe`, `tribal-duel`, `inter-tribe-leaderboard`). Points can be self-awarded (P0-8).
- **Defects.** Cheatable scoring; competition without learning value.
- **Elevation. Circle Clash**: weekend team events where points come only from verified learning (reviews done, lessons passed, scored speaking). Server-side scoring; fair leagues by size.
- **UX and render.** Event banner, live team bars, highlights reel at the end. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Circles → Clash; Today → Events.

Global Chat

- **What it is.** Public chat rooms per language.
- **Code.** `screens/chat/global_chat_screen_material3.dart`, `screens/chat/global_chat_screen.dart`, `screens/chat/community_chat_screen.dart`, `screens/chat/community_chat_screen_material3.dart`. Scan: 4 file(s), 2,206 lines; 4 silently swallowed catch(es); 35 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `global_chat_screen.dart`, `community_chat_screen.dart`, `community_chat_screen_material3.dart`.
- **Current state.** Works (route `global_chat`). Variants: `global_chat_screen.dart` and community chat screens unrouted (CONSOLIDATE into the village and global chat).
- **Defects.** P0-4/5/6 socket flaws; moderation cosmetic (P0-11); open to minors (P0-10).
- **Elevation. Language Squares**: public rooms per language and level with a "target-language only" mode, Tone Lens on messages, tap-to-translate, and one-tap "correct me" requests. Real moderation and slow-mode.
- **UX and render.** Chat with level badges and pinned daily topic. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Square → Rooms.

Private Chat (DMs) *(revived)*

- **What it is.** One-to-one messaging with search and contact info.
- **Code.** `screens/chat/private_chat_list_screen.dart`, `screens/chat/private_chat_screen.dart`, `screens/chat/private_chat_screen_material3.dart`, `screens/chat/chat_search_screen.dart`, `screens/chat/contact_info_screen.dart`. Scan: 5 file(s), 2,853 lines; 3 mock/fake marker(s); 3 silently swallowed catch(es); 22 `Text(` widgets vs 1 l10n lookups; not reachable from `main.dart`: `private_chat_screen_material3.dart`, `chat_search_screen.dart`, `contact_info_screen.dart`.

- **Current state.** List and chat work (route `private-chat-inbox`). **`chat_search_screen.dart` and `contact_info_screen.dart` (block/report) are unrouted (REVIVE)**; the material3 chat variant is unrouted (CONSOLIDATE).
- **Defects.** P0-4/5/6; no reachable block/report UI; open to minors.
- **Elevation. Language-exchange DMs**: each message can be written in the target language with an inline "check my message" (AI suggestion before sending) and a partner correction feature. Revived chat search finds words and media; revived contact info gives block, report, mute and shared-words history.
- **UX and render.** WhatsApp-familiar layout made learning-native; correction bubbles in a distinct colour. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Chats.

Call History

- **What it is.** Log of voice/video calls.
- **Code.** `screens/chat/call_history_screen.dart`. Scan: 1 file(s), 268 lines; 5 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `call-history`).
- **Defects.** Calls carry no learning value afterwards.
- **Elevation.** Each call entry shows minutes spoken in the target language and words learned (opt-in transcript), and offers "save phrases from this call".
- **UX and render.** List with minutes chip. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Chats → Calls.

Chat Welcome & Community Guidelines *(revived)*

- **What it is.** Welcome screen for the chat section.
- **Code.** `screens/chat/welcome_screen.dart`. Scan: 1 file(s), 332 lines; 1 mock/fake marker(s); 2 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `welcome_screen.dart`.
- **Current state. Unrouted** (REVIVE).
- **Defects.** Community norms never shown.
- **Elevation.** Revive as the **first-entry pledge**: 3 friendly rules (be kind, help learners, no harassment) with an illustrated agreement, shown once before the first message, plus a link in settings.
- **UX and render.** Short illustrated cards; one-tap agree. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village first entry; Me → Safety.

Moderation Tools *(revived)*

- **What it is.** Tools for moderators.
- **Code.** `screens/chat/moderation_tools_screen.dart`, `screens/social_audio/moderation_screen.dart`. Scan: 2 file(s), 531 lines; 25 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `moderation_tools_screen.dart`.
- **Current state.** `chat/moderation_tools_screen.dart` **unrouted** (REVIVE); social-audio moderation works.
- **Defects.** No moderation surface for chat.
- **Elevation.** Revive as **Community Stewards** tools for circle leaders and trusted volunteers: reported queue, mute/timeout, slow mode, pinned rules, with escalation to staff. Training module required.
- **UX and render.** Queue cards with context and quick actions. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.

- **Lives in.** Village → Circles → Steward tools (role-gated).

Social Hub & Connections

- **What it is.** Friends, connections, friend profiles and people search.
- **Code.** `screens/social/social_hub_screen.dart`, `screens/social/user_connections_screen.dart`, `screens/social/friend_profile_screen.dart`, `screens/chat/user_search_global_id_screen.dart`, `screens/social/global_people_search_screen.dart`. Scan: 5 file(s), 2,427 lines; 1 mock/fake marker(s); 42 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `global_people_search_screen.dart`.
- **Current state.** Works (routes `social-hub`, `connections`, `friend-profile`). `global_people_search_screen.dart` unrouted (CONSOLIDATE into the global-ID search).
- **Defects.** Find-anyone search with no age or privacy controls.
- **Elevation. Language partners**: suggested partners who speak the language you learn and learn the one you speak (tandem matching), with friend profiles showing languages, level and shared goals. Under-18s never appear in adult discovery.
- **UX and render.** Partner cards with "speaks / learns" pills. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → People.

Friend Quests & Challenge a Friend

- **What it is.** Create quests with friends and challenge them.
- **Code.** `screens/social/friend_quests_screen.dart`, `screens/social/create_friend_quest_screen.dart`, `screens/social/challenge_friend_screen.dart`. Scan: 3 file(s), 1,875 lines; 35 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (routes `friend-quests`, `create-friend-quest`, `challenge-friend`).
- **Defects.** Challenges are score-only, and scores can be faked (P0-8, P0-18).
- **Elevation. Buddy quests**: shared weekly goals (both practise five days), async challenges on real skills (tone duel, review race), and nudges that don't nag.
- **UX and render.** Paired progress bars; challenge cards in chat. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → People → Quests; Today.

Share Progress

- **What it is.** Share achievements externally.
- **Code.** `screens/social/share_progress_screen.dart`. Scan: 1 file(s), 417 lines; 2 TODO(s); 15 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `share-progress`).
- **Defects.** Generic share cards.
- **Elevation. Beautiful share cards**: the learner's avatar saying a phrase they learned (short clip), Passport stamps and streak milestones in the target language. Privacy-checked (no exact location, kids can't share publicly).
- **UX and render.** Story-format cards (9:16) with the brand system. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me → Progress → Share; after milestones.

Social Gifting

- **What it is.** Send gifts to friends.

- **Code.** `screens/social/social_gifting_screen.dart`. Scan: 1 file(s), 379 lines; 20 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works as UI; opened from a friend profile (`friend_profile_screen.dart:367`). Currency can be self-minted (P0-7).
- **Defects.** Cheatable economy.
- **Elevation. Gift a lesson / streak freeze / cosmetic** using server-validated currency; send a voice blessing with the gift ("Ẹ ku odun!").
- **UX and render.** Gift-wrap animation; message composer. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → People → Gift.

Community Corrections

- **What it is.** Native speakers correct learners' submissions.
- **Code.** `screens/social/community_corrections_screen.dart`. Scan: 1 file(s), 336 lines; 9 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works as UI; opened from the Social Hub (`social_hub_screen.dart:125`).
- **Defects.** No reputation or quality control for correctors.
- **Elevation. Native Corrections**: learners post short writing or audio; verified natives correct with inline edits and audio; reputation for helpful correctors; best corrections feed the Living Dictionary as usage examples (with consent).
- **UX and render.** Diff-style inline edits; audio replies. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Corrections.

Ancestral Tree

- **What it is.** Family tree feature.
- **Code.** `screens/social/ancestral_tree_screen.dart`. Scan: 1 file(s), 419 lines; 10 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `ancestral_tree`).
- **Defects.** Not connected to learning.
- **Elevation. Family Tree in your language**: heritage learners build their family tree with kinship terms (ìyá àgbà, nne nne, ugogo), record relatives' names with correct tones, and invite family to add voice notes. Private by default.
- **UX and render.** Tree with avatar nodes; kinship terms on edges. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me → Heritage; Learn → Culture.

Micro-Mentors

- **What it is.** Short sessions with mentors.
- **Code.** `screens/community/micro_mentor_hub_screen.dart`, `screens/community/micro_mentor_session_detail_screen.dart`. Scan: 2 file(s), 830 lines; 38 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `micro-mentor-hub`).
- **Defects.** No verification, safeguarding or payment flow visible [verify].
- **Elevation. Book a native** for 10–20 minute focused sessions (pronunciation clinic, heritage conversation, exam prep), with verified mentors, ratings, safeguarding rules, and payouts in local currency. Mentors earn from LingAfriq: jobs for native speakers.
- **UX and render.** Mentor profiles with sample audio; booking sheet. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.

- **Lives in.** Village → Elder Hut → Mentors.

Live Classroom

- **What it is.** Live online classes with lobby, notes, speaker queue, roster, assignments and privacy.
- **Code.** `screens/chat/live_classroom_screen_material3.dart`, `screens/classroom/classroom_lobby_screen.dart`, `screens/classroom/classroom_notes_screen.dart`, `screens/classroom/speaker_queue_screen.dart`, `screens/classroom/classroom_roster_screen.dart`, `screens/classroom/classroom_assignments_screen.dart` (+5 more). Scan: 11 file(s), 5,676 lines; 1 direct Groq-client call site(s) (P0-3 dependency); 2 `Random` use(s); 4 silently swallowed catch(es); 112 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `live_classroom_screen.dart`, `live_classroom_mobile_screen.dart`, `classroom_session_screen.dart`, `classroom_chat_livekit_screen.dart`.
- **Current state.** Works (routes `live-classroom`, `classroom-*`, `speaker-queue`). Four older classroom screens unrouted (CONSOLIDATE into `live_classroom_screen_material3.dart`).
- **Defects.** Five parallel implementations; safeguarding for minors not visible.
- **Elevation. School in the Village**: scheduled live classes by level with a teacher, raise-hand speaker queue, shared whiteboard with Tone Lens, auto-notes (transcript → key phrases), and assignments that land in the learner's Review. Recordings opt-in.
- **UX and render.** Classroom grid; hand-raise queue; notes panel. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → School.

Teacher Classroom Dashboard *(revived)*

- **What it is.** Dashboard for teachers to manage a class.
- **Code.** `screens/gamification/classroom_dashboard_screen.dart`. Scan: 1 file(s), 739 lines; 19 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `classroom_dashboard_screen.dart`.
- **Current state. Unrouted** (REVIVE).
- **Defects.** Teachers have no management surface.
- **Elevation.** Revive as **LingAfriq for Schools**: class roster, assign units, see weak skills per student, run a live game (Conversation Relay, Liar Liar) on a projector. A B2B channel with Nigerian, Kenyan and South African schools (and diaspora Saturday schools).
- **UX and render.** Desktop/tablet-friendly layout; class heatmap. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me → Teaching (role-gated); web.

Social Audio Rooms *(revived)*

- **What it is.** Clubhouse-style audio rooms: discover, create, room detail, following and schedule.
- **Code.** `screens/social_audio/room_discovery_screen.dart`, `screens/social_audio/create_room_screen.dart`, `screens/social_audio/room_detail_screen.dart`, `screens/social_audio/following_screen.dart`, `screens/social_audio/scheduled_sessions_screen.dart`. Scan: 5 file(s), 1,669 lines; 6 mock/fake marker(s); 39 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `following_screen.dart`, `scheduled_sessions_screen.dart`.
- **Current state.** Discovery, create and detail are reachable (not orphans; no named route, entry point unverified). **`following_screen.dart` and `scheduled_sessions_screen.dart` are unrouted (REVIVE).**
- **Defects.** No scheduling or following, so rooms are empty; moderation for minors.
- **Elevation. Village Radio**: scheduled audio shows hosted by natives (folktale night, Pidgin news, Zulu music talk), learner rooms by level, and revived following and schedule so people come back. Live captions with Tone Lens; replays become listening lessons.

- **UX and render.** Schedule cards; speaker stage; captions bar. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Radio.

Status (WhatsApp-style)

- **What it is.** 24-hour status updates.
- **Code.** `screens/wa/wa_status_list_screen.dart`, `screens/wa/wa_status_create_screen.dart`, `screens/wa/wa_status_view_screen.dart`. Scan: 3 file(s), 758 lines; 16 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (routes `wa-status`, `wa-status-create`, `wa-status-view`).
- **Defects.** Status events broadcast globally (P0-6).
- **Elevation. Language Status**: post a daily status in the target language (text, photo or voice) with a word-of-the-day sticker; friends react with corrections or "I learned this". Prompt: "Say what you ate today in Igbo".
- **UX and render.** Familiar ring UI; stickers with the learner's avatar. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Status.

Starred Messages & Media Gallery

- **What it is.** Starred messages and shared media.
- **Code.** `screens/wa/wa_starred_messages_screen.dart`, `screens/wa/wa_media_gallery_screen.dart`. Scan: 2 file(s), 441 lines; 15 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (routes `wa-starred`, `wa-media-gallery`).
- **Defects.** Starred items aren't reusable for learning.
- **Elevation. Starred = Saved phrases**: starring a message adds it to My Words with audio; the media gallery doubles as a personal phrasebook of voice notes.
- **UX and render.** Grid with phrase overlays. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Chats → Saved.

Snaps & Streaks (Snapchat-style)

- **What it is.** Disappearing photo/video snaps, stories and streaks.
- **Code.** `screens/snap/snap_camera_screen.dart`, `screens/snap/snap_inbox_screen.dart`, `screens/snap/snap_story_feed_screen.dart`, `screens/snap/snap_streaks_screen.dart`, `screens/snap/snap_viewer_screen.dart`. Scan: 5 file(s), 629 lines; 13 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (routes `snap-*`). Snaps never actually disappear (P0-12).
- **Defects.** Privacy promise broken; content risks for minors.
- **Elevation. Clips**: send a photo or 10-second video with a phrase overlay in the target language (camera labels from Point & Say suggest words); streaks count only when both sides use the language. Real deletion, kids restricted to circles.
- **UX and render.** Camera with phrase chips; AR text in correct diacritics. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Clips; Lens camera.

Community Feed (X-style)

- **What it is.** Twitter/X-style public feed: home, compose, threads, explore, lists, profiles, notifications.
- **Code.** `screens/feed/x_feed_home_screen.dart`, `screens/feed/x_compose_screen.dart`, `screens/feed/x_post_detail_screen.dart`, `screens/feed/feed_post_thread_screen.dart`, `screens/feed/x_explore_screen.dart`, `screens/feed/x_lists_screen.dart` (+3 more). Scan: 9 file(s),

2,183 lines; 4 mock/fake marker(s); 1 silently swallowed catch(es); 46 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `post_detail_screen.dart`, `post_detail_clean_screen.dart`.
- **Current state.** Works (routes `x-feed-*`). Older post-detail screens unrouted (CONSOLIDATE).
- **Defects.** Moderation cosmetic (P0-11); generic feed with no learning hook.
- **Elevation. The Square feed**: posts in the target language with Tone Lens and tap-to-translate; native "verified speaker" badges; threads for questions ("How do you say...?") that resolve into the Living Dictionary; lists by language and topic.
- **UX and render.** Feed cards with language tag; translate toggle. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Square.

Explore, Search & Community Profile

- **What it is.** Community discovery, search and profiles.
- **Code.** `screens/feed/explore_community_screen.dart`, `screens/feed/search_community_screen.dart`, `screens/feed/community_profile_screen.dart`. Scan: 3 file(s), 324 lines; 3 mock/fake marker(s); 7 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (routes `explore-community`, `search-community`, `community-profile`).
- **Defects.** Overlaps with the X explore screen.
- **Elevation.** Explore becomes the **Square's discover page** (trending phrases, events, top correctors); search folds into the global Search; community profiles show languages, contributions and badges.
- **UX and render.** Discover grid. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Square → Discover; global Search.

Create Content (UGC)

- **What it is.** Users create lessons, quizzes and stories, with validation feedback.
- **Code.** `screens/ugc/ugc_hub_screen.dart`, `screens/ugc/create_lesson_screen.dart`, `screens/ugc/create_quiz_screen.dart`, `screens/ugc/create_story_screen.dart`, `screens/ugc/ugc_validation_feedback_screen.dart`, `screens/ugc/create_lesson_screen_enhanced.dart` (+2 more). Scan: 8 file(s), 2,318 lines; 57 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `create_quiz_screen_enhanced.dart`, `create_story_screen_enhanced.dart`.
- **Current state.** Works (route `ugc`). Enhanced variants unrouted (CONSOLIDATE, harvesting their features).
- **Defects.** No editorial pipeline; content quality unknown.
- **Elevation. Creator Studio**: templates for mini-lessons, quizzes and stories; AI checks for diacritics and accuracy; native-review queue; published content credited to the creator, with revenue share for top creators.
- **UX and render.** Editor with live preview; review status chips. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Create (unlocks at B1).

Voice Contribution

- **What it is.** Record voice samples for the dataset.
- **Code.** `screens/voice_contribution/voice_contribution_screen.dart`. Scan: 1 file(s), 514 lines; 1 silently swallowed catch(es); 18 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `contribute_voice`).
- **Defects.** Consent and licensing of recordings unclear [verify].
- **Elevation. Lend your voice**: natives record prompted sentences with explicit consent and licence choice (AI stack §5), get credited, and earn currency or cash. Feeds the TTS/ASR data flywheel.

- **UX and render.** Prompt card, quality meter, consent sheet. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Contribute.

Native Speaker Contribution & Lesson Item Verification *(revived)*

- **What it is.** Natives contribute content and verify lesson items.
- **Code.** `screens/contribute/native_speaker_contribution_screen.dart`, `screens/contribute/lesson_item_verification_screen.dart`. Scan: 2 file(s), 1,500 lines; 41 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `native_speaker_contribution_screen.dart`, `lesson_item_verification_screen.dart`.
- **Current state. Both unrouted** (REVIVE).
- **Defects.** The human-validation loop the language factory needs (§6.4.1) is unreachable.
- **Elevation.** Revive as the **Verify queue**: native speakers swipe through AI-drafted lesson items (correct / fix / reject), add audio, and earn reputation. This powers the fast-scaling waves.
- **UX and render.** Swipe cards with audio; fix-inline editor. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Contribute → Verify (verified natives).

Native Content Review

- **What it is.** Review queue for native content.
- **Code.** `screens/content/native_content_review_screen.dart`. Scan: 1 file(s), 258 lines; 8 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `native-content-review`).
- **Defects.** Overlaps with the revived verification screens.
- **Elevation.** Becomes the **reviewer dashboard** above the Verify queue: stats, disagreements and escalations to linguists.
- **UX and render.** Dashboard with queue counts. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Village → Contribute → Review.

18.7 Me hub (15 features)

Profile

- **What it is.** The learner profile, with editing.
- **Code.** `screens/tabs_view/profile/profile_tab_material3.dart`, `screens/profile/profile_screen_material3.dart`, `screens/tabs_view/profile/profile_edit_screen.dart`, `screens/settings/edit_profile_screen.dart`, `screens/profile/user_profile_screen.dart`. Scan: 5 file(s), 2,302 lines; 5 mock/fake marker(s); 9 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `user_profile_screen.dart`.
- **Current state.** Works (Profile tab). `user_profile_screen.dart` unrouted (CONSOLIDATE). Two edit-profile screens.
- **Defects.** Duplicate edit flows; profile is stats-heavy, identity-light.
- **Elevation. My LingAfriq identity**: avatar (§4), languages and levels, heritage story (optional), Passport stamps, contributions and circles. One edit flow.
- **UX and render.** Avatar hero; language pills; stamp shelf. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me (avatar, top-left).

Global Handle (@id) *(revived)*

- **What it is.** Pick a unique @handle used across chat and the feed.
- **Code.** `screens/profile/global_handle_edit_screen.dart`, `screens/profile/profile_global_id_section.dart`. Scan: 2 file(s), 666 lines; 18 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `global_handle_edit_screen.dart`, `profile_global_id_section.dart`.
- **Current state. Unrouted** (REVIVE). The live chat search already uses global IDs (`user_search_global_id_screen.dart`).
- **Defects.** People can be searched by global ID but can't set one.
- **Elevation.** Revive in onboarding (after account creation) and profile: choose an @handle, with suggestions in the target language (e.g. @ade_learns_igbo) and safety checks.
- **UX and render.** Inline availability check. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me → Profile → Handle; onboarding.

Progress Dashboard

- **What it is.** Charts of learning progress.
- **Code.** `screens/progress/progress_dashboard_screen.dart`, `screens/global/global_progress_screen.dart`. Scan: 2 file(s), 904 lines; 1 mock/fake marker(s); 11 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `global_progress_screen.dart`.
- **Current state.** Works (route `progress`). `global_progress_screen.dart` unrouted (CONSOLIDATE as a tab).
- **Defects.** Vanity metrics (XP) rather than ability.
- **Elevation. Can-do progress**: CEFR can-do statements ticked with evidence, skill radar (listen, speak, read, write, tone), words known, speaking minutes and forecast ("A2 in about 6 weeks at this pace"). Tabs: Me, Languages, Global.
- **UX and render.** Calm charts; evidence links to recordings. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me → Progress.

Achievements & Badges

- **What it is.** Achievements and a badge collection.
- **Code.** `screens/achievements/achievements_screen.dart`, `screens/gamification/badge_collection_screen_material3.dart`, `screens/gamification/badge_collection_screen.dart`. Scan: 3 file(s), 1,115 lines; 20 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (routes `achievements`, `badges`).
- **Defects.** XP awards can be self-awarded (P0-8); badges for volume, not mastery.
- **Elevation. Cultural badges** earned by real milestones: first proverb used correctly, 100 tones nailed, first native conversation. Artwork from regional craft traditions (adire, kente, Zulu beadwork), reviewed. Server-verified.
- **UX and render.** Badge cabinet with stories behind each design. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me → Achievements.

Leaderboards

- **What it is.** Global and country leaderboards.
- **Code.** `screens/gamification/leaderboard_screen.dart`, `screens/tabs_view/standings/standings_tab_material3.dart`. Scan: 2 file(s), 782 lines; 9 `Text(` widgets vs 0 l10n lookups.

- **Current state.** Works (route `leaderboard`, standings tab).
- **Defects.** Cheatable (P0-8); demotivating for beginners.
- **Elevation. Leagues** of 30 similar learners with weekly promotion, an opt-out, and circle and country boards; points only from verified learning.
- **UX and render.** League ladder; your row pinned. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me → Leagues; Village → Circles.

Magic Items

- **What it is.** Power-ups and items.
- **Code.** `screens/gamification/magic_items_screen.dart`. Scan: 1 file(s), 491 lines; 20 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `magic_items`).
- **Defects.** Economy self-mintable (P0-7).
- **Elevation. Cosmetic and helper shop**: avatar outfits from regional designers, streak freezes and hint tokens; server-validated currency; nothing pay-to-win.
- **UX and render.** Shop grid with try-on on the avatar. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me → Wardrobe & Shop; Village → Market.

Settings, Privacy & Appearance

- **What it is.** App settings, privacy, password change, dark mode, account deletion.
- **Code.** `screens/settings/settings_screen_material3.dart`, `screens/settings/settings_screen.dart`, `screens/settings/privacy_settings_screen.dart`, `screens/settings/change_password_screen.dart`, `screens/tabs_view/profile/change_password_screen.dart`, `screens/tabs_view/profile/delete_account_dialogue.dart`. Scan: 6 file(s), 3,184 lines; 1 mock/fake marker(s); 32 `Text(` widgets vs 1 l10n lookups; not reachable from `main.dart`: `settings_screen.dart`.
- **Current state.** Works (route `settings`; dark mode toggle in drawer). Older settings screen unrouted (CONSOLIDATE). Two change-password screens.
- **Defects.** Privacy controls thin for minors; duplicates.
- **Elevation.** Grouped settings: Learning (daily goal, reminders, voice), Privacy & Safety (who can message me, discoverability, data export/delete per NDPA), Appearance (theme, text size, reduced motion), Account.
- **UX and render.** Standard grouped list; search in settings. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me → Settings.

Subscription & Plans *(revived)*

- **What it is.** Plans and purchase.
- **Code.** `screens/subscription/subscription_screen.dart`. Scan: 1 file(s), 286 lines; 5 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `subscription_screen.dart`.
- **Current state. Unrouted** (REVIVE). No live paywall in the app (rg for Paywall/RevenueCat/purchasePackage in live files: none); purchases not verified server-side (P0-27).
- **Defects.** No monetisation surface.
- **Elevation.** Revive as **LingAfriq Plus**: free core learning forever; Plus adds unlimited Polie voice, offline packs, mentor credits and family sharing. Local pricing (NGN, KES, ZAR) and mobile-money options; server-verified receipts.
- **UX and render.** Honest comparison table; no dark patterns. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.

- **Lives in.** Me → Plus.

Family / Guardian Dashboard *(revived)*

- **What it is.** Family plan management.
- **Code.** `screens/subscription/family_dashboard_screen.dart`. Scan: 1 file(s), 529 lines; 10 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `family_dashboard_screen.dart`.
- **Current state. Unrouted** (REVIVE).
- **Defects.** No guardian controls although onboarding collects "child" (P0-10).
- **Elevation.** Revive as the **Guardian hub**: link child profiles, see progress, set safe-social rules (circles only, no DMs), screen time and a weekly report. Required for kids' mode.
- **UX and render.** Child cards with progress and controls. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me → Family (guardians).

About & Policy *(revived)*

- **What it is.** About the company and legal policies.
- **Code.** `screens/tabs_view/profile/about_us_screen.dart`, `screens/tabs_view/profile/app_policy_screen.dart`. Scan: 2 file(s), 102 lines; not reachable from `main.dart`: `about_us_screen.dart`.
- **Current state.** About is **unrouted** (REVIVE); policy works (route `policy`).
- **Defects.** No mission story in the app.
- **Elevation.** Revive About as **Our Story**: the mission, the native-speaker community, language-credit roll (contributors) and AI transparency (which models, how data is used).
- **UX and render.** Editorial page with photos of contributors (with consent). Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me → About.

Sign in, Sign up & Email Verification

- **What it is.** Authentication flows.
- **Code.** `screens/auth/world_class_login_screen.dart`, `screens/auth/world_class_signup_screen.dart`, `screens/auth/forgot_password_screen.dart`, `screens/auth/email_verification_screen.dart`, `screens/auth/login_screen.dart`, `screens/auth/sign_up_screen.dart`. Scan: 6 file(s), 2,972 lines; 30 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `login_screen.dart`, `sign_up_screen.dart`.
- **Current state.** Works. Older login/sign-up unrouted (CONSOLIDATE). Login enumeration and no lockout (P0-25); every release logs users out (P0-20).
- **Defects.** P0-20, P0-25.
- **Elevation.** Try-before-signup (first lesson without an account), then passkey, Google/Apple, phone OTP (key for Nigeria/Kenya) and email; generic error messages; lockout.
- **UX and render.** Short, branded flows. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Before the app; Me → Account.

Onboarding & Placement

- **What it is.** First-run onboarding and placement test.
- **Code.** `screens/onboarding/unified_onboarding_screen.dart`, `screens/onboarding/placement_test_screen.dart`, `screens/onboarding/kijiji_onboarding_screen.dart`,

`screens/onboarding/enhanced_onboarding_flow_screen.dart`, `screens/onboarding/modern_onboarding_screen.dart`, `screens/onboarding/onboarding_screen.dart` (+2 more). Scan: 8 file(s), 8,766 lines; 1 mock/fake marker(s); 110 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `kijiji_onboarding_screen.dart`, `enhanced_onboarding_flow_screen.dart`, `modern_onboarding_screen.dart`, `onboarding_screen.dart`, `placement_quiz_screen.dart`.

- **Current state.** Unified onboarding works; answers stored as `Map.toString()` (P0-21). Five other onboarding variants and placement quiz unrouted (CONSOLIDATE; harvest Kijiji's village story).
- **Defects.** P0-21; long forms before value.
- **Elevation.** See §5: 90 seconds to the first spoken word. Choose language and why, speak one greeting, meet your avatar, get a 3-minute adaptive placement. Kijiji's village-arrival story becomes the narrative frame.
- **UX and render.** Full-screen illustrated steps; skip always available. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** First run.

Splash

- **What it is.** Launch screen.
- **Code.** `screens/splash/splash_screen.dart`. Scan: 1 file(s), 13 lines.
- **Current state.** Works.
- **Defects.** Plain wait screen [verify content].
- **Elevation.** A 1-second greeting in the learner's target language, with the avatar waving (Rive), cached for offline.
- **UX and render.** Brand mark + greeting. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Launch.

Features Guide, Navigation Hub & First-use Tips *(revived)*

- **What it is.** A guide to app features, a navigation hub, and per-feature intro screens.
- **Code.** `screens/help/features_guide_screen.dart`, `screens/help/stitch_navigation_hub_screen.dart`, `screens/help/feature_preloader_screen.dart`, `screens/help/games_preloader.dart`, `screens/help/gamification_preloader.dart`, `screens/help/polie_preloader.dart`. Scan: 6 file(s), 1,353 lines; 1 mock/fake marker(s); 9 `Text(` widgets vs 0 l10n lookups; not reachable from `main.dart`: `feature_preloader_screen.dart`, `games_preloader.dart`, `gamification_preloader.dart`, `polie_preloader.dart`.
- **Current state.** Guide and Stitch hub work (routes `features_guide`, `stitch-hub`). **The four preloaders are unrouted (REVIVE as contextual tips).**
- **Defects.** Features are explained in one long list instead of in context.
- **Elevation.** Becomes the **All features** directory (A–Z, searchable, with locked items showing "unlocks at…" and a preview), which replaces the drawer so nothing is lost. The revived preloaders become **first-use tips**: a 3-card intro the first time someone opens Polie, games or gamification.
- **UX and render.** Directory grouped by hub with icons; tips as bottom sheets. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me → All features; contextual first use.

Staff & Micro-mentor Reports

- **What it is.** Staff view of mentor reports.
- **Code.** `screens/staff/micro_mentor_reports_screen.dart`. Scan: 1 file(s), 340 lines; 15 `Text(` widgets vs 0 l10n lookups.
- **Current state.** Works (route `staff-micro-mentor-reports`; drawer item).

- **Defects.** Staff tooling in the consumer app; role checks need verification [verify].
- **Elevation.** Keep in-app for staff on the go, behind server-checked roles, with a fuller version in the admin console (§14).
- **UX and render.** Role-gated; compact queue. Rendered on the single unified design system (§3.4), with tone-safe Noto fonts, 48 dp targets, dark mode and l10n.
- **Lives in.** Me → Staff (role-gated).

---

*Method note: every finding cites file and line in the read-only copy dated as provided on 17 Oct 2026. Items marked **[unverified]** or **[verify]** could not be confirmed from source alone (server configuration, deployed environment variables, store-console settings, or legal thresholds). No `.riv` files, audio or `.git` history were present in the copy, so audio quality and the secret history in git were not inspected directly. The Rive avatars have not been built yet (confirmed by Tope), so §4 is a forward brief, not an audit.*